

Experimental apparatus for universal spin-motion control in multi-ion registers



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0.1 Overview

Proposed thesis structure

1. Introduction

2. Theory

- Ion trap theory needed.
 - Trapping fields
 - QP transitions + group theory on this
 - Ion-light interactions. With spin, with motion.
 - MS gates + SDFs
- Noise (Look at Harty + Amy thesis)
 - Model for general noise. Using error channels + PTM + Choi matrices. Discuss classifications of noise channels.
 - How to measure noise. Tomography, RB.
 - How to simplify noise using RB.
 - How k-copy RB simplifies noise but retains structure.
 - Group + representation theory for RB
 - Reducing physical noise. Coherent with H engineering, incoherent with fast gates.
 - Reducing the impact of noise by error mitigation techniques.
 - Virtual distillation on spatially correlated noise.

3. **Experimental Apparatus** – Description of the new experimental apparatus.

- Intro for why we built a new apparatus. What we wanted to achieve with it.
- Block diagram of whole experiment highlighting automation.
- See current chapter for most sections.

- Ground loops and how we found them. Suggest improvements to current system.
- Single addressing system. Chat to IRO about what minimal info to include.
- Section on Pulse shaping with phaser, sinara? Link this to requirements/theory for frequency selectivity.
- Verdi noise spectra
- 729 laser characterisation. Including linewidth, cavity ringdown, noise spectra. FNC noise reduction. Keeping it alive? Autolock feature.

4. **Experimental Characterisation** – spin/motion/spin-motion elementary interactions that are used throughout the thesis. This is the chapter submitted for confirmation of status.

- Spin coherence times with and without box, and with and without echo sequence.
- 729 laser characterisation. Including linewidth, cavity ringdown, noise spectra. FNC noise reduction. Keeping it alive?
- New mode stability with enclosed resonator
- Motional coherence time
- Going to long chains
- micromotion compensation
- All mode frequencies. Using James.
- All modes SBC. Here also a discussion on sub-doppler cooling options.
- MS gates improved + Walshing
- Error budget for MS gates
- QP matrix elements -> polarisations -> Stark shifts?

5. **k-copy Randomised Benchmarking** – Presentation of the theory and experimental results demonstrating parallel single-qubit randomised benchmarking to distinguish local and global noise contributions on each qubit. This is the paper extended.

- How k-copy RB simplifies noise but retains structure.
- Group + representation theory for k-copy RB to derive correlated populations
- Shared global fields leading to spatially correlated noise. Error channel for this.
- Optimal sampling for multi-exponential
- Parameter range
- Noise injection calibration
- Stark shift calibration
- Extracting errors for large parameter space
- Parity measurements
- Pairwise measurements

6. **Motional Control** – Presentation of the experimental results demonstrating motional control

- Describe Two-SDF toolkit
- Describe tomography method for motional states. Could show Fock state here.
- Results:
 - Squeezed state
 - Beam splitter
 - Parity operation
 - Fredkin gate

7. **Virtual Distillation** – Presentation of the experimental results demonstrating virtual distillation.

- two copy, B matrix demonstration
- Virtual distillation on spatially correlated noise
- any-to-any two-qubit gate
- $\pi/4$ calibration
- Motional virtual distillation. What errors does this mitigate? How to implement it? Results?

8. **Conclusion and Outlook**

1

Introduction

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1.1 Motivation

The following chapter details the characterisation work performed during the commissioning of a new ion trap apparatus. Over the past two years, my colleagues and I have been constructing this system, which is now approaching readiness for our desired experiments. In this *Motivation* section, I will provide a high-level overview of the physics we want to explore, state the unique aspects of the apparatus that enable these experiments, and finally, give a timeline for planned future work. The complexity of our system necessitates collaboration, and so I would like to acknowledge the rest of the *FastGates* team, for their extensive contributions to bringing the system online.

There are two research directions that we are pursuing with this apparatus. Both involve the interplay between the “spin” electronic degrees of freedom and the motion of the ions which we describe by quantum harmonic oscillators.

The first is the demonstration of fast, high-fidelity two-qubit entangling gates between trapped ions. High-fidelity entangling gates are critical for developing a quantum computer which can perform useful calculations with high success probability [1]. Gate errors associated with decoherence of the quantum state are often dominated by incoherent processes, such as spontaneous emission, or interactions between the qubit and the environment [2]. These errors accumulate with time, and so may be mitigated by increasing the algorithm “clock-speed”, hence our desire for fast entanglement [3, 4]. Typically in ion traps, local two-qubit gates are performed via the ion’s shared motion, which may be excited by interaction with a laser [5, 6]. As the effective Rabi frequency approaches that of the frequency separation of the motional modes, the dynamics between spin and motion become complicated by multiple simultaneous mode excitations [4]. Our apparatus is designed to provide the required laser intensities and mode frequencies such that we can explore this regime, and demonstrate the required control to tame these multiple mode excitations and perform high-fidelity entangling gates.

The second direction is the use of the motion of the ion directly as a quantum resource [7, 8]. This falls into the continuous variable quantum computing paradigm (CVQC), where instead of discrete qubits, the quantum state is described by bosonic “qumodes” [9]. CVQC reduces the technical overhead of scaling discrete systems, as larger Hilbert spaces are easily accessed due to the many level nature of the individual qumodes. We want to demonstrate a universal set of gates for CVQC, which includes, single-mode Gaussian and non-Gaussian gates (demonstrated in [10]), and two-mode entangling gates. Further, we want to demonstrate the ability to simultaneously leverage both bosonic and spin degrees of freedom of our ions to efficiently encode problems in the full hybrid system we have access to [11, 12]. Our apparatus has both precise motional control and stability through our trap choice, and the individual addressing system that is required when interacting with many-ion chains.

Details of the apparatus are presented in the Appendices. Summarised here are the unique aspects enabling the desired experiments:

- **High laser intensities:** The system is designed to provide high laser intensities, which are required to drive fast entangling gates. This is achieved through the use of a 5 W Ti:Sapph system (section ??) and two high numerical aperture ($\text{NA} = 0.6$) objectives (section 3.4).
- **Single ion addressing:** Using crossed acousto-optic deflectors, we can selectively address individual ions in a multi-ion chain (section 3.6.5). This enables fast entangling gates between ions in a long chain, and the ability to perform motional interactions on selected modes.
- **Standing wave addressing:** The system is designed to provide standing optical waves at the ion position. Standing waves are required for some fast gate schemes [13].
- **Motional control:** The Paul trap used in the system, produced by *National Physical Laboratory* in the UK, has low heating rates (section 4.3.2), deep trapping potentials relative to surface style ion traps, and flexible control over DC confinement (section 3.2) (e.g. ion shuttling). Low heating rates are required for long motional coherence times (section 4.3.3).

Here, the characterisation of the spin qubits is in the context of discrete variable quantum computing. We are interested in the fidelity of qubit state-preparation (section 4.2.1), measurement (section 4.2.2), single- (section 4.2.5) and two-qubit gates (section 4.4.2). These individual elements comprise longer circuits, and so all contribute to the overall fidelity of any chosen algorithm.

The motional characterisation is currently focused on the stability of the modes (section 4.3.3), and the ability to perform spin-dependent forces, SDFs (section 4.4.1). SDFs are the base interaction on which many of the desired CVQC gates are comprised [14], and so these must be well characterised and understood in our system.

To push the system forwards, in the next 6 months, we will focus on commissioning the single addressing system, and on demonstrating squeezing of the motional

state of the ion.

As described above, the single addressing system is a critical component for many of our planned experiments. Section 3.6.5, and [15] describe the system in detail. The immediate next steps are to align the constructed optical system to the ion chain, and to implement the logical and RF control system required to drive the acousto-optic deflectors. Once this is complete, we will characterise the crosstalk of the system by performing single addressing of ions in a chain and measuring spin populations. We will then be able to demonstrate fast entangling gates between ions in a chain, and the ability to perform motional interactions on selected modes. Squeezing of the motional mode via the method outlined in [14], is a significant step towards demonstrating the universal set of gates for CVQC. The constituent interactions of the effective squeezing gate (two SDFs), are the same interactions needed for trisqueezing [10, 16] — a non-Gaussian gate, and for two-mode squeezing [17] — an entangling gate. Furthermore, once we have demonstrated generating squeezing characteristic functions, there is a clear path to experimentally demonstrating the extraction of Feynman diagram terms with the hybrid system being used as a quantum simulator [11].

2

Theory

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2.1 Trapping Ions

2.1.1 Trapping Potential

From Earnshaw's theorem, $\nabla^2 V = 0$, a stable stationary point in 3D cannot be realised using only static electric potentials, V , as if the potential is confining in two dimensions, it will be anti-confining in the third. Therefore, to achieve stable trapping, either an oscillating electric field (Paul trap [18]), or a static magnetic field (Penning trap) is used.

Will include pseudopotential relating V_{RF} and Ω_{RF} to the radial mode frequencies ω_m (see Vera eqn 3.4),

$$\omega_r \propto \frac{V_{RF}}{M\Omega_{RF}}. \quad (2.1)$$

Further, will use James to give ion spacing and non-COM mode frequencies as will be relevant for single addressing, fast gates (if discussed), and for motional mode excitati as will be relevant for single addressing, fast gates (if discussed), and for motional mode excitation.

2.2 Light Matter Interactions

2.2.1 Rabi Flopping

2.2.2 Motional Coupling

2.2.3 Geometric Phase Gates

3

Ion Trap Apparatus

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After 70 years of ion confinement development (since the quadrupole mass spectrometer in the 1950s [18]), and the myriad use-cases from time standards, to analytic chemistry, to quantum computing, there are naturally many systems and these apparatus chapters in existence today. As such, we structure this chapter in the following way: Following on from the standard requirements of an ion trap

system, provided in the introduction section 2.1, we discuss the unique motivation and design specifications of our system; We then present an abstract schematic of the system in its current form, highlighting the constituent subsystems, and the optical and electrical signal paths; The remaining sections of the chapter contain the technical detail, perhaps more relevant to the practicing experimental student, and the hardware enthusiastic.

3.1 Specification

Ion-trap experiments are exceptionally versatile in their application, however in the vast search space of quantum-information applications, we want a system particularly tailored for three avenues of study:

1. Error channels on multi-qubit chains.
2. Continuous variable quantum computing and simulation utilising the spin and motional degrees of freedom of the confined ions.
3. Fast two-qubit entangling gates via phase stabilised optical standing waves.

These latter two research directions were explored within our group on a previous apparatus, leading to state-of-the-art two-qubit entanglement rates on the ion trap platform [4, 13], creation of highly non-classical motional states [10, 19], and a simulation of Z2 gauge theories encoded via the spin and motional degrees of freedom of a two-ion chain [saner2025real-time, 20]. However, hard limitations of both scaling these demonstrations to longer ion chains, and of interaction strength due to available optical intensity, were often restrictive. To solve both these issues, a new apparatus has been designed, constructed, and characterised, motivated by these applications. Colloquially the system is named “*FastGates*”, owing to its experimental objectives, and we shall refer to it as such throughout this chapter.

These objectives give the following specific control requirements:

- Storage of medium length (5-10) qubit registers, section 3.2.

- Individual internal state control of each qubit, section 3.6.5.
- Stable motional modes for storing quantum information, section 3.2.1.
- Dual high numerical aperture (NA) optical access for high-intensity standing waves, section 3.4.

As the lifetime of such an apparatus extends past a single PhD generation, this thesis serves as a point-in-time description of the system, and where applicable, the planned future extensions will be highlighted as such. An abstract schematic of the current configuration is shown in figure 3.1 with blocks resembling either electronic, mechanical, or optical subsystems.

3.2 The Ion Trap

Recently, the micro-fabricated surface style linear Paul trap has gained popularity due to the maturity of chip fabrication technologies [22] and the potential route to scalability this offers [23]. In the surface trap, the 3D radial and axial electrodes of a “macro” trap are effectively projected onto a 2D surface. The stable confining point of such a trap is typically designed to be on the order of 50 – 100 μm above the chip surface. The maturity of chip fabrication has enabled the manufacturing of complex multi-zone devices with many DC electrodes, amenable to ion shuttling [**<empty citation>**], and hence furthering progress towards Quantum CCD type architectures [23]. However, this surface style geometry typically comes with two costs: the depth of the trapping potential is often greatly reduced due to anharmonicity [**<empty citation>**], and the close proximity of the surface to the ion can be a large contributor to motional heating rates [**haeffner**, 2, 24].

Our trap, provided by the *National Physical Laboratory* in the UK, brings together the advantages of chip fabrication as well as the low heating rates and high trapping depths of a macro 3D style trap through utilising a micro-fabricated 3D trap geometry with greater ion-surface distances than typical planar traps. Details on its design and characterisations can be found in Ref. [21, 25, 26]. Figure 3.2 shows

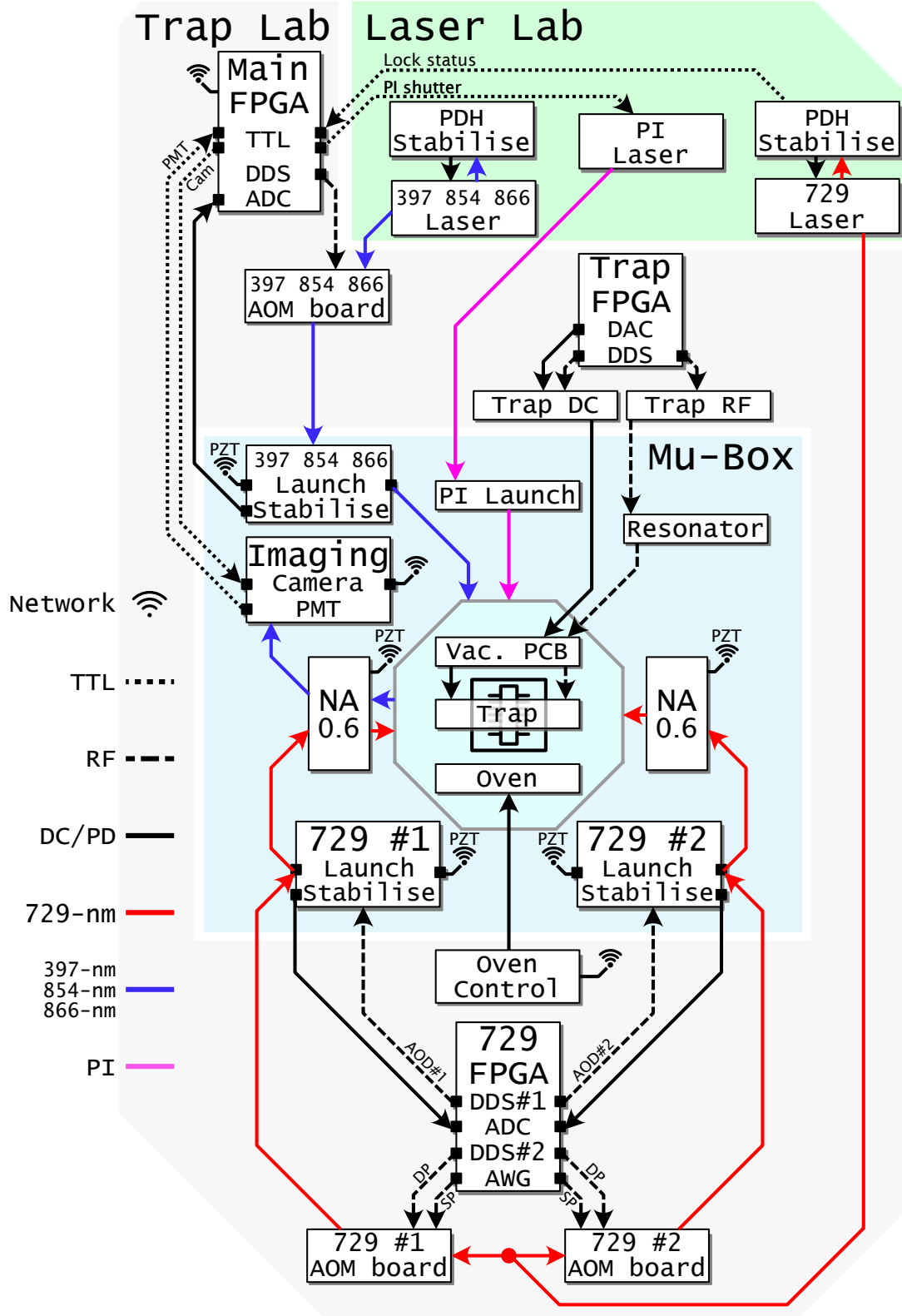


Figure 3.1: Block diagram schematic of the *FastGates* apparatus. Blocks resemble discrete subsystems, which shall be discussed within this chapter. Lines resemble relevant optical (free-space or in-fibre), RF, DC, and communication paths between subsystems. Block location signifies whether the device is in-vacuum, within the magnetic shielded (MuMetal) box, within the “trap-lab”, or within the “laser-lab”.

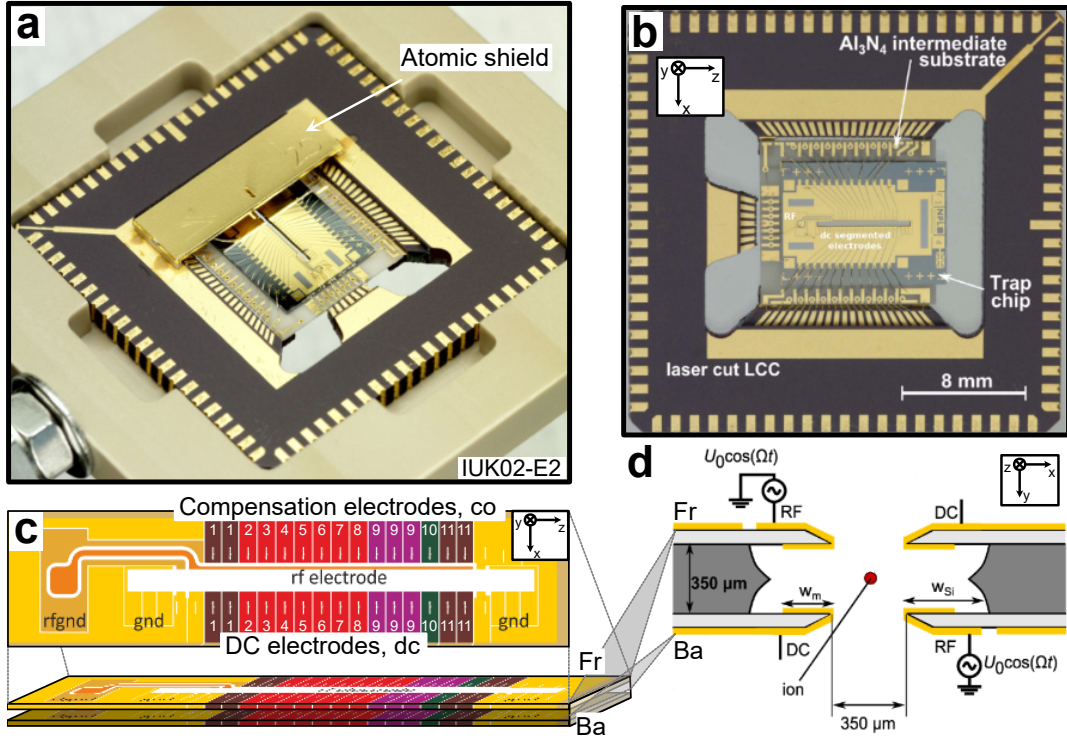


Figure 3.2: Schematics of the *NPL* trap chip used in this experiment. **a)** A photograph of the trap chip used in the *FastGates* experiment before being placed under vacuum. The top “atomic shield” can be seen here. **b)** A front view of the trap chip. Axial confinement is provided by a subset of the segmented DC electrodes. Radial confinement is provided by the RF rails. The trap chip and leadless chip carrier (LCC) are highlighted. **c)** A schematic of the electrode geometries and labelling for the front-face of the trap chip. A similar arrangement exists for the back-face however with the RF rail on the diagonally opposite electrode. **d)** A slice view of the trap. The red point here represents the ions, the gold are the RF and DC electrodes, and the grey is the silica substrate. It can be seen that the ion is $350 \times \sqrt{2}/2 \simeq 250 \mu\text{m}$ away from the nearest electrode. The numerical aperture of the ion plane due to the electrodes is ~ 0.71 . Figures b) and d) adapted from Ref. [21], other panels adapted from private communication with the *NPL* team.

the electrode geometry of the trap and the relevant length scales. The trap package consists of an outer leadless chip carrier (LCC) with outer pads routing DC, RF, and ground connections to an intermediate Al₃N₄ substrate. The connection to the front and back surfaces of the trap chip is made through vias and wire bonds to the intermediate substrate [wilpers2013compact]. The trap chip is a monolithic double-sided silicon device with gold plated electrodes on the front (Fr) and back (Ba) surfaces. These two layers are shown schematically in the bottom panel of (c). The electrode schematic for the front surface is shown, and the electrode indexing

convention is highlighted. Electrodes recessed from the ion due to the RF rail are named “compensation”-electrodes (co), while electrodes on the opposite side provide the strong DC potential required for axial confinement¹. We typically trap ions centered on electrode dc5. Panel (d) shows a slice view of the trapping region, highlighting the ion position between the front and back surface electrodes. The dual RF rails and DC electrodes are shown in gold, and the undercut silicon bulk is highlighted in dark grey.

The next two subsections describe the source of RF and DC voltages to the trap electrodes. The physical mounting of the trap in the vacuum system is discussed in section 3.4, and the optical access to the trap is discussed in section 3.4.

3.2.1 Trap RF Chain

To utilise radial motional modes for low error quantum gates, we require the radio frequency (RF) field that produces the desired trapping pseudopotential to be both frequency and amplitude stable. Furthermore, as the natural time scale for two-qubit gate durations is $\tau = 2\pi/\omega_m$ (see section 2.2.3), where ω_m is the motional mode angular frequency, then greater motional mode frequency is desired for fast entanglement. From equation 2.1, it can be seen that radial motional mode frequency is proportional to RF voltage, and inversely proportional to the frequency. The *NPL* trap used in the *FastGates* apparatus is rated for a max peak-to-peak voltage of 400 Vpp on the RF electrodes. To target high motional mode frequencies, saturating this maximum voltage is desirable, however, for safe operation, typically the trap is provided a maximum RF voltage $V_{RF} = 360$ Vpp. The RF chain which provides this signal is schematically shown in figure 3.1, beginning with the *Trap FPGA*. Each element of the chain is designed to provide a low-noise, high-voltage RF signal to the trap electrodes, and is discussed in turn below.

The frequency source is a direct-digital-synthesiser (DDS), named *Urukul* [kasprowicz2022urukul],

¹Note: Compensation electrodes co1 and co11 on both front and back surfaces are grounded on the trap package, and so are not available for use. There is a continuity fault on front dc11 and so this electrode is also not available for use.

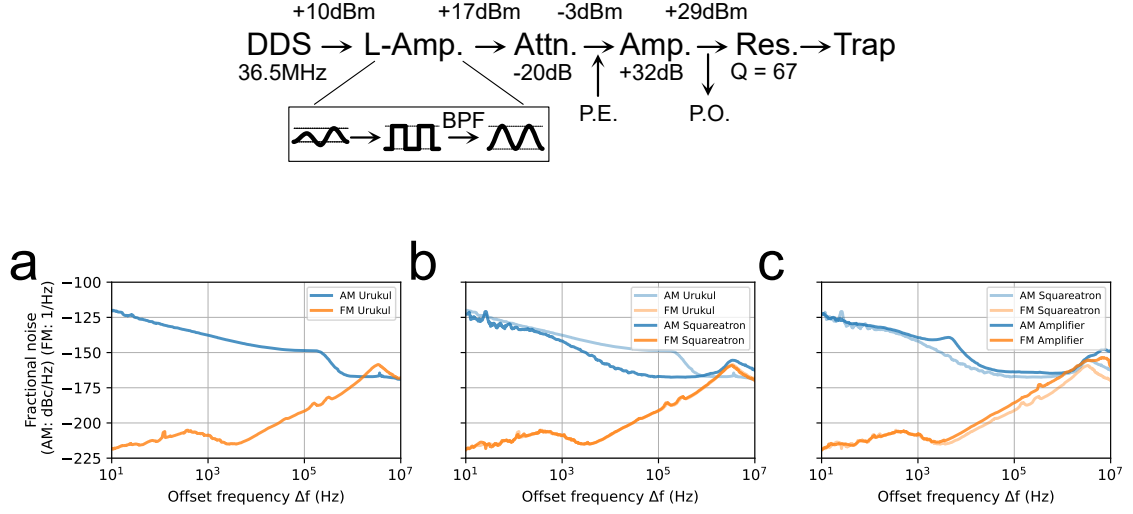


Figure 3.3: Characterisation of the RF chain. **a)** SSB spectra for AM and PM, given as fractional noise, measured after the *Urukul*. **b)** Measured after the *Squareatron*. **c)** Measured after the amplifier.

part of the *Artiq Sinara* [27] hardware ecosystem. *Urukul* is used extensively in our lab for driving acousto-optic modulators (AOMs) and here used as the source drive of our trap RF. The *Urukul* is operated at maximum output power, +10 dBm, and with a frequency ~ 35 MHz (set by resonator, discussed below). As can be seen in equation 2.1, amplitude and frequency instability will directly lead to radial mode instability. Expanding for small amplitude, dV_{RF} , and drive frequency, $d\Omega_{RF}$, errors we find the motional mode frequency, ω'_r , is related to the ideal frequency, ω_r , by

$$\omega'_r = \omega_r(1 + dV_{RF}/V_{RF} - d\Omega_{RF}/\Omega_{RF}), \quad (3.1)$$

where we can see both errors affect the motional mode frequency linearly. The instability of the oscillator can limit motional mode coherence time (see section 4.3.4 for characterisation with the ion). We measure noise originating from both amplitude modulation (AM) and frequency modulation (FM) across the RF chain using a phase noise analyser². This was done to verify low-noise operation of the individual devices, and to estimate expected motional mode errors due to the inherent noise on the RF chain.

²Rhode and Schwartz FSWP PNA

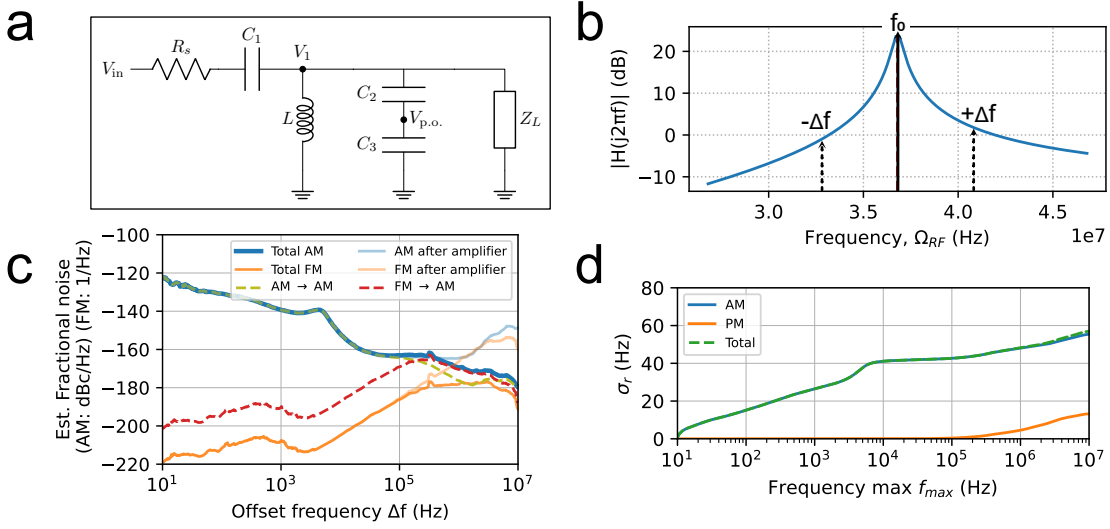


Figure 3.4: Expected SSB after the resonator, and estimated mode frequency noise. **a)** Resonator circuit diagram. **b)** Resonator lineshape, given by magnitude of the transfer function $|H(f)|$. **c)** Expected AM and FM fractional noise spectra after the resonator. **d)** Integrated RMS noise of radial motional mode frequency due to the RF chain.

We quote FM as the fractional frequency stability power spectral density (PSD),

$$S_y(f) = \frac{f^2}{f_0^2} S_\phi(f), \quad (3.2)$$

where $S_\phi(f)$ is the phase noise PSD (which is the Fourier transform of the Autocorrelation function) f is the offset frequency, and f_0 is the carrier frequency. $S_y(f)$ has units of $\left[\frac{1}{\text{Hz}}\right]$. AM noise is quoted as the single sideband (SSB) power, S_α , in a 1 Hz bandwidth at an offset frequency, f , from the carrier, normalised to the total power of the carrier, and is here given in units of $\left[\frac{\text{dBc}}{\text{Hz}}\right]$.

Figure 3.3, panel (a) displays the *Urukul* AM and FM spectra. The *Urukul* is phase referenced to the master FPGA board (*Kasli*), which is in turn phase locked to an external 10 MHz atomic clock³. It can be seen that AM noise dominates up to ~ 1 MHz, suggesting that this will be the limiting error source for motional mode stability. The AM noise may be reduced by using a non-linear voltage clamp. We use an “ultra-low noise limiting amplifier” named *Squareatron* developed previously in the group, and further refined by *Oxford Ionics*. *Squareatron* takes an input RF

³**ToDo: Abaqus atomic clock PN**

signal and converts to a stable amplitude square-wave output which is subsequently bandpassed around the desired RF frequency to recover the sinusoidal signal, but with significantly reduced AM noise. The AM and FM after the *Squareatron* is shown in panel (b). It can be seen that between 1 kHz and 1 MHz, AM is considerably suppressed. There appears to be an increase in AM at higher frequencies **ToDo: explore this further**. The *Squareatron* has a 17 dBm output power, which is attenuated by -20 dB before an amplification stage to reach the desired power for the trap. Highlighted in the above chain schematic is the option to input a parametric excitation signal (P.E.), which we shall discuss further in section 4.2.3. For the following measurements, the P.E. branch was removed. AM and FM introduced by the low-noise amplifier⁴, can be seen in panel (c). Most notably an increase in AM is seen between 1 kHz and 10 kHz.

To impedance match the $50\ \Omega$ line from the amplifier to the small capacitance of the trap electrodes and vacuum feedthrough, a resonant matching circuit is used. This “resonator” has three main effects on our system:

1. The source $50\ \Omega$ is matched to the (almost) purely capacitive load.
2. The voltage step up provides the necessary amplitude for high motional mode frequencies.
3. The input RF is bandpassed around the desired trapping frequency.

The latter two points are practically useful, especially the bandpass effect of the resonator attenuating noise on the RF line, however, changes to the resonators Q factor and central frequency due to thermal and mechanical noise can be a significant source of low frequency motional mode instability. Opting for good passive stability, a PCB based design⁵ with good thermal management was seen as preferable to large helical resonators, which are often used in ion trap experiments for their high Q factors. Aiming for a moderate Q factor of ~ 70 allows for a good

⁴Mini-Circuits ZHL-1-2W-S+

⁵This style PCB resonator has been used around the Oxford Ion Trapping Group for some time, and is based originally on a design by Tom Harty

voltage step up, and some bandpass filtering, without being overly sensitive to thermal and mechanical noise. The circuit diagram of the constructed resonator ⁶ is shown in figure 3.4 (a). The inductor consists of XX(**ToDo gauge and oxygen?**) gauge Copper was made to have, $L \sim 0.8 \mu\text{H}$, with toroidal iron powder core⁷, and C0G (low tempco) rated ceramic capacitors, $C = 5.4 \text{ pF}$. The vacuum can feedthrough and trap were measured to present an $\sim 18 \text{ pF}$ load. The trap itself presents a $C_T = 4 \text{ pF}$ [**wilpers2013compact**], the cable between the resonator and the vacuum feedthrough contributes $\sim 4 \text{ pF}$, and the remaining presented capacitance is expected to originate from the feedthrough and in-vacuum RF line. In vacuum, the RF line is a *Kapton* coated 0.4 mm diameter single strand copper wire, which connects to an in-vacuum distribution PCB. The RF connection to the trap LCC is made by *fuzz button* vias.

The resonator had a measured quality factor of $Q = 66.7(X)$ (ToDo: get quality factor plot), and a central frequency of $36.5(X) \text{ MHz}$. The effect of the resonator on the AM and FM noise can be calculated through the transfer function, $H(f)$, of the resonator circuit. Capacitors ($C2$ and $C3$) are used to provide a pick-off (p.o.) voltage which can be monitored, and if required, used in a closed feedback loop. We however have not attempted this as of yet, and shall ignore the effect of the pick-off capacitors for this analysis. The transfer function of the simplified circuit is then given by,

$$H(f) = \frac{V_{out}}{V_{in}} = \frac{Z_p}{Z_s + Z_p}, \quad (3.3)$$

where

$$\begin{aligned} Z_p &= \left(\frac{1}{j2\pi f L} + \frac{1}{Z_L} \right)^{-1}, \\ Z_s &= R_s + \frac{1}{j2\pi f C_1}. \end{aligned} \quad (3.4)$$

Given the approximate capacitance and inductance values of the resonator and trap, the analytic $|H(f)|$ is plotted in figure 3.4 panel (b), and shows good

⁶Credit to Emily Hirsch for construction of the recent resonator.

⁷Micrometals T80-6

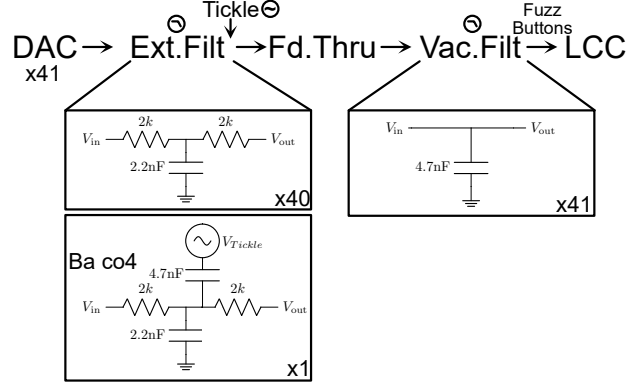
agreement with the measured quality factor and central frequency (here $Q = 69$ and $f_0 = 36.8\text{MHz}$). Final AM and FM spectra are calculated through considering both upper ($+\Delta f$) and lower sidebands ($-\Delta f$), as shown in panel (b), following reference [danielsonAM2018]. We find that for both AM $\rightarrow$ AM and FM $\rightarrow$ FM (where " $\rightarrow$ " represents the transform from input noise type to output noise type) the input PSD are low pass filtered with a cut-off frequency of $\sim 0.25\text{ MHz}$. We also consider the cross term, FM $\rightarrow$ AM, which here we expect to be non-negligible at higher offset frequencies due to the asymmetry of the resonator lineshape. Panel (c) shows the expected fractional noise after the resonator, with total FM noise (orange line), total AM noise (blue line), and contributions from AM $\rightarrow$ AM (dashed green), FM $\rightarrow$ AM (dashed red). Integrating the AM and FM fractional noise spectra between some chosen range of offset frequencies, f_{min} and f_{max} , gives the expected fractional RMS noise on the radial mode frequency of the ions. Assuming a mode frequency of $w_r = 3\text{ MHz}$ the RMS, σ_r , is shown in figure 3.4 panel (d) as a function of f_{max} , with $f_{min} = 10\text{ Hz}$. It can be seen that the expected noise is dominated by the AM contribution, and in the frequency range considered, corresponds to a mode frequency instability of $\sigma_r = 60\text{ Hz}$. This analysis excludes any noise less than 10 Hz , and environmental noise coupled in by the resonator, which may be a significant contribution at low frequencies due to thermal and mechanical noise. The motional modes will also be subject to noise from other coupling mechanisms, such as voltage noise on the DC electrodes, or fluctuating stray electric fields, which we do not consider here.

Here we have characterised both AM and FM noise on the RF chain and estimated the expected contribution to motional mode instability. Further characterisations of the motional mode stability using the ion as a sensor is discussed in section 4.3.3.

3.2.2 Trap DC Voltages

Static fields are required for reshaping the trapping psuedo-potential. This is useful for setting desired mode frequency splittings, for shuttling ions, and for compensating

stray electric fields. As shown in figure 3.2, the *NPL* trap has 40 independently controllable DC electrodes. A further DC bias voltage can be applied directly to the silicon bulk. DC Voltage is supplied to the LCC through the following chain:



and embedded within the schematic 3.1. Two multi-channel DACs, known as *Fastino*, part of the *Artiq Sinara* hardware ecosystem, provide the initial 41 DC channels. The Fastino outputs are filtered by a passive cascaded RC low pass filters with resistances, $R = 2k$, and capacitances $C_1 = 2.2 \text{ nF}$ and $C_2 = 4.7 \text{ nF}$ ⁸ yielding a -3 dB “cut-off” frequency of 7.4 kHz . The low pass filter circuit is shown in the above schematic, with the first stage on an external to vacuum PCB (Ext. Filt), and the final shunt capacitor on an in-vacuum PCB (Vac. Filt)⁹ within close proximity of the trap chip to mitigate RF pickup on the DC lines. The feedthrough to the vacuum system has 51 pins¹⁰ with non-DC pins used to provide ground connections. On the external filter board, one channel, corresponding to electrode *Ba co4*, has a capacitively coupled in AC signal. This originates from a DDS (*Urukul*) and is used to resonantly excite the motional modes of the ion chain for characterisation and calibration purposes (see section 4.3.3). The in-vacuum PCB is connected to the trap chip via a custom PEEK interposer with embedded *Fuzz buttons* vias¹¹. The ion is confined in the “operation” zone, seen in figure 3.2 (c). The axial trapping is provided by electrodes *Ba dc5* and *Fr dc5*, where *Ba* and *Fr* correspond to the

⁸For in-vacuum use. 1206 NPO 4.7nF 100V, SRT 1206A472JBCB

⁹Vacuum compatible PCB using Rogers RT6002. Filter boards were designed by Mariella Minder.

¹⁰*Allectra* 51-way Micro-D

¹¹*Custom Interconnects* Au/BeCu 0.025" diameter x 0.104" length Fuzz Button

back and front plane of the trap respectively. The DAC can provide up to 10 V to these electrodes to produce an axial frequency of 1.6 MHz.

3.2.3 System Grounding

Thinking of removing this section. Poor ground design (or no design at all).

RF signal and return in vacuum may be important.

3.3 Magnetic Field

Due to the fragility of the internal states of the ion (these are *literally* state-of-the-art magnetic field sensors), we must take great care in isolating the ion from a noisy environment. We do not have access to magnetic field insensitive transitions in $^{40}\text{Ca}^+$, therefore uncontrolled Zeeman shifts of the internal atomic energy levels must be suppressed through enforcing a stable external magnetic field. This magnetic field stability is required for long spin coherence times (see section 4.2.6), and for low error single- and two-qubit gates. A permanent magnet array of Samarium Cobalt¹², $\text{Sm}_2\text{Co}_{17}$, is used in a Helmholtz configuration to create a stable magnetic field of $\simeq 5 \text{ G}$ ¹³. Samarium Cobalt was chosen for its low temperature coefficient of remenance, $-0.03\%/K$.

The ion is shielded from unwanted external magnetic fields by a custom designed MuMetal enclosure¹⁴, with an internal dimension of $650 \times 650 \times 650 \text{ mm}$. The shielding consists of two layers of MuMetal shielding with internal shield thickness of 1.5mm and external shield thickness of 2.0mm. The factory quoted shielding factor at the center of the box is $\times 546$ for DC fields. Spin coherence time comparisons with and without magnetic shielding are shown in section 4.2.6.

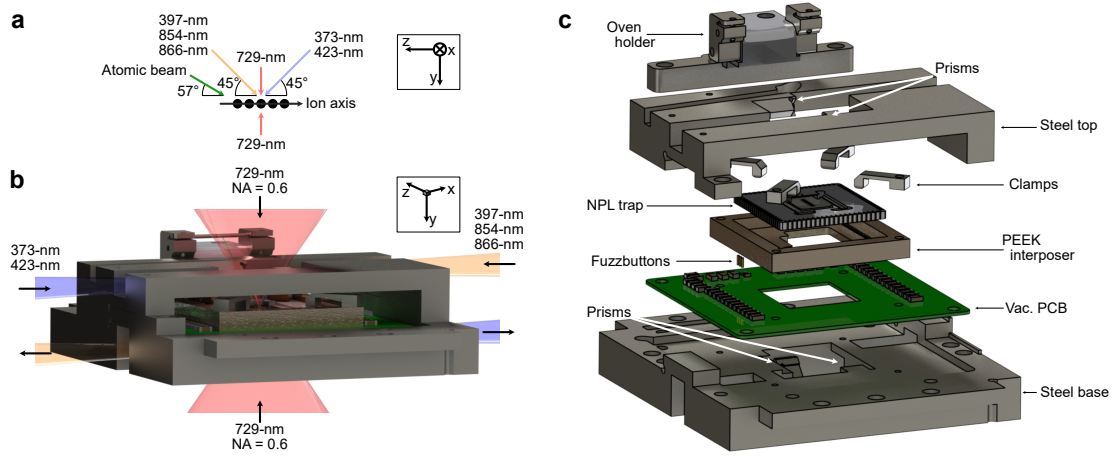


Figure 3.5: a) Laser and atomic beam geometries with respect to the ion chain. b) Compact trap housing with dual NA=0.6 access. The size of the high NA objectives restricts access for other beams, and so in-vacuum mirrors are used to redirect low-NA beams (373, 397, 423, 854, 866-nm) to the trap center. c) Exploded view of the trap housing with components labelled.

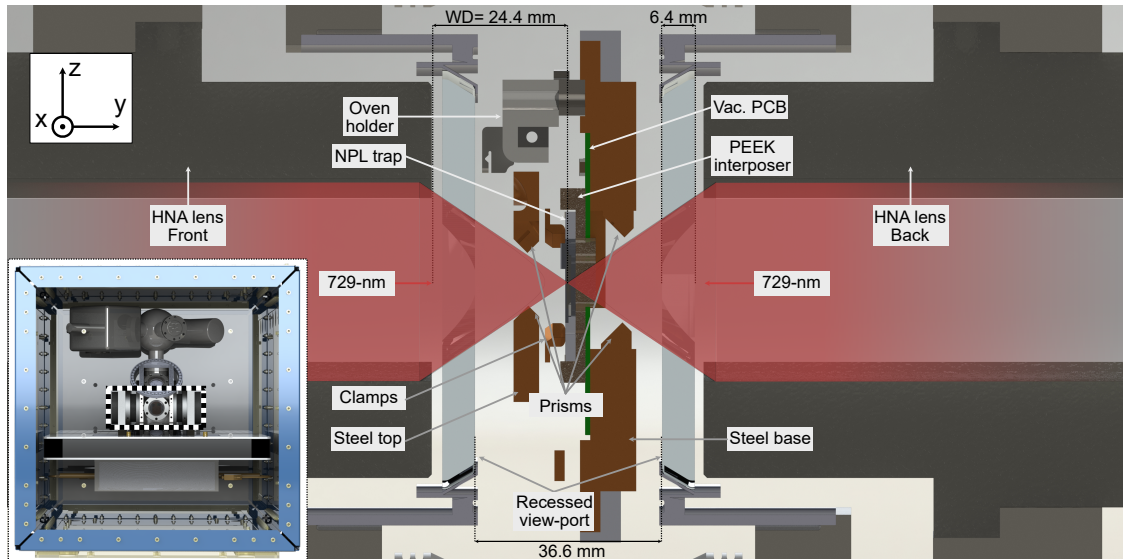


Figure 3.6: Cross section of the vacuum chamber and dual high NA objective sandwich. Lower right inset shows the vacuum chamber housed in the MuMetal enclosure, with dashed lines indicating the cross section region. Custom recessed CF-100 viewports are used to provide optical access for the high NA objectives (working distance = 24.4 mm).

3.4 Vacuum System and Beam Geometry

Ultra High Vacuum (UHV) is required to prevent ion loss. However, UHV equipment is often bulky, and puts constraints on the access and visibility of the ion chain. Here, we describe the custom vacuum system, allowing for multiple beam access geometries including dual high numerical aperture (NA) objectives required for single addressed optical standing waves. The vacuum system was designed by Sebastian Saner, Chris Ballance, and Mariella Minder, and constructed by Sebastian Saner, Fabian Pokorny, and myself.

A residual pressure of $< 10^{-11}$ mbar is desired. For this strict requirement, care must be taken in selecting in-vacuum materials, and thorough cleaning and baking procedures must be followed. A summary of tactics that were useful in the construction of the vacuum system can be found in [28, 29].

The system consists of a 6" spherical-octagonal experimental chamber¹⁵ connected to a spherical chamber with ion pump¹⁶, ion gauge¹⁷ and Titanium sublimator pump (TSP)¹⁸ attached. After the final system bake, a He leak test was performed and no external leaks were found. We do observe a gradual pressure increase that we suspect is either due to the in-vacuum optics, the optics epoxy adhesive¹⁹, or the PCB solder. Regardless, the ion pump and TSP maintain UHV at the desired $< 10^{-11}$ mbar (the TSP is fired every 4 weeks at 41 A for 60 seconds to provide a fresh Ti layer).

Figure 3.5 (a) shows the beam geometries with respect to the ion chain axis. The quadrupole 729-nm beam has high NA (NA = 0.6) access perpendicular to the ion chain axis, while 373-nm, 397-nm, 423-nm, 854-nm and 866-nm beams have low NA (NA < 0.05) access at an angle of 45° to the ion chain axis. The atomic beam from the oven is at an angle of $\sim 57^\circ$ to the ion chain axis. As such, there is a doppler

¹²*Bunting Magnets* E643SM and E644SM.

¹³Designed and constructed by Sophie Howarth.

¹⁴*Magnetic Shields* custom design.

¹⁵Kimball MCF600-SphOct-F2C8

¹⁶Agilent VacIon Plus 20 Pump

¹⁷Agilent UHV-24P Ion gauge

¹⁸Scanwel custom housing

¹⁹EPO-TEK 353ND

shift of the atomic beam with respect to the 423-nm photoionisation laser, which is accounted for in the set laser frequency (given 12° angle between atomic and 423-nm beam, and 500 K mean temperature, expect +126 MHz frequency shift). Figure 3.5 (b) shows the custom trap housing including NA=0.6 access for the 729-nm beams, and in-vacuum mirrors (prisms) to redirect the low NA beams to the trap center. The low NA beams enter the housing via one of the four beam access channels parallel to the trap chip plane, and are reflected through the trap slit. The exiting beam is reflected by a second mirror out of the housing to prevent scattered light incident on the ion chain. Figure 3.5 (c) shows an exploded view of the trap housing with components labelled. The trap chip is interfaced to an in-vacuum PCB (described in section 3.2.2) via a custom PEEK interposer with embedded *Fuzzbutton* vias (one for each DC and RF channel). The trap chip is held in place by 4 steel clamp arms, which attach to a steel base plate which provides mechanical stability and houses the lower in-vacuum prisms. A top steel plate houses the upper prisms. The tube oven is mounted on two steel clamps which are electrically isolated from the base plate by a PEEK spacer. The trap housing is fastened to the vacuum can by 4 steel arms attached to alternating side ports of the octagonal chamber²⁰. Figure 3.6 shows a cross section of the vacuum chamber and recessed dual high NA objectives. Two recessed CF100 viewports²¹ are required to bring the custom objectives²² to the designed working distance of 24.4 mm from the ion chain.

3.5 Imaging System

To image the ion chain, and readout the qubit states, fluorescent scatter of incident 397-nm light is collected. Here we briefly describe the imaging system, while section 4.2.2 describes the qubit readout procedure and fidelity. The imaging system is designed to provide high spatial resolution of the ion chain, while also providing a large field of view to accommodate long ion chains. The imaging system must also be compatible with the high NA objectives used for single addressed optical

²⁰Using *Kimball Groove grabbers* P/N 53-482610

²¹*UK Atomic Energy Authority* P/N VPR100015

²²*Photon Gear, inc.* Custom 18WD Atom Imager

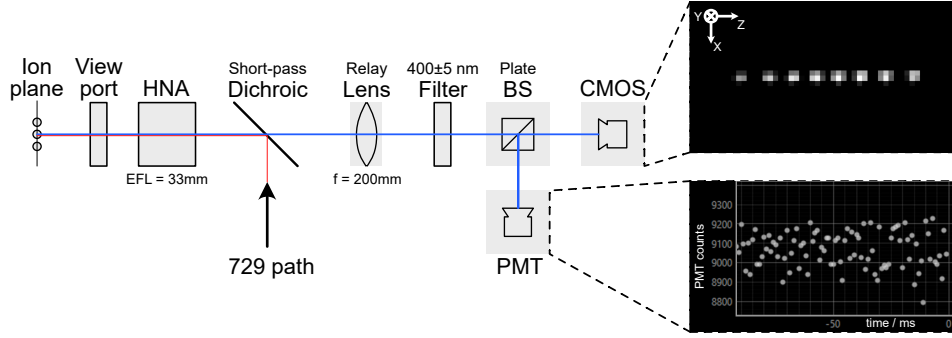


Figure 3.7: Optical path for fluorescent imaging of the ions using both a CMOS camera and a Photon-multiplier tube. Insets on right hand side show typical signal for: spatial resolution of eight ions in a linear chain (top), time resolution of detected photons from the ion chain (bottom).

standing waves. Figure 3.7 shows the optical path schematic of the imaging system, with both a high quantum-efficiency ($\simeq 70\%$ at 397 nm) CMOS camera²³ for spatially selective qubit readout on a multi-ion chain, and a photon multiplier tube (PMT)²⁴, for experiments requiring time resolved photon counting (e.g. micromotion compensation, section 4.2.3). Typical signals of both devices can be seen in right panels of figure 3.7. The optical path consists of a high numerical aperture objective²⁵ (effective focal length, $f = 33$ mm), $NA = 0.6$, a dichroic mirror²⁶ (transmission: 397 nm; reflection: 729 nm), a relay lens²⁷ ($f = 200$ mm), a 90 : 10 plate beamsplitter²⁸, and the camera and PMT.

The objective lens is custom designed to compensate for aberrations due to the recessed vacuum viewport, however, due to the large NA, accurate alignment of all degrees of freedom of the objective is critical. As such, the NA is attached to a 5-axis alignment stage²⁹, with z-axis focus controlled by a piezo micrometer³⁰ for fine adjustment of the ion focus with the MuMetal box closed (requires $< \mu\text{m}$ precision).

²³Hamamatsu Orca-Fusion BT C15440-20UP

²⁴Hamamatsu Photon counting head H10682-210

²⁵Photon Gear, inc. Custom 18WD Atom Imager

²⁶**Thorlabs TODO**

²⁷Thorlabs Achromat Doublet, ACT508-200-A-ML

²⁸**Thorlabs TODO**

²⁹Newport ULTRAlign M-562-XYZ, M-36, SM-13

³⁰Newport CONEX-TRA12CC

Given the chosen lens focal lengths, we expect a magnification of $\times 6$ from the ion plane to the imaging plane. With an expected ion separation of $5\text{ }\mu\text{m}$, and the camera pixel pitch of $6.5\text{ }\mu\text{m}$, we expect around 4.5 pixels between ions, which, if the system is well aligned, is more than sufficient for low cross-talk spatial readout of qubits.

3.6 Ca^+ Laser Systems

Singly-charged group 2 elements are popular in ion trap experiments due to their single outer electron resulting in Hydrogen-like energy levels. In this thesis we use $^{40}\text{Ca}^+$.

$^{40}\text{Ca}^+$ has no nuclear spin giving the (relatively) simple level structure shown in figure 4.1 with the relevant laser transitions used in our apparatus are indicated. Details on the ion choice, and utility of the lasers is given in section 4.1, here we only describe the hardware and feedback control to provide this stabilised light.

The relevant wavelengths are the following: The qubit is defined by the quadrupole transition at 729-nm. Access to other excited levels, outside of our defined two level system, are crucial for ion trap quantum logic to enable qubit readout via state selective fluorescence (397-nm, and 866-nm transitions), and state preparation via optical pumping (866-nm, and 854-nm transitions). Access to two additional transitions in neutral calcium for isotope selective ionisation, 423-nm and 373-nm, are also required [30].

This section begins with a description of laser frequency and power stabilisation techniques common to multiple laser systems used in this thesis. Following this, the specific systems are described, starting with the diode lasers, which drive all but the 729-nm transition. Subsequently, the Ti:Sapph 729-nm source is discussed. Finally, the global and single addressed beam arrangements for the 729-nm system are described.

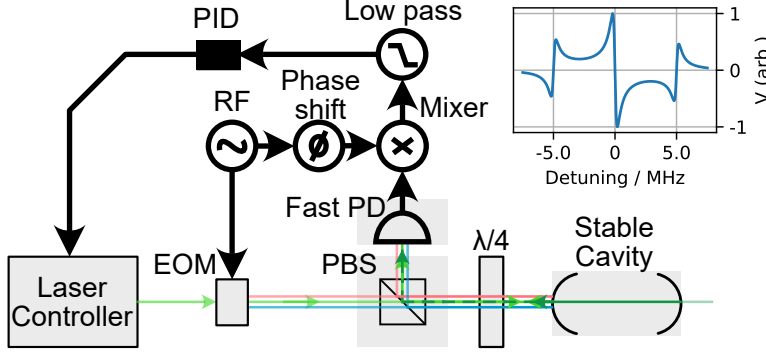


Figure 3.8: PDH optical and control schematic. Green optical path represents the carrier frequency, blue and orange lines represent the upper and lower sidebands respectively, generated by an EOM. The reflected light is directed onto a fast photodiode, and the signal is demodulated with the same oscillator signal used to drive the EOM, but delayed by some chosen phase. The signal is finally low pass filtered to recover the error signal, an example of which is seen in the upper right panel for a finesse $F \sim 4000$, a free spectral range of $FSR = 1.5$ GHz, and a modulation frequency of $f_{mod} = 5$ MHz.

3.6.1 Frequency Stabilisation

All lasers exhibit frequency noise due to unavoidable environmental coupling (e.g. temperature fluctuations, and acoustics), and to instabilities in control signals (e.g. voltage and current controls). For the lasers corresponding to dipole transitions used in this thesis, we are relatively lenient and require slow frequency excursions of < 1 MHz with respect to the qubit transition, and laser linewidths less than the natural linewidth of the dipole transitions, which are often multiple MHz. However, for the quadrupole transition at 729-nm, we require significantly more stringent frequency stability, with laser linewidths of < 100 Hz, and negligible phase modulation of the laser light over a few MHz of the qubit transition to ensure spectator transitions are not excited during qubit operations. Although the order of magnitude of the required frequency stability is different for the dipole and quadrupole transitions, the concepts and techniques used to achieve this stability are similar. Generally, a closed loop consisting of measurement of the optical frequency of the laser and feedback to the laser control is required. In this work, stable external optical cavities are used for referencing the laser frequency. Specifically, we use the Pound-Drever-Hall (PDH) techniques to generate

the error signal for the feedback loop. The PDH technique is well described in the literature [oswald2022characterization, black2001introduction, 31], and so here we only give a brief overview of the technique to contextualise the required hardware introduced in the following sections.

The conceptual goal of the PDH technique is to measure the frequency difference between the laser and a resonant frequency of a stable optical cavity to generate an error signal, preferably linear over typical frequency excursions (while *locked*). This error signal can then be modified by some controller (typically PID), to directly infer how to tune the laser controls to minimise this error signal, often termed *locking* the laser to the cavity. Further, the error signal should be insensitive to amplitude noise on the laser light, to avoid amplitude to frequency noise conversion due to the feedback loop (in contrast, side-of-fringe stabilisation is sensitive to amplitude drifts).

Practically this scheme consists of phase modulation of the incident laser light, here by electro-optical modulators (EOMs), to generate sidebands, which are used as phase references to compare the phase of the incoming laser field to that of the phase of the field inside the cavity. Figure 3.8 shows a schematic of the PDH scheme, highlighting the beam paths and closed loop. The phase modulated light is directed onto the cavity, with the total return signal being some combination of the reflected sidebands, carrier, and the cavity leakage field, with the latter providing the phase of the electric field inside the cavity. The reflected light is directed onto a fast photodetector, and the signal is demodulated with the same oscillator signal used to drive the EOM, but delayed by some chosen phase. The resulting signal is low pass filtered to remove high frequency components, yielding the final error signal, an example of which is seen in the upper right panel of Figure 3.8. The central region is anti-symmetric and linear, centered on the desired lock point. Using a tuned controller for feedback to the laser controller, this provides a stable lock of the laser to the cavity.

As the cavity is used as a frequency reference, both a high finesse and stable

optical lengths are desirable. The gradient of the PDH error signal is proportional to finesse, leading to increased signal to noise with higher finesse (see Ref. [oswald2022characterization] for more details). Optical length deviations are minimised through mounting the cavity in a vacuum chamber, and taking care in isolating it from temperature fluctuations and vibrations.

Particular details of the PDH hardware and control loops are given in the following sections for each laser system.

3.6.2 Intensity Stabilisation

The light-matter interactions considered in this thesis are all proportional either to intensity or amplitude of the optical electric field at the ion's position. Intensity fluctuations over time therefore lead to fluctuations in the interaction strength, which are detrimental to the fidelity of quantum operations. Intensity noise originates either from total beam power fluctuations, or due to beam pointing drift with respect to the ion position. The former is mitigated through a closed feedback loop between a pick-off photodiode signal before the ion monitoring the optical power, and the drive RF amplitude of an upstream AOM. For all beams, this optical pick off is one of the last elements before the beam enters the vacuum system to reduce the impact of lossy elements between the stabilise point and the ion position. The low-noise photodiodes used were designed by David Nadlinger (*QLNPD*), and the feedback system controller, known as *SUServo*, was written by Peter Drmota. Beam pointing drifts are mitigated through careful mechanical design of the optical paths for passive stability, and through automated intermittent beam pointing calibrations using the ion as a sensor (see section 4.2.4).

3.6.3 Diode Lasers

For transitions 397-nm, 854-nm, and 866-nm, we require light that is frequency stable to < 1 MHz level, well below the natural line widths of the dipole transitions,

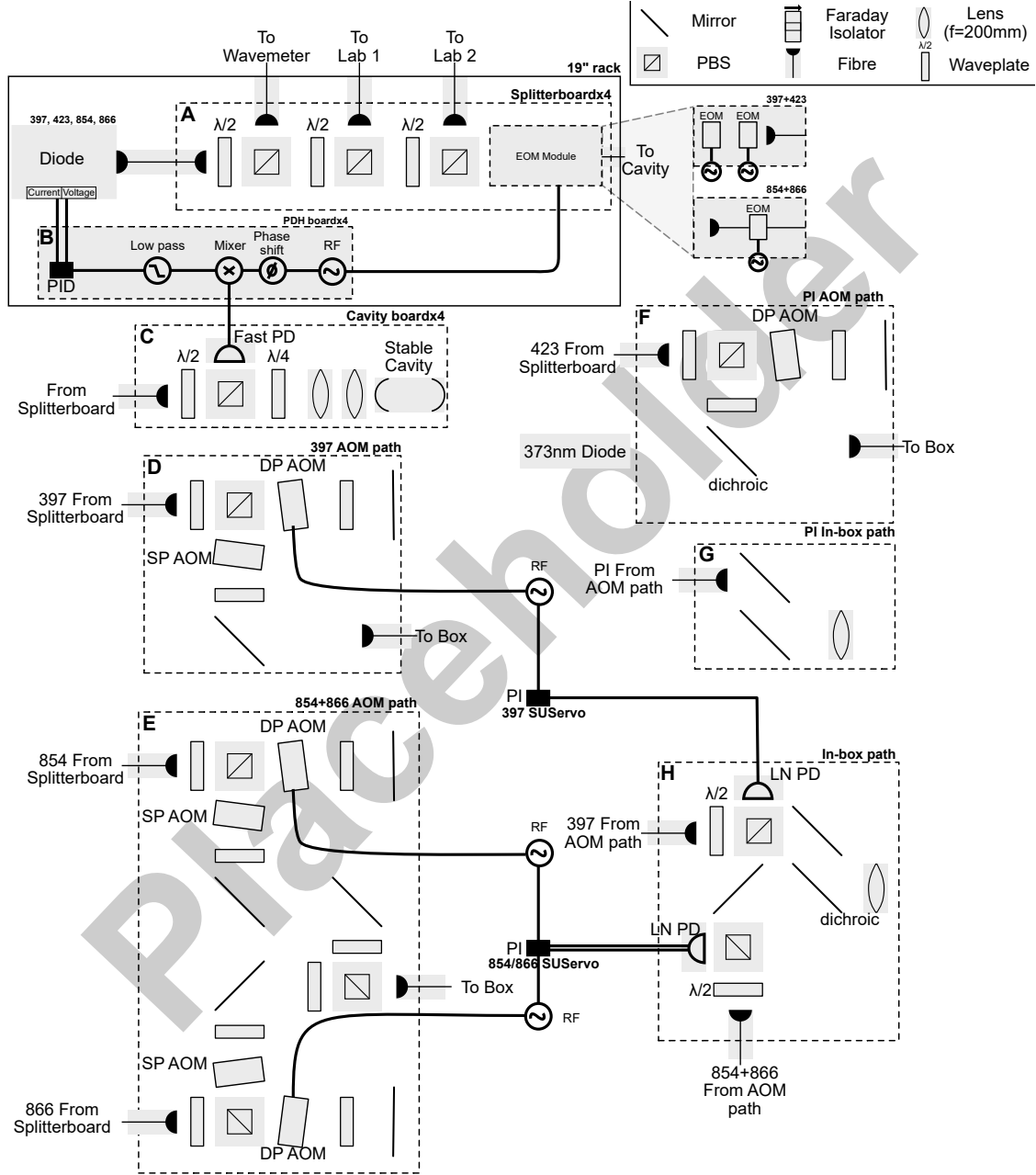


Figure 3.9: All beam paths for diode laser systems. Dotted boxes resemble modular optical paths with fibre and control lines in and out only. Frequency and optical power control loops are included.

and power stable to $< 1 \mu\text{W}$. All beam paths and relevant control loops for the diode laser systems are shown in figure 3.9. The laser diode heads³¹, controllers³², and splitterboards³³ (fig 3.9 (a)) are housed in a single 19" rack. The splitterboard distributes the light between two ion-trap experiments, the PDH lock cavity (fig 3.9 (c)), and a diagnostic wavemeter for checking wavelength and lock status³⁴. The splitterboard also houses an Electro-optic modulator (EOM) module for applying both offset and PDH sidebands for an offset PDH lock. The offset sidebands may be arbitrarily set (within the GHz range of the EOM) to provide a continuous choice of laser lock frequencies between cavity modes, rather than maintaining a fixed offset between the laser output frequency and the discrete cavity mode frequencies. For 397-nm and 423-nm, resonant free space EOMs³⁵ are used, for 854-nm and 866-nm broadband fibre EOMs³⁶ are used for convenience. The path on the reference cavity board (fig 3.9 (c)) consists of a four-bore reference cavity³⁷ under vacuum and temperature control, two lens for matching the collimated input beam to the cavity mode, and a beamsplitter and $\lambda/4$ waveplate for directing the reflected cavity light to a fast photodiode³⁸. The beatnote due to the carrier and PDH sidebands are mixed down and low-passed by the *Toptica* PDH module to produce the signature antisymmetric PDH trace. Gains for PID controllers of both the diode current and voltage are set to stabilise the optical frequency.

The light from the “laser” laboratory is carried to the “trap” laboratory using 16 m polarisation maintaining fibres³⁹. In the “trap” laboratory we use AOMs in a double pass configuration for both frequency shifting and power stabilising the light (fig 3.9 (d) and (e)).

As space inside the MuMetal enclosure is constrained, it is advantageous to couple

³¹*Toptica* diodes. 854/866: MDL DL pro; 397: MDL DL pro HP; 423: DL pro

³²*Toptica* DLC Pro with PDH boards

³³Designed by *OITG Cavities* group

³⁴*HighFinesse* WS7

³⁵*Qubig* PM7-VIS-25 and PM9-VIS-2.25

³⁶*EOSPACE* PM-0s5-05-PFA-PFA-850

³⁷*Stable Laser Systems* SLS-6010-1-4bore

³⁸*Thorlabs* PDA10A2

³⁹*Schafter-Kirchoff ...*

multiple beams in a single fibre before the enclosure. The second stage photo-ionisation light, at 373-nm⁴⁰, is combined with the 423-nm beam by a dichroic (fig 3.9 (f)) before being coupled to a 16 m fibre that takes the light directly into the MuMetal enclosure. The PI light uses a parabolic mirror outcoupler⁴¹ inside the MuMetal box, and care was taken for the beam propagation to be perpendicular to both the lens and vacuum viewport window as it was found that the difference in refractive index of the optical materials at these two wavelengths was significant enough to cause the two beams to be separated by more than one waist at the ion plane after initial rough alignment. The 854-nm and 866-nm light is combined after their respective double and single pass AOMs, and coupled into a 3 m PM⁴² fibre carrying the light into the enclosure.

Light from both the PI and 397/854/866-nm in-box path is directed onto the ions via the in-vacuum prisms, discussed in section 3.4. The final mirrors prior to the vacuum system are motorised⁴³, so that beam alignment to the ions can be optimised when the MuMetal enclosure is closed. Practically we find that these beams rarely need to be realigned (few months), unless there is a significant ambient temperature change of a few degrees.

3.6.4 Narrow Linewidth 729 Laser

Lasers are a key tool for creating the highly localised, strong electric field amplitudes and gradients needed to drive both carrier and sideband transitions of the trapped ion.

As shown in Figure 4.1, two sublevels within the $4S_{1/2}$ to $3D_{5/2}$ manifolds define the qubit. This is an electric quadrupole transition as $\Delta l = 2$. For the Calcium ion this transition is at 729-nm, and so a near resonance 729-nm laser is used to implement single- and multi-qubit gates (sections 4.2.5 and 4.4.2). This transition is also used after Doppler cooling for resolved sideband cooling to prepare the motional

⁴⁰*Toptica* iBEAM-SMART-375-S.

⁴¹*Thorlabs* RC02APC-F01

⁴²*SK* ...

⁴³*Newport Picomotor* piezo mirror mount P/N 8821

mode close to its ground state (as discussed in section 4.3.1). This transition is narrow linewidth due to the long lived $3D_{5/2}$ level, and so, for power efficiency, a narrow linewidth laser must be used. For the “fast entangling gates” use-case, high intensity electric-fields are required at the ion to drive sideband transitions, this will be discussed further in the single-addressing and two-qubit gate sections, but here it is sufficient to say we require >100 mW of light at the ion plane. Here we describe the 729-nm system consisting of a Ti:Sapph laser system pumped with by an Nd:YAG 532-nm laser.

An *M2 Solstis* Ti:Sapph is pumped via 18 W of 532-nm light from a *Coherent Verdi-V* system to produce around 5 W of 729-nm light. The Ti:Sapph is engineered to operate with a stable < 50 kHz linewidth. Ti:Sapph crystals have broadband gain profiles, which are often exploited in research environments to create frequency tunable laser systems. We however want a narrow linewidth, single frequency laser. The *Solstis* has multiple intracavity frequency selective elements which consist of (in order of coarse frequency selectivity), a birefringent filter, a tunable Fabry-Pérot etalon, and the surrounding bow-tie cavity. For stable single mode operation, the *Solstis* employs an active “dither” servo to lock the peak of etalon transmission to one of the cavity longitudinal mode. This dither consists of periodically varying the etalon spacing with a frequency of around 20 kHz. We must be aware of this dither frequency as the phase modulation leads to the creation of sidebands on our light which can interact with the ion causing unexpected errors in gates. This dither frequency (and harmonics of it), can be observed via composite pulse experiments on the ion, however it is currently not expected to be a limiting source of error in any of the interactions we study.

As mentioned, the *Solstis* alone can operate with linewidths of < 50 kHz, however this linewidth is pushed down further by referencing the Ti:Sapph output to an ultra high finesse cavity⁴⁴ and applying a PDH lock (see above section). A schematic of the 729-nm system is shown in Figure 3.10. As opposed to the simpler controller implemented for the diode lasers, the 729-nm uses a cascaded PID lock with a fast

⁴⁴*Stable Laser Systems* P/N 6020-4

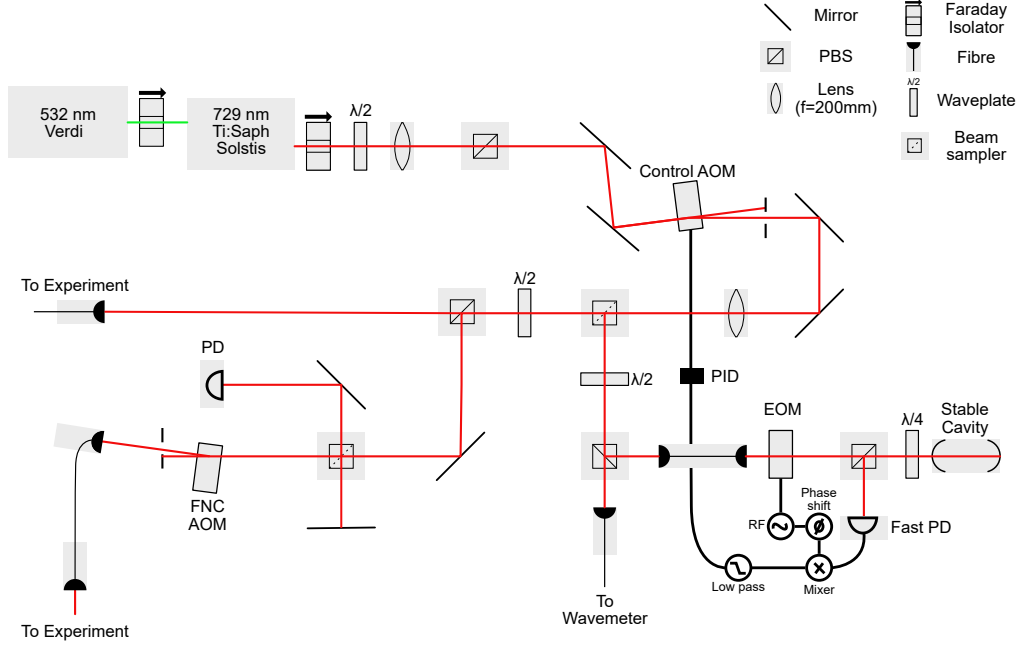


Figure 3.10: The 729-nm system. A Ti:Sapph laser tuned to 729-nm is pumped by a 532-nm source. Light is picked off at the first beam sampler to stabilise by PDH locking to a cavity.

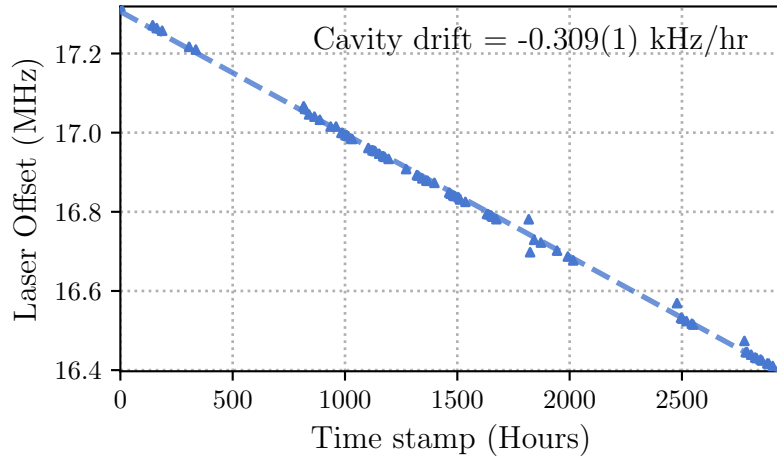


Figure 3.11: Cavity drift over 125 days measured by reference to the ion transition. The laser offset here is with respect to running all 729-nm AOMs at their central frequencies. Outliers are due to bad fits of the ion transition frequency, not due to changes in the cavity resonance.

feedback loop to the AOM situated after the Solstis, and an intermediate and slow feedback loop to piezos controlling the Solstis cavity length. With this system, a linewidth of $\ll 1$ kHz for the 729-nm light is expected. The electronics for this control loop are provided also by *Stable Laser Systems* in the form of their *FPGA Servo* lock box. For an effective PDH lock, both short and long term stability of the reference cavity is required. To ensure the cavity is insensitive to the environment, it is made of an ultra low expansion material, and mechanically isolated from the environment by being mounted on a vibration isolation platform. The cavity is housed in a vacuum system to prevent refractive index fluctuations due to air, and to further reduce thermal coupling to the environment. The cavity is temperature stabilised at the zero crossing temperature of 30.6°C. Long term cavity drifts of 309(1) Hz/hr are measured over 125 days, seen in figure 3.11. This measurement uses the ion as a frequency reference to probe the cavity frequency.

Figure 3.10 displays the other beam paths for our 729-nm system. Some light is picked off and sent to a wavemeter to continuously monitor the frequency. However, the majority is coupled to two output fibres for our experiment and another within the group. The 729-nm light is transported from a dedicated laser lab to the trap apparatus lab by a 10 m single mode polarization maintaining fibre⁴⁵. The fibre is beneficial in cleaning up the mode from the Ti:Sapph, however it can introduce phase noise due to mechanical and thermal effects along the 10 m length. To remove this introduced noise, passive stabilisation in the form of thick foam tubing along the fibre length, as well as active stabilisation by a fibre-noise-cancellation (FNC) technique [32], is used⁴⁶. A comparison of spin coherence times is shown in section 4.2.6 with and without fibre noise cancellation enabled.

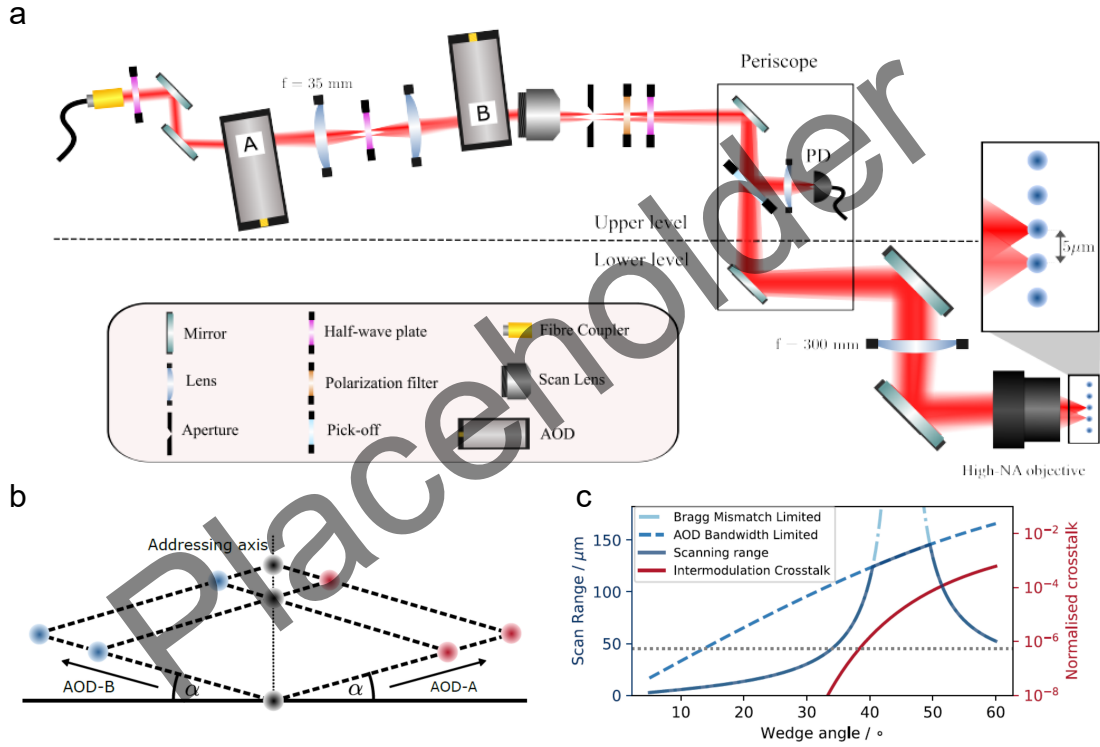


Figure 3.12: AOD setup

3.6.5 Single Addressing System

Single ion addressing is a requirement for executing arbitrary circuits on a register of ion-qubits. Furthermore, it is advantageous to focus the optical drive onto the ion, increasing the intensity of light, and so driving faster gates. A unique design feature of our system is the ability to produce single ion addressing standing waves whose spatial phase can be controlled individually for each ion, thus enabling any-to-any connectivity for fast entangling gates. In the future we particularly wish to extend the fast two-qubit entangling gate work demonstrated in [13] on a multi-ion chain where multiple motional modes are present and contributing to the dynamics.

The design and construction of the single addressing system was led by Iver Øvergaard and, as of writing, has been in constant use in the experiment for around 1 year. Here we shall quote the design specification and relevant details of

⁴⁵*SK ...*

⁴⁶FNC control consists of *Sinara Stabilizer* and *Pounder* mezzanine board with PID software developed by Ayush Agrawal.

the single addressing system. In section ?? we present the characterisation data upon installation. Further technical information can be found in IRO’s Transfer report [oevergaard2025transfer], available upon request.

Multiple methods for single ion addressing have been developed in the ion trap community, including waveguide arrays [integrated, 33], micro-mirrors [<empty citation>], spatial-light-modulators [<empty citation>], and acousto-optic deflectors [<empty citation>]. Each method has its own advantages and disadvantages in terms of speed, resolution, ease of integration, and scalability, and so choice of hardware is often a matter of experimental preference and constraints. We require:

- Addressing of up to two ions in a ~ 10 ion chain with low crosstalk to neighbouring ions.
- Fast reconfigureability of the beam position within a single experimental shot (i.e. between different gates within a single circuit execution).
- Good optical power efficiency to ensure high intensity electric fields at the ion.
- Flexibility in beam positioning to allow for fine tuning of the overlap of two counterpropagating beams used to create the standing wave.
- Individual phase control of each addressed beam.
- Beam position stability to avoid power fluctuations.
- A compact design to fit within the magnetic shielded region of the experiment.
- Good telecentricity (fixed beam k-vector direction across the scanned area) to avoid unwanted travelling wave component in a standing wave gate scheme.
- Zero frequency shift of the beam at the ion plane.

Crossed Acousto-Optic Deflectors (cAODs) are a popular choice for single ion addressing, offering fast rise times ($\sim 10 \mu\text{s}$), and good resolution over a large scanning area (~ 500 resolvable spots with 2° angular range) with zero frequency

shift along the addressing axis. AODs operate through the same mechanism as AOMs, but are optimised for beam deflection rather than frequency modulation. As an addressing system cAODs have poor scaling of power efficiency with the number of addressed spots, and so are not ideal for large scale systems. As we are interested in addressing ≤ 2 ions at a time, this is not a significant issue for our system.

Figure ?? (a) shows a schematic of the constructed addressing system. Due to space constraints within the magnetic shielded region, the optical elements are located on two height levels with a periscope steering the beam between them. The beam is delivered via single-mode polarisation maintaining fibre to the upper level. The beam is steered through the first AOD, with beam angle optimised for maximum diffraction efficiency into the -1^{st} order. Two lenses in a 4f configuration re-image the deflection origin of the first AOD onto the second AOD. The second AOD is aligned for maximum diffraction efficiency into the $+1^{st}$ order. The cAOD arrangement is termed “crossed” as the second AOD is rotated by some angle α with respect to the first, so that the deflection from the second AOD has some orthogonal component to that of the first. When the two AODs are driven with RF tones of the same frequency, the output orders are listed: (0,0), (-1, +1), (-1,0), (0,+1). The second is the desired order for single ion addressing as it has zero frequency shift yet has some overall drive-frequency dependent deflection angle. To translate this deflection angle into some physical displacement at the ion plane, a telecentric scanning lens is used. The telecentricity ensures that the beam k-vector direction is fixed across the scanned area, which is important for standing wave gate schemes to avoid unwanted travelling wave components. At the intermediate focus after the scanning lens, the unwanted orders are blocked by an adjustable slit aperture. The beam polarisation is also cleaned and selected at this stage using a polarisation filter and half-wave plate. A plate beam splitter is used within the periscope to pick off $\sim 5\%$ of the light for monitoring on a photodiode, which is used in a feedback loop to stabilise the intensity at the ion plane (the same method as discussed in section ??). The scan lens, in combination with a relay lens, forms a beam expander to increase the beam NA for the final objective lens to focus

to a beam waist of $\approx 1.5 \mu\text{m}$ at the ion plane. Characterisation of the system (details given in section ??) showed that the addressing beam could be focused to a Gaussian waist of $1.50(1) \mu\text{m}$, measured by scanning a single trapped ion with Rabi experiments. The same beam was then tested in a five-ion chain, where the nearest-neighbour intensity crosstalk was measured to be $4.16(2) \times 10^{-4}$.

3.6.6 Global Addressing System

The single addressing provides the utility of spatially selecting a few ions to drive with the 729-nm light, however, the power efficiency scales as $O(n^2)$, with n being the number of addressed spots. As such, we limit the single addressing to simultaneously illuminate only one or two ions at a time. It is therefore convenient to also have a global beam that has dimensions larger than the ion chain to address all the ions in parallel. This global branch is currently counterpropagating to the single addressing path, and shares the same optical path as the imaging system (see figure 3.7). To increase the beam intensity at the ions, the global branch uses an anamorphic prism pair⁴⁷ to elliptically shape the beam, with a 4:1 aspect ratio, with major axis along the ion chain direction. The resulting major axis has a beam waist radius of $52(3) \mu\text{m}$ ⁴⁸. As with the other beams directed to the ion, we use a final piezo controlled mirror⁴⁹ for fine alignment and control when the MuMetal enclosure is closed.

3.7 Control System

3.8 Ion Loading

Atoms are provided to the trapping region by heating Calcium flakes housed in the resistive tube oven described in section 3.4. The thermocouple, tag welded onto the tube oven, provides a temperature reading for current feedback to reach a target temperature of $\simeq 200^\circ\text{C}$. This requires $\simeq 2 \text{ A}$ across the 0.5Ω load of the oven

⁴⁷*ThorLabs* PS883-B

⁴⁸The beam profile was measured by extracting per-ion Rabi frequencies while driving on resonance quadrupole transitions (see section 4.2.5) and assuming a Gaussian cross section.

⁴⁹*ThorLabs* POLARIS-K2S2P

tube. The tube oven has a pinhole that is aligned to the ion flux aperture, shown in figure ??, and the atomic shield pinhole shown in figure 3.2. These apertures were included in the design to improve the lifetime of the trap by ensuring neutral Calcium metal cannot build up on the trap surface and cause shorts between RF, ground, and DC electrodes. The aperture design *luckily* failed. It failed as we observe atomic flux at all electrode locations suggesting the aperture is too large. It was *lucky* as the planned trapping region of dc10 became non-viable after electrode Fr dc11 developed a continuity fault (which we believe is in-vacuum). As such, ions are loaded at dc5.

The neutral atom flux is ionised using a two-step photoionisation scheme [**<empty citation>**].

The first stage involves a frequency stabilised 423-nm beam that couples the 4S->XX transition in neutral calcium. The isotope shift is XX, which is much larger than the transitions natural linewidth and the lasers linewidth, as such the desired isotope can be selectively loaded. We use natural abundance Calcium, with typical isotope proportions of XX% : ^{40}Ca , XX% : ^{42}Ca , XX% : ^{43}Ca . In this work we use exclusively $^{40}\text{Ca}^+$, and so set the 423-nm to excite neutral ^{40}Ca only⁵⁰. The second ionisation stage couples XX->continuum using a 373-nm UV laser source (although the exact wavelength does not matter as long as <xxx-nm).

The 373-nm laser can cause charge build-up on the trap electrodes, and so we use a physical shutter to block the beam when not actively loading ions.

⁵⁰We practically see no loading of other Calcium isotopes, and only see rare non-fluorescent ions (once every few weeks) after collisions with residual gas (we suspect this is CaH^+).

4

Experimental Characterisation

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Before we can proceed with our planned experiments involving the motion and spin of the atoms, the apparatus must be characterised. This characterisation serves two purposes: it allows us to benchmark the performance of our system against state-of-the-art results, and it reveals any current limitations that must be addressed before we can successfully carry out the experiments.

As discussed above, we are interested in two paradigms: discrete variable quantum

computing (DVQC), that concerns the manipulation of qubits (or spins), and continuous variable quantum computing (CVQC), that concerns the manipulation of qumodes (or harmonic motion).

For DVQC, we must have access to the following elementary operations:

- Qubit state preparation, here via optically pumping the electronic state of the ion into one sublevel of the $4S_{1/2}$ manifold — the lower level of our qubit, section 4.2.1.
- Measurement of the qubit state, here via state-dependent fluorescence, section 4.2.2.
- Single qubit rotations, here via interaction with the 729-nm laser, section 4.2.5
- Two qubit entangling-gates, here via spin-dependent forces (SDFs), using the 729-nm laser and the motional modes of the ion, section 4.4.2.

For CVQC, we need:

- Motional state preparation, here via Doppler cooling and sideband cooling, section 4.3.1.
- Measurement of the motional state, which is not yet implemented on our system but will follow [34].
- Single and two qumode gates, which are also not yet implemented, but will utilise SDFs following [14], section 4.4.1.

We split this chapter into interactions involving only the spin, then those focussed only on the motion, and finally we describe spin-motion interactions. First, however, we describe the ion, its level structure, and the relevant lasers to manipulate its state.

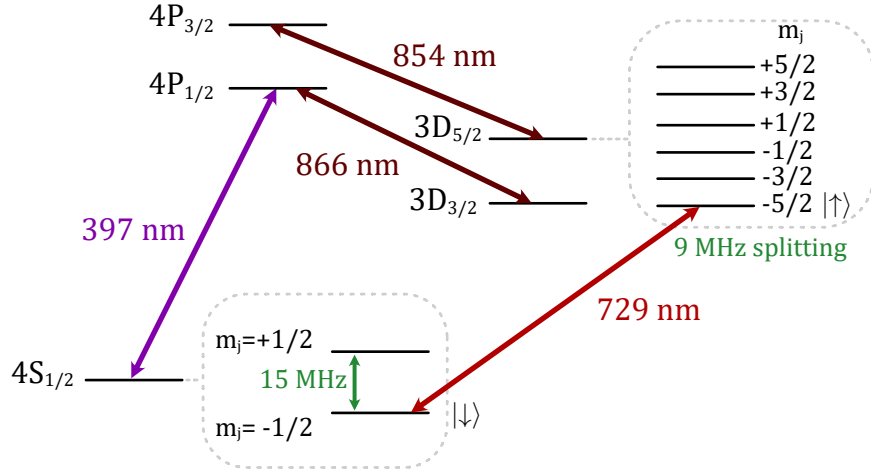


Figure 4.1: Electronic energy levels of $^{40}\text{Ca}^+$, which will be used in this thesis. The levels are split by the Zeeman effect due to a 5.4 G external magnetic field (they are shown explicitly only for the qubit manifolds). The transitions marked are required for cooling and control over the ion. The chosen qubit levels are labelled with $|\downarrow\rangle$, $|\uparrow\rangle$.

4.1 The Ion

The $^{40}\text{Ca}^+$ ion has a rich level structure, shown in figure 4.1, with many accessible quadrupole transitions $4S_{1/2} \leftrightarrow 3D_{5/2}$. The quadrupole transition is ideal for use as a qubit due to the long-lived metastable level ($\tau = 1.1$ s) [35], and the ability to optically couple the transition via a laser. As mentioned in section ??, the 729-nm beam is used for this purpose. By selecting the appropriate frequency of the 729-nm beam, qubit operations such as state preparation (section 4.2.1) and single qubit gates (section 4.2.5) may be driven, or by setting the detuning such that a motional sideband is near resonance, to couple the spin and motional degrees of freedom (section 4.4.1). Here, we discuss which quadrupole transitions are used, and the rationale behind their choice.

Zeeman splitting of the $4S_{1/2}$ and $3D_{5/2}$ states leads to 2 and 6 non-degenerate sublevels, respectively. Due to the magnetic field strength of 5.4 G (section 3.3), the S -levels are split by ~ 15 MHz, and the D -levels by ~ 9 MHz. This splitting is considerably larger than both the natural linewidth of the 729-nm transition ($\sim \text{Hz}$), and the Ti:Sapph laser linewidth (< 1 kHz), meaning the transitions can be selectively addressed by frequency tuning of the laser. There is some freedom

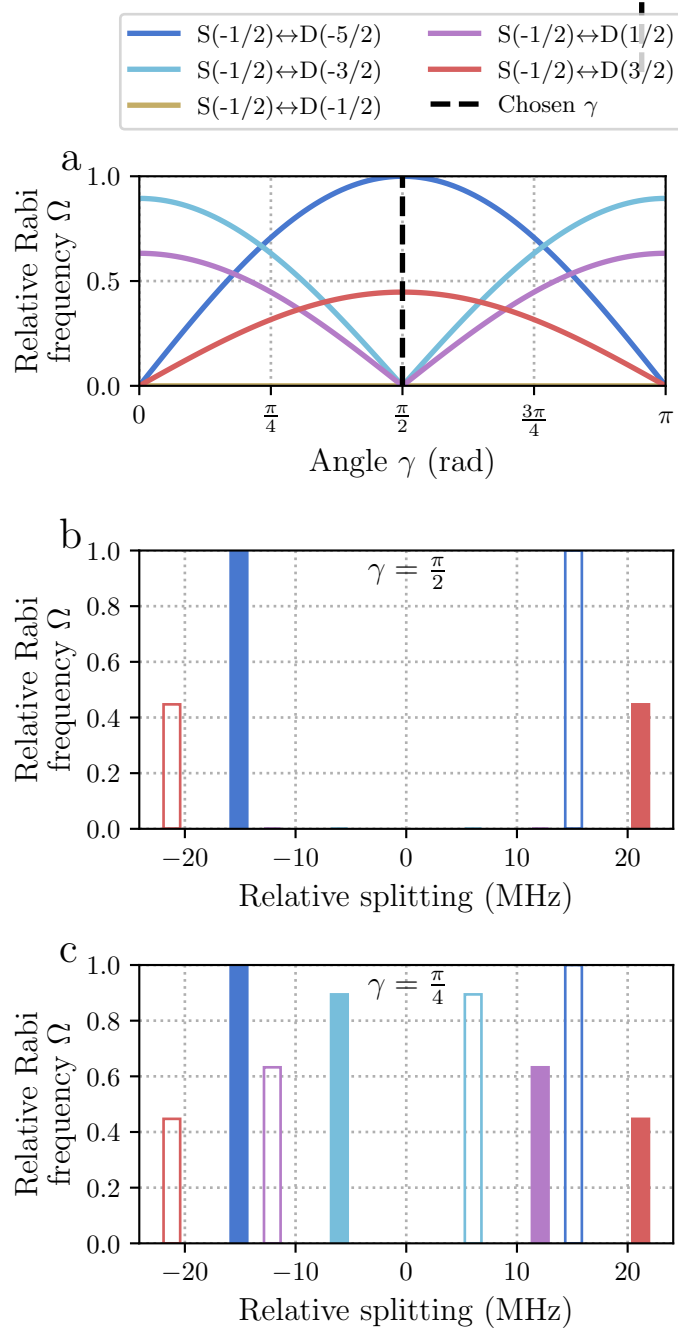


Figure 4.2: Normalised quadrupole Rabi frequencies, Ω , and frequency splittings for the $4S_{1/2} \leftrightarrow 3D_{5/2}$ Zeeman sublevels in $^{40}\text{Ca}^+$. Solid bars correspond to transitions with lower state $|4S_{1/2}, m_j = -1/2\rangle$, while open bars correspond to transitions with lower state $|4S_{1/2}, m_j = +1/2\rangle$. **a)** Relative Rabi frequencies as angle γ between the polarisation vector and the magnetic field is varied. The k-vector of the light is assumed to be perpendicular to the B-field. **b)** Relative frequency splittings and strengths when $\gamma = \pi/2$. This is the polarisation chosen for the current and future experiments to reduce the number of transitions which can cause off-resonant interactions when the qubit defined by $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ is used. **c)** Relative frequency splittings and strengths when $\gamma = \pi/4$. At this angle, all transitions are accessible apart from those with $\Delta m_j = 0$. This is the polarisation chosen for our initial experiments to allow access to all transitions for characterisation.

in choosing which of these 10 possible (with $\Delta m_j \leq 2$) quadrupole transitions we define as our qubit. Each transition has a varying coupling strength, here denoted by the relative Rabi frequency, Ω , depending on the polarisation and k -vector of the interacting light with respect to the B-field axis. The desired criteria for qubit and polarisation choice are as follows:

- Access to a closed-cycle transition for sideband cooling, (for details on this, see section 4.3.1). Closed-cycle here means $|4S_{1/2}, m_j = -1/2\rangle \rightarrow |3D_{5/2}\rangle \rightarrow |4P_{3/2}, m_j = -3/2\rangle \rightarrow |4S_{1/2}, m_j = -1/2\rangle$.
- The qubit transition needs a large quadrupole matrix element, leading to a high Rabi frequency, Ω , to enable laser-power-efficient, fast, gates. Polarisation, γ , may be optimised to increase this coupling.
- Ideally the qubit transition should have a low magnetic field sensitivity, to reduce the effect of magnetic field noise. Using a MuMetal enclosure around the experiment does relax this constraint, see section 3.3.
- To reduce the effects of off-resonant interactions due to nearby transitions, it is desirable to limit the number of accessible transitions near to the qubit transition. This is done by selecting a polarisation, γ , such that unwanted transitions are suppressed.

Given these criteria, we use the transition $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ with magnetic field sensitivity $-0.446 \text{ MHz G}^{-1}$. Closed-cycle sideband cooling is possible with this same transition, and state preparation via optical pumping is done via $|4S_{1/2}, m_j = +1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ with magnetic field sensitivity $-0.624 \text{ MHz G}^{-1}$.

Figure 4.2 *a*, shows the normalised Rabi frequencies of each transition as the angle γ between polarisation vector and B-field is varied. Here only linearly-polarised light with k -vector perpendicular to the B-field direction is considered. The polarisation of the 729-nm laser is set to $\gamma = \pi/2$ (black dashed line), which maximises the Rabi frequency of the qubit transition, while suppressing nearby

Parameter	Value
729-nm laser power	1000 μW
729-nm laser detuning	0 MHz*
729-nm pulse duration	5 μs
854-nm laser power	60 μW , $250 \times I_{\text{SAT}}$
854-nm laser detuning	0 MHz
854-nm pulse duration	40 μs
866-nm laser power	15 μW , $\sim 70 \times I_{\text{SAT}}$
866-nm laser detuning	0 MHz
866-nm pulse duration	30 μs

Table 4.1: * With respect to qubit transition frequency.

Experimental parameters used for state preparation of the ion into the $|\downarrow\rangle = |4S_{1/2}, m_j = -1/2\rangle$ state.

transitions apart from that needed for state preparation. Figure 4.2 *b*, shows the relevant frequency splitting of the transitions at $\gamma = \pi/2$, which is the set polarisation for all experiments apart from initial quadrupole characterisation. For initial characterisation of the quadrupole transitions, the polarisation was set to $\gamma = \pi/4$, which allowed access to all transitions apart from those forbidden by the beam geometry, $\Delta m_j = 0$, see figure 4.2 *c*.

Other required lasers for manipulating the ion are: the 397-nm and 866-nm lasers used to form a closed-cycle for fluorescence, the 854-nm laser used to deshelve the ion from the $3D_{5/2}$ level for qubit state preparation. Details on these lasers can be found in section 3.6.

4.2 Spin

Discrete variable quantum computing consists of manipulating many two-level systems, which we refer to either as spins, or qubits. In this section the methods used to manipulate these spins are described, and their performances are benchmarked.

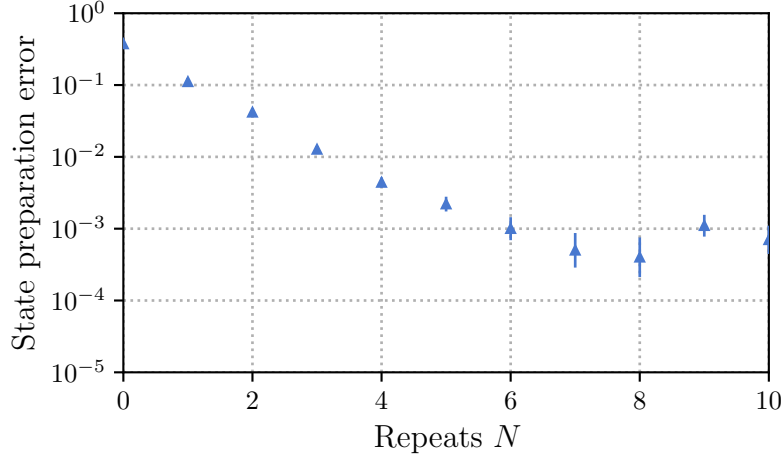


Figure 4.3: State preparation error as a function of the number of state preparation pulses, N . The error is defined as population not in the $|\downarrow\rangle$ state after the preparation sequence has been applied. The error is reduced to below 10^{-3} after $N = 7$ pulses, which is used in our experiments.

4.2.1 State Preparation

To utilise two levels of the ion as a qubit, the electronic state must first be prepared into the qubit manifold. Here, the ion is prepared into the $|\downarrow\rangle = |4S_{1/2}, m_j = -1/2\rangle$ state. This is done via effective optical pumping by pulsing the quadrupole transition $|4S_{1/2}, m_j = +1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$, followed by deshelling pulses using the 854-nm, 866-nm lasers on resonance. The 729-nm laser pulse duration and intensity is set to invert the populations, i.e. perform a π -pulse. Details for laser parameters are summarised in table 4.1. These three pulses are repeated N times sequentially. Figure 4.3 shows the state preparation error with increasing N . The error is found by applying a π -pulse on the state preparation transition directly after the state preparation sequence and looking at the population in the $|\downarrow\rangle$ state. Any reduction from $|\downarrow\rangle = 1$ indicates residual population in the $|4S_{1/2}, m_j = +1/2\rangle$ state, or the D levels. After $N = 7$ pulses, which is used in our experiments, there is a state preparation error of $< 10^{-3}$. This is further verified by randomised benchmarking in section 4.2.5. The state preparation error is likely limited by accidental excitation of the $|4S_{1/2}, m_j = -1/2\rangle$ state to the $D_{5/2}$ level. This may

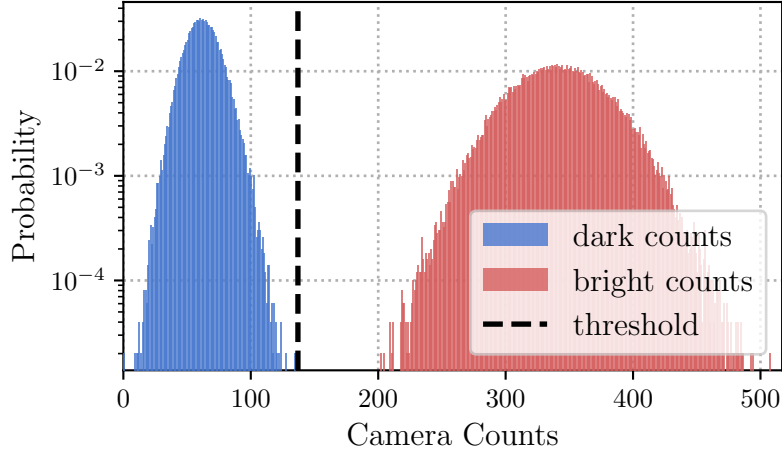


Figure 4.4: Readout histogram with 100,000 individual measurements for bright (red) and dark (blue) populations. The camera counts are presented as a histogram and fitted to Poissonian distributions to calibrate a threshold for discerning if the ion is bright or dark with the lowest probability for false measurement. Dark counts are mainly due to 397-nm light scattered from the nearest trap electrodes. With 100 μ s readout duration, and laser parameters set to Doppler cooling parameters, a threshold count of 137 photons is found, giving an expected statistical error of 6×10^{-9} given the fitted Poissonians.

be due to off-resonant excitation of another transition, or due to spectral purity of the 729-nm laser.

Another protocol for state preparation is optical pumping on the dipole transition (397-nm), which often is much faster. However, in this setup it is impractical as polarisation selectivity of the 397-nm transition is not possible due to the constraint of beam geometry from the in vacuum optics, see section 3.4, and frequency selectivity is not possible due to the low magnitude fixed B-field providing Zeeman shifts less than the natural linewidth of the 397-nm transition, see section 3.3.

4.2.2 Measurement

To measure the qubit state, the 397-nm and 866-nm lasers are applied, and the scattered 397-nm photons are imaged by the objective onto a camera (see section 3.5). 397-nm light will only be scattered if the qubit is in $|\downarrow\rangle$. Use of a high numerical aperture ($\text{NA} = 0.6$) objective greatly increases the collection efficiency. This efficiency allows the 397-nm beam to be red-detuned to Doppler cooling parameters

Parameter	Value
397-nm laser power	3 μ W, $1 \times I_{\text{SAT}}$
397-nm laser detuning	-17.5 MHz
866-nm laser power	15 μ W, $\sim 70 \times I_{\text{SAT}}$
866-nm laser detuning	0 MHz
Doppler cooling duration	1000 μ s
Readout duration	100 μ s
Camera threshold	137 counts

Table 4.2: Experimental parameters used for Doppler cooling and state-selective fluorescence of the ion.

(see section 4.3.1), while still maintaining low readout duration (100 μ s) and low readout error. Cooling during readout increases the experiment's repetition rate by reducing the amount of dedicated cooling time required.

The parameters used for readout are summarised in table 4.2, and a typical histogram of readout counts for one ion can be seen in figure 4.4. The readout threshold is calibrated by taking bright counts with the 397-nm and 866-nm lasers on, and dark counts with the 397-nm laser on and the 866-nm laser off. This assumes dark counts are due to 397-nm scattering from the trap electrodes. Fitting Poissonian distributions to the bright and dark measurements, an optimal count threshold of 137 is found, shown by black dashed line. This threshold gives an expected statistical readout error of $\epsilon_{\downarrow} = \epsilon_{\uparrow} = 6 \times 10^{-9}$, where $\epsilon_{\downarrow, \uparrow}$ are the expected error associated with measuring bright or dark. This measurement ignores errors due to the finite lifetime of the metastable $3D_{5/2}$ level, which dominate $\epsilon_{\uparrow}$. The expected error due to the finite lifetime, τ , is $\epsilon_{\text{decay}} = 1 - e^{-t/\tau}$, where t is the readout duration. For $^{40}\text{Ca}^+$, $\tau \sim 1.1$ s, $\epsilon_{\uparrow} \approx \epsilon_{\text{decay}} = 10^{-4}$. Assuming the metastable lifetime is limited only by spontaneous decay overlooks leakage from the 854-nm deshelving beam, which could significantly shorten τ [33].

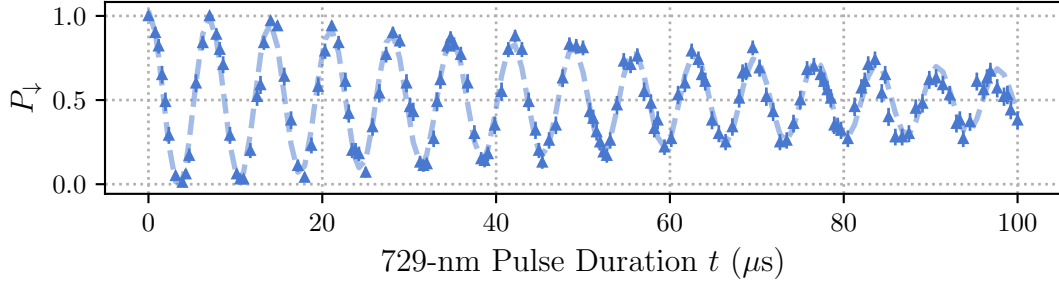


Figure 4.5: Long duration, t , Rabi flop on the qubit transition to extract Rabi frequency. Estimated probability of being in $|\downarrow\rangle$, $P_{\downarrow}$, at each time step using 200 individual experimental shots. Fitted dotted line using decaying oscillation model. Extracted Rabi frequency $\Omega/2\pi = 0.2864(2)$ MHz, and decay rate $\lambda = 0.0107(7) \mu\text{s}^{-1}$.

4.2.3 Micromotion Compensation

4.2.4 Beam Pointing Calibration

4.2.5 Single Qubit Rotations

A qubit is fully described by $|\psi\rangle = \alpha|\downarrow\rangle + \beta|\uparrow\rangle$, where α and β are complex numbers satisfying $|\alpha|^2 + |\beta|^2 = 1$. The qubit state can be manipulated by applying single qubit rotations, $R_{x,y}(\theta)$, commonly referred to as gates. In the Bloch sphere, x - and y -rotations are performed by perturbations resonant with the qubit transition, here, using the 729-nm laser. The rotation angle, θ , is given by $\theta = \Omega t/2$, where Ω , is the Rabi frequency, and t is the pulse duration. The rotation axis, is determined by the relative phase of the light field and the qubit. We define the initial relative phase to correspond to a y -rotation.

Gate Time Calibration

The pulse duration, for a $R(\pi/2)$ ($\pi/2$ -pulse), is calibrated by Rabi oscillations. Figure 4.5 shows the population in $|\downarrow\rangle$, $P_{\downarrow}$, calculated from 200 experimental shots. To find the Rabi frequency, Ω , the data is fitted to a decaying oscillation model,

$$P_{\downarrow} = \frac{1 + e^{-\lambda t} \cos(\Omega t/2)}{2}, \quad (4.1)$$

where λ is the decay rate due to decoherence effects [2]. The Rabi frequency is found to be $\Omega/2\pi = 0.2864(2)$ MHz, and the decay rate $\lambda = 0.0107(7) 1/\mu\text{s}$. From

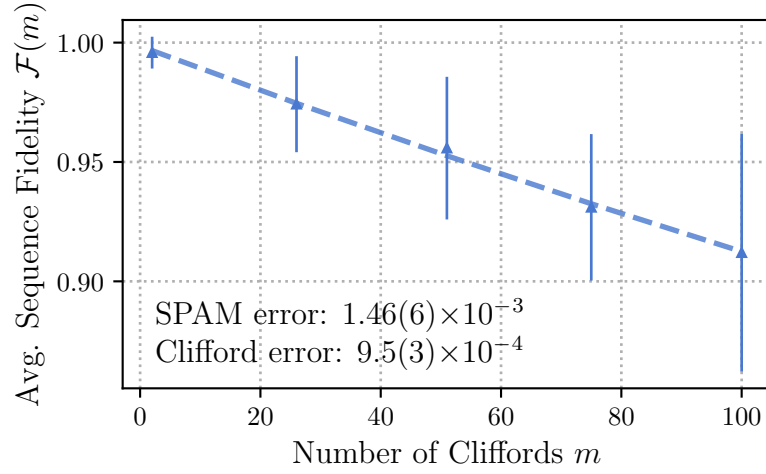


Figure 4.6: Estimating single qubit Clifford fidelities, $\mathcal{F}$, and errors due to state preparation and measurement, using randomised benchmarking. The data points are the average survival population after applying a sequence of m random Clifford gates, and a final inverting Clifford gate. The dashed line is the fit to the decay model given by equation 4.2. The error bars are given by the standard deviation of the survival populations.

this $R(\pi/2) = 1.7 \text{ } \mu\text{s}$ can be estimated. These measurements were taken with 1 mW of 729-nm power, with beam radius $\sim 15 \mu\text{m}$ at the ion.

Randomised Benchmarking

To evaluate the quality of both our state preparation and single qubit rotations, randomised benchmarking (RBM) [36] is employed. RBM consists of applying random combinations of gates from the single-qubit Clifford group to estimate an average error per gate. The single-qubit Clifford group is the set of unitaries which map the Pauli matrices to one another through conjugation. The RBM protocol used here is described in [37]. The sequence of operations is as follows:

1. Prepare the qubit in $|\downarrow\rangle$ via optical pumping, section 4.2.1.
2. Apply a random sequence of m single qubit Clifford gates, where m is the length of the sequence.

3. Apply a final “inverting” Clifford gate, which is chosen such that the full sequence performs the identity operation.
4. Measure the population in $|\downarrow\rangle$, $P_\downarrow$, via state-selective fluorescence, section 4.2.2.
5. Repeat steps 1-4 multiple times to find the average survival population, $\langle P_\downarrow \rangle$.
6. Repeat steps 1-5 for many different random sequences with a range of sequence lengths, m .
7. Fit the decay of the average survival population, $\mathcal{F}(m) = \langle P_\downarrow(m) \rangle$, to a decay model to find the average error per Clifford gate.

The decay model used to find the fidelity versus number of Clifford gates is given by [37],

$$\mathcal{F}(m) = \frac{1}{2} (1 + (1 - 2\epsilon_{\text{SPAM}})(1 - 2\epsilon_c)^m), \quad (4.2)$$

where $\mathcal{F}(m)$ is the fidelity of the sequence of length m , ϵ_{SPAM} is the state preparation and measurement error, and ϵ_c is the average error per Clifford gate. The Clifford gates are decomposed into sequences of $\pi/2$ - and π -pulses about either the x - or y -axes. Up to $m = 100$ Clifford gates are probed, and the decay of the fidelity, $\mathcal{F}(m)$, is fit using the above model. The error per Clifford is found to be $\epsilon_c = 9.5(3) \times 10^{-4}$, while the SPAM error is $\epsilon_{\text{SPAM}} = 1.46(6) \times 10^{-3}$. The decay plot for this RBM sequence can be seen in figure 4.6. The independently characterised state preparation (section 4.2.1) and measurement (section 4.2.2) errors of $\sim 10^{-3}$, and $\sim 10^{-4}$ respectively, are consistent with the measured ϵ_{SPAM} .

4.2.6 Spin Coherence Times

Decoherence, as can be seen in figure 4.5, can be caused by laser frequency and power fluctuations, magnetic field noise, and spontaneous emission from the $3D_{5/2}$ level. The coherence time of the qubit is a useful diagnostic for finding unwanted noise sources, and for benchmarking the performance of the qubit as a store of quantum information.

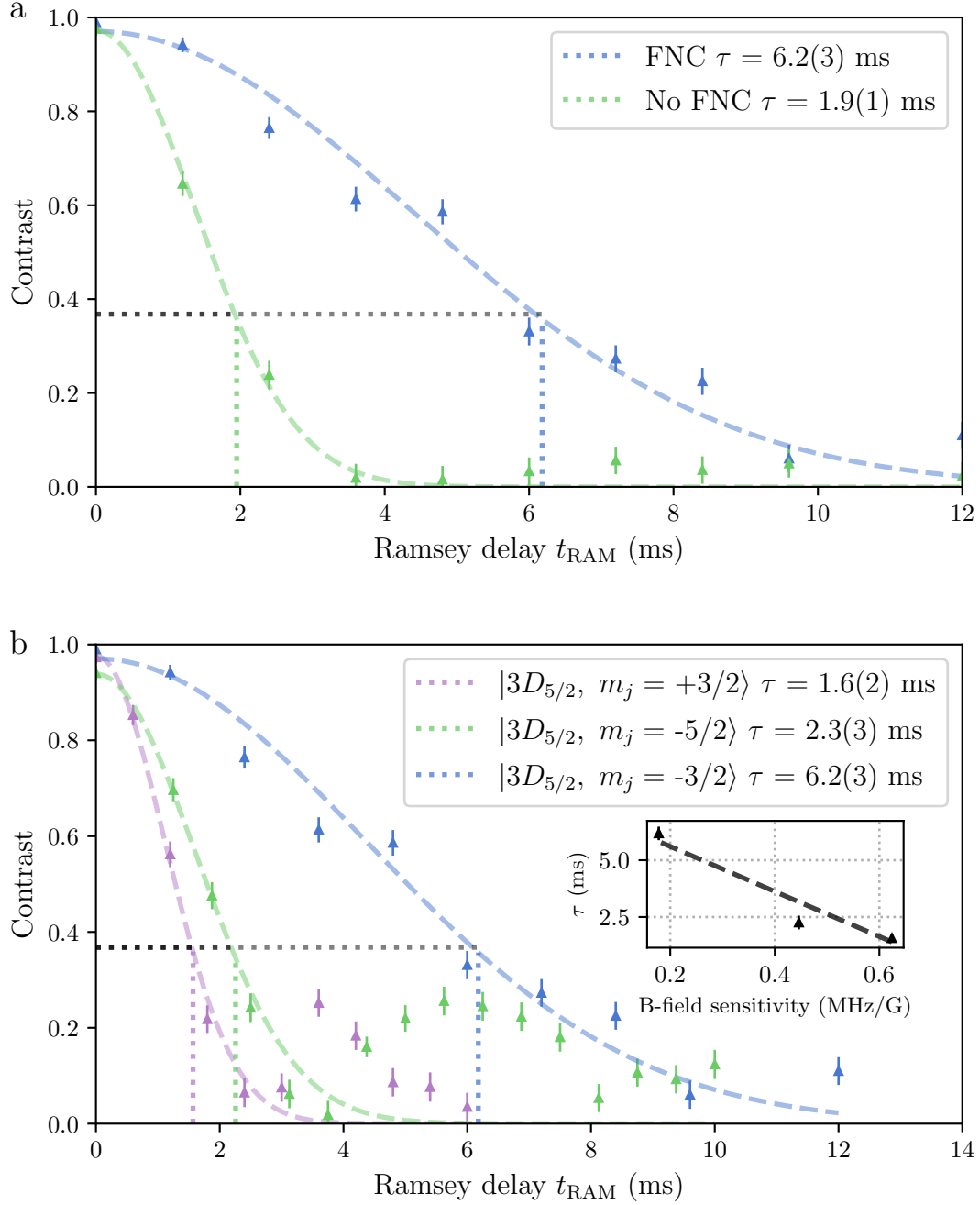


Figure 4.7: Coherence time measurements. **a)** Coherence time of the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ transition with and without fibre noise cancellation (FNC). **b)** Coherence time of three quadrupole transitions with varying magnetic field sensitivities. The inset plot shows a linear fit between the coherence time and the magnetic field sensitivity of the transition.

To measure the coherence times, we perform Ramsey sequences with varying delay durations. The Ramsey sequence is as follows:

1. Prepare the qubit in $|\downarrow\rangle$ via optical pumping, section 4.2.1.
2. Apply a $R_y(\pi/2)$ pulse to prepare the qubit in $|+\rangle = \frac{1}{\sqrt{2}}(|\downarrow\rangle + |\uparrow\rangle)$ superposition state.
3. Wait for a time t_{RAM} .
4. Apply a second $R_\phi(\pi/2)$ -pulse, with a variable axis of rotation, ϕ .
5. Measure the population in $|\downarrow\rangle$, $P_\downarrow$, via state-selective fluorescence, section 4.2.2.
6. Repeat steps 1-5 for different values of ϕ and fit the resulting fringe to find the contrast.
7. Repeat steps 1-6 for different values of t_{RAM} to find the decay in contrast with time. The coherence time is defined as the time at which the contrast has decayed to $1/e$ of its initial value.

With this tool, the efficacy of the fibre noise cancellation (FNC, see section ??) can be measured. Figure 4.7 *a* shows the coherence time of the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -3/2\rangle$ transition with and without FNC. The coherence time is increased from 1.9(1) ms to 6.2(3) ms with FNC, showing that without FNC, the coherence times are limited by laser phase noise.

With FNC enabled, to see if the coherence time is now limited by magnetic field noise, transitions with varying magnetic field sensitivities are compared. Figure 4.7 *b* shows the coherence time of three quadrupole transitions with varying $D_{5/2}, m_j$ numbers: $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = \{-5/2, -3/2, +3/2\}\rangle$. These transitions have magnetic field sensitivities $-0.446, -0.178, +0.624$ MHz G⁻¹ respectively. The contrast decays for all three qubits are shown in the figure. The inset plot shows a well correlated linear fit between the coherence time and the magnetic field sensitivity, suggesting that the coherence time is limited by magnetic field noise. For both $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ and

$|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = +3/2\rangle$, revivals in contrast can be seen which suggest that the dominant noise source is oscillatory and coherent over the Ramsey scan duration.

These measurements were taken without the MuMetal enclosure (section 3.3) fully assembled around the vacuum system, which we expect will greatly reduce external magnetic field noise, and therefore increase the coherence times.

4.3 Motion

Motion of ions trapped in the same potential can be described by a set of normal modes [38]. These modes are labelled with respect to the ion chain geometry: axial modes oscillate parallel with the chain, and radial modes oscillate perpendicular to the chain. Due to the geometry of the 729-nm beam access with respect to the ion chain (see section 3.4), only the radial modes can be addressed in our system. For the *NPL* trap (see section 3.2), the radial modes frequencies are non-degenerate. In our system, for a single ion, the radial mode frequencies are $\omega_{\text{upper}}/2\pi = 4.0$ MHz, and $\omega_{\text{lower}}/2\pi = 2.5$ MHz, and the axial mode frequency is $\omega_{\text{axial}}/2\pi = 1.6$ MHz. All work here is using the upper radial mode.

The motional mode, i , leads to sidebands in the 729-nm spectrum at $\omega_0 \pm \omega_i$ [39], where ω_0 , the carrier, is the frequency of the 729-nm laser when on resonance with the qubit transition. The $+\omega_i$ sideband is referred to as the blue sideband (BSB), and the $-\omega_i$ is the red sideband (RSB). The RSB, BSB, and RF fields near to resonance with the motional modes, are the tools used in this section to manipulate the motion of the ion.

The motional state preparation, heating rate, frequency stabilities, and coherence times are characterised in this section.

4.3.1 Cooling

For any interaction that involves the motion of the ion, it is essential to both prepare and measure the motional state with high fidelity. For all motional interactions considered in this thesis, the ion is initially prepared as close as possible to the

motional ground state, with a mean phonon number $\bar{n} \approx 0$. This is done by an initial stage of Doppler cooling, followed by pulsed sideband cooling on the 729-nm transition. The final motional state is characterised by measuring the population in Fock states using RSB and effective BSB pulses.

Doppler Cooling

Doppler cooling relies on the fact that light incident on a moving ion appears frequency-shifted in the ion's rest frame. For Doppler cooling of $^{40}\text{Ca}^+$, both the 397-nm and 866-nm lasers are applied. The 397-nm laser is red-detuned from the dipole $|4S_{1/2}\rangle \leftrightarrow |4P_{1/2}\rangle$ transition, leading to preferential scattering of 397-nm photons that have a net cooling effect [40]. The excited $4P_{1/2}$ level decays via two channels: emission of a 397-nm photon back to $|4S_{1/2}\rangle$, or an 866-nm photon to $|3D_{3/2}\rangle$, with a branching ratio of 14.5 [41]. Since many photon scattering events are required, the cooling cycle must be closed. To prevent population trapping in the metastable $|3D_{3/2}\rangle$ level, the 866-nm laser is applied on resonance during cooling.

Equilibrium is reached when the cooling rate balances photon recoil heating. Assuming the 397-nm transition is broadened only by its natural linewidth γ , the minimum Doppler temperature is,

$$T_{\text{Doppler}} \approx \frac{\hbar\gamma}{2k_B}, \quad (4.3)$$

where $\hbar$ is the reduced Planck constant, and k_B is Boltzmann's constant [40].

For $^{40}\text{Ca}^+$, the natural linewidth of the 397-nm transition is $\frac{\gamma}{2\pi} = 21$ MHz[42], giving $T_{\text{Doppler}} = 0.5$ mK. For a radial mode frequency of $\frac{\omega}{2\pi} = 4$ MHz, the expected mean phonon number is,

$$\bar{n} = \frac{1}{e^{\hbar\omega/k_BT} - 1}, \quad (4.4)$$

yielding $\bar{n} = 2.3$ at the Doppler limit. The parameters used here for Doppler cooling are summarised in table 4.2.

Parameter	Value
729-nm laser power	3.0 mW
729-nm laser detuning	-4.0 MHz*
729-nm pulse duration $N = 1$	2.5 μ s
729-nm pulse duration $N = N_{\max}$	33 μ s
Repumping 866-nm duration	14 μ s
Deshelving 854-nm duration	5 μ s
Number of cooling cycles $N_{\max}$	50

Table 4.3: *729-nm detuning is relative to the $|4S_{1/2}, m_j = -1/2\rangle \leftrightarrow |3D_{5/2}, m_j = -5/2\rangle$ transition frequency.

Experimental parameters used for sideband cooling. Other parameters for the repump and deshelve pulses are the same as for state preparation, see table 4.1. The 729-nm laser is detuned to the RSB resonance for the upper radial mode, $\omega_{\text{upper}}/2\pi = 4.0$ MHz, and the pulse duration for $N = N_{\max}$ is chosen to be the expected RSB $R(\pi)$ -pulse duration for $n = 1$.

Sideband Cooling

To further cool the ions toward their motional ground state, resolved sideband cooling is used. As mentioned above, the motion of the ion modulates the absorption spectrum of the ion, leading to red and blue sidebands (RSB and BSB). For the $|4S_{1/2}\rangle \leftrightarrow |3D_{5/2}\rangle$ transition, at appropriate motional mode frequencies, interaction strengths, and using a narrow linewidth laser (section ??), these sidebands can be resolved spectroscopically. The pulsed sideband technique employed consists of RSB pulses on the 729-nm transition, followed by deshelving, and repumping pulses on the 854-nm and 866-nm transitions respectively. For efficient cooling, initial short 729-nm pulses are applied to preferentially excite the thermal population with high phonon number, as the pulse duration for RSB $R(\pi)$ is proportional to $\sqrt{n}$, where n is the phonon number. The RSB pulse length is linearly increased to the expected RSB $R(\pi)$ -pulse duration for $n = 1$.

An example pulse sequence for a single experimental shot using sideband cooling, can be seen in figure 4.8 *a*. Experimental parameters used are summarised in table 4.3.

To verify the efficacy of our sideband cooling, thermometry experiments are performed by probing Fock state populations. This is done using on resonance RSB

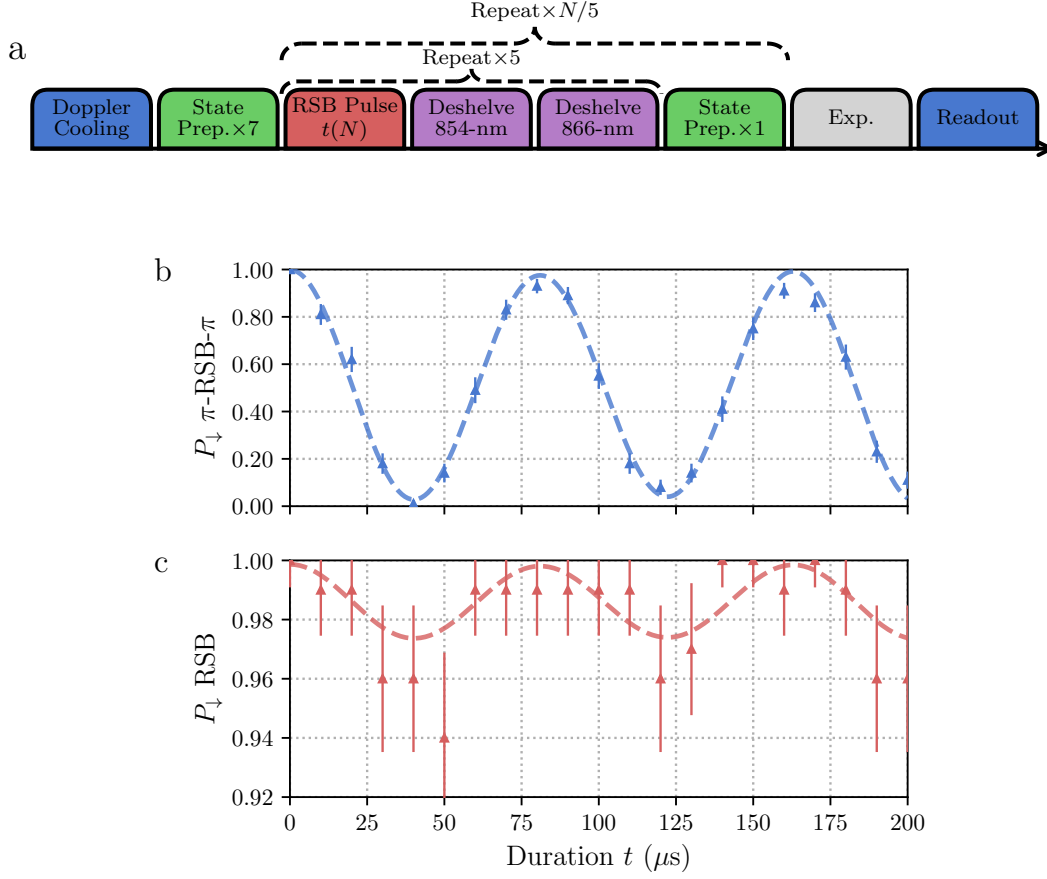


Figure 4.8: **a)** Example pulse sequence for one experimental shot including pulsed sideband cooling. The *Exp.* block represents an arbitrary sequence of operations, for the experiment under study, such as a thermometry sequence. The blue blocks represent Doppler cooling and readout (section 4.2.2), the green blocks represent qubit state preparation (section 4.2.1), the red blocks are RSB pulses with varying durations $t(N)$, and the purple blocks are deshelling pulses. After every 5 sideband cooling pulses, there is an interleaved state preparation sequence to ensure there is no population trapped outside of the closed cooling cycle. The RSB length is linearly increased with pulse number N , upto the expected RSB $R(\pi)$ -pulse duration for $n = 1$. **b, c)** Thermometry scans after Doppler and sideband cooling. The red points are the populations measured via on-resonance RSB pulses, and the blue points are the populations after a sequence of $R(\pi)$ -RSB- $R(\pi)$ with varying RSB probe durations. The dashed lines are fits to a thermal Fock state distribution, with truncation at $n = 100$. The extracted mean occupation number is $\bar{n} = 0.03(1)$, and $\eta\Omega/(2\pi) = 11.6(1)$ kHz.

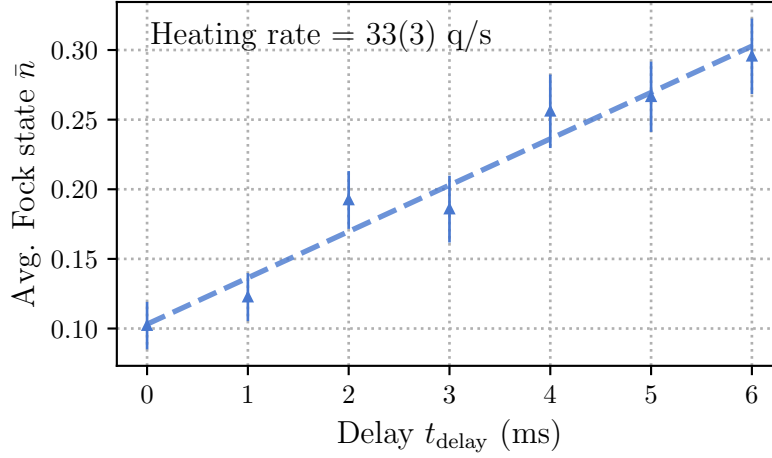


Figure 4.9: Heating rates of upper radial mode, measured by varying the delay between cooling and thermometry pulses. The dashed line is a linear fit to the data, with slope 33(3) quanta/s. The error bars are given by the standard deviation of the fitted $\bar{n}$.

pulses and effective BSB pulses composed of a π -RSB- π pulse sequences. The time dynamics of population flopping is measured as RSB pulse length, t , is varied. In the case of Fock state $|0\rangle$, there should be full contrast “BSB” oscillations, and no visible oscillations when probed with RSB pulses. A thermal Fock state distribution [43] (with truncation at Fock state = 100) is fitted to these signals to extract the mean occupation number, and $\eta\Omega$, the carrier Rabi frequency multiplied by the Lamb-Dicke parameter. A typical thermometry scan after Doppler and sideband cooling can be seen in figure 4.8 *b, c*, where *b* shows the population after effective “BSB” pulses, and *c* shows the population after RSB(t) pulses with duration t . These oscillations are fit to thermal Fock state distributions, shown by the dashed lines. The mean occupation number after sideband cooling is found to be $\bar{n} = 0.03(1)$, and $\eta\Omega = 11.6(1)$ kHz.

4.3.2 Heating Rates

After cooling the motion, the ion will return to a temperature in equilibrium with the environment. However, this process can be slowed by adequate isolation of the ion.

In UHV ion trap systems (see section 3.4), motional heating is predominantly caused by electric field noise from the ion trap [2]. Noise due to thermal charge fluctuations at the trap surface can be mitigated by increasing ion-electrode distances. The *NPL* trap (section 3.2), used here, has an ion-electrode distance of 250 μm , larger than most surface traps, but less than that of a macroscopic blade or rod style trap [44]. Heating rates were probed by a series of thermometry experiments, explained above, with varying delay times, t_{delay} , between cooling and thermometry pulses. A typical plot can be seen in figure 4.9. The heating rate of the system was found by a linear fit, shown by the dashed line, to be 33(3) quanta per second on the upper radial 4 MHz mode on one ion. As will be discussed in section 4.3.4, the heating rate sets a limit for the motional coherence time.

4.3.3 Motional Mode Stability

Due to thermal drifts and microphonics introducing amplitude and frequency noise on the RF chain, drifts of the radial motional mode frequencies are seen over time. As will be discussed in the following section 4.3.4, this can lead to dephasing of the motional state. Here, the current motional mode stability is characterised to estimate its contribution to dephasing, and to set a benchmark for future stability improvements.

The motional mode frequency is measured via an RF “tickle” experiment. This consists of short ($\sim 5 \mu\text{s}$) RF field pulses applied to one of the trap DC electrodes which has a non-zero electric field projection to the mode we wish to probe. The frequency of the RF field is scanned around the expected motional mode frequency, and when resonant, drives the motion of the ion. This resonance is measured by observing an increase in fluorescence counts using a 397-nm laser red-detuned by 100 MHz. To measure mode frequency drift, this sequence is repeated every 10 seconds over multiple hours. A typical plot can be seen in figure 4.10 *a*. Over this two hour period, the frequency had an absolute drift of 1.5 kHz. Typically, we calibrate the mode frequency every 10 minutes, which is indicated with the red

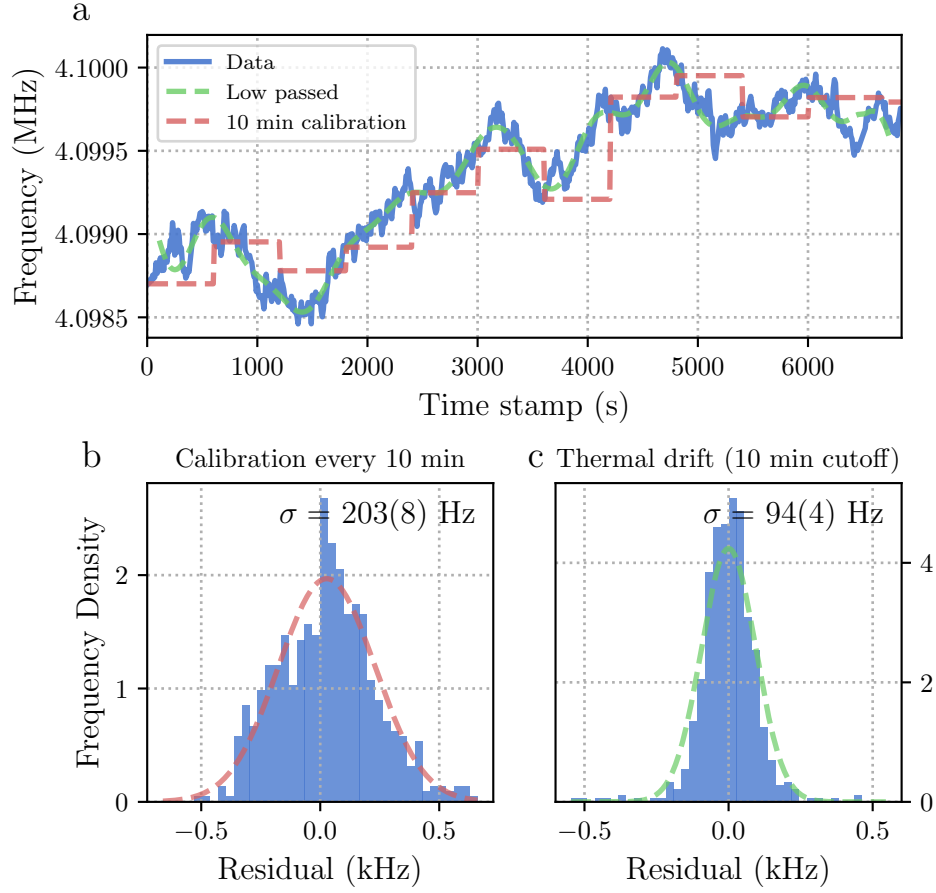


Figure 4.10: **a)** Motional mode frequency drift over time. The radial motional mode frequency is measured by exciting motion using an RF field in resonance with the motional frequency, and looking at increased fluorescence. The resulting frequency trace is shown in blue. The green line shows a low pass filter of the data, with cut-off frequency of 1.67 mHz, corresponding to an assumed 10 minutes slow thermal drift. The motional mode frequency is typically calibrated every 10 minutes, which is indicated by the red line. **b)** Residuals from the measured data and expected calibrated mode frequencies. The standard deviation of these residuals suggests the typical mis-set detuning of the mode frequency in current experiments with automated mode frequency calibrations every 10 minutes. **c)** Residuals from the measured data and the low passed trace. The standard deviation of these residuals are due to noise with period less than 10 minutes.

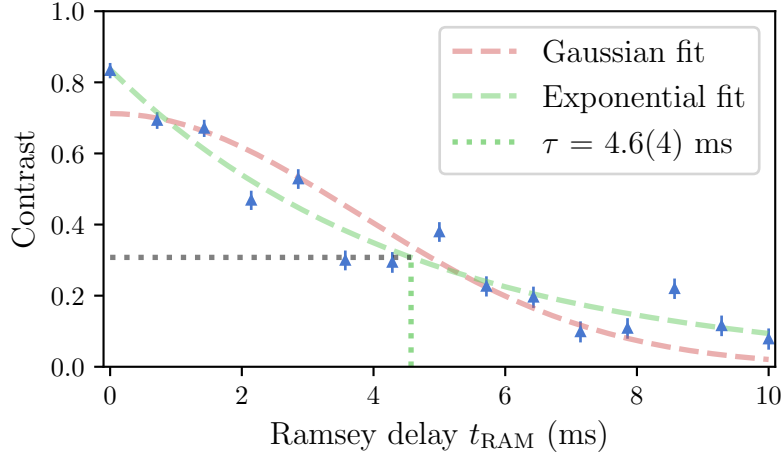


Figure 4.11: Motional coherence time measured by Ramsey sequence on the motional state. The blue points are the population in $|\downarrow\rangle$ after a Ramsey sequence with varying delay time, t_{RAM} . The dashed line is an exponential decay fit, with a coherence time $\tau = 4.6(4)$ ms.

dashed line. Residuals between this calibrated line and the data, shown in *b*, have a standard deviation of $\sigma/(2\pi) = 203(8)$ Hz, giving the expected uncertainty of the mode frequency detuning given our current periodic calibration routine. A slow drift with frequency cut-off of 1.67 mHz is also fitted to the data, shown in green. The residuals from this fit, shown in *c*, have a standard deviation of $\sigma/(2\pi) = 94(4)$ Hz. This suggests that the motional mode frequency is relatively stable over time scales shorter than 10 minutes, but has a slow drift on longer time scales, which is likely due to thermal fluctuations.

4.3.4 Motional Coherence Times

We will utilise the motional modes of the ion to store quantum information. As with the section on spin coherence times 4.2.6, the fidelity of operations, and success probability of algorithms will be limited by the coherence time of the motional state. To measure motional coherence, a Ramsey sequence is performed between states $|\downarrow, 0\rangle$ and $|\downarrow, 1\rangle$, where the first element is the qubit state, and the

second element is the motional Fock state. To prepare these states the following pulse sequence is applied:

1. State prepare system to $|\downarrow, 0\rangle$ via optical pumping and sideband cooling.
2. Apply $R_y(\pi/2)$ -pulse on carrier transition to prepare the qubit in $(|\downarrow, 0\rangle + |\uparrow, 0\rangle)/\sqrt{2}$.
3. Apply $R(\pi)$ -pulse on the RSB to prepare the motional state in $(|\downarrow, 0\rangle + |\downarrow, 1\rangle)/\sqrt{2}$.
4. Delay for some time t_{RAM} .
5. Apply $R(\pi)$ -pulse on the RSB to map accumulated phase onto the qubit state.
6. Apply $R_\phi(\pi/2)$ -pulse on the carrier, with variable axis of rotation, ϕ .
7. Measure the population in $|\downarrow\rangle$ via state-selective fluorescence, section 4.2.2.
8. Repeat steps 1-7 for different values of ϕ to find the contrast of the Ramsey fringe.
9. Repeat steps 1-8 for different values of t_{RAM} to find the decay in contrast with time. The coherence time is defined as the time at which the contrast has decayed to $1/e$ of its initial value.

There are two main mechanisms for motional decoherence[45]: motional heating, as characterised in section 4.3.2, and motional dephasing due to mode frequency instability, as discussed in section 4.3.3.

A heating rate dominated coherence time is given by, $\tau_{\text{HEAT}} = (\sqrt{e} - 1)/\dot{n}$ [45], where $\dot{n}$ is the heating rate. A dephasing dominated coherence time, if noise is correlated on timescales long compared to the Ramsey delay [46], is characterised by a Gaussian decay profile with a coherence time of $\tau_{\text{DEPH}} = \sqrt{2}/\sigma_\omega$.

Evaluating both of these models with $\dot{n} = 33(3)$ q/s, and $\sigma_\omega/(2\pi) = 94(4)$ Hz, the expected motional coherence times of $\tau_{\text{HEAT}} \approx 20$ ms and $\tau_{\text{DEPH}} \approx 2.3$ ms respectively, are found. From these auxiliary characterisations of the motional

modes, it is expected that the coherence time will be limited by motional dephasing. Figure 4.11 shows the decaying coherence when probed by the above sequence. It can be seen that the contrast has a good fit assuming an exponential decay (green dashed line), defined by $C(t_{\text{RAM}}) = C(0)e^{-t_{\text{RAM}}/\tau}$, where $C(t_{\text{RAM}})$ is the contrast at time t_{RAM} , and τ is the coherence time. The fit yields a motional coherence time of $\tau = 4.6(4)$ ms, defined as the time at which the contrast decays to $1/e$ of its initial value. An exponential decay suggests that the noise source causing the dephasing is correlated only for durations much shorter than the Ramsey delay times [46]. The Gaussian decay, (red dashed line), which was appropriate for the previous spin coherence time measurements in section 4.2.6, is a poor fit here.

4.4 Spin-Motion

For full control of the spin-motion hybrid system, we require interactions that couple the two. Perhaps the simplest of this class of interactions are the red- and blue-sidebands (RSB and BSB respectively). The RSB interaction was used previously in the thermometry and sideband cooling sections 4.3.1. The RSB (BSB) consists of a single frequency laser tuned at the carrier frequency minus (plus) the motional mode frequency, ω_m . The interaction is well described by the (Anti-) Jaynes-Cummings Hamiltonian, and effectively couples spin flips with the addition or subtraction of a motional quanta depending on the initial spin state. Here, another such interaction coupling spin and motion, known as the spin-dependent force, SDF, is introduced. While the RSB and BSB interactions couple the spin and motion in the motional Fock basis, the SDF couples them through spin-dependent displacements, thus in a coherent state basis. The direction of the displacement is determined by the spin state and an effective detuning parameter.

4.4.1 Spin-Dependent Forces

The optical Mølmer-Sørensen (MS) scheme [6] is used to generate the SDF via a bichromatic laser field. Bichromatic refers to the simultaneous application of two

tones symmetrically detuned around the qubit carrier frequency, with absolute detuning approximately equal to the motional mode frequency, $\delta \approx \omega_m$. The resulting interaction, when ignoring off-resonant and higher order couplings, is given by,

$$\hat{H}_{MS} = i\hbar\eta\Omega\hat{\sigma}_\phi \cos(\delta t) \left(\hat{a}e^{-i\omega_m t} + \hat{a}^\dagger e^{i\omega_m t} \right), \quad (4.5)$$

where η is the Lamb-Dicke parameter, Ω is the carrier Rabi frequency, $\hat{a}(\hat{a}^\dagger)$ is the lowering (raising) operator, and $\hat{\sigma}_\phi$ is the Pauli operator with $\hat{\sigma}_\phi = \cos(\phi)\hat{\sigma}_x + \sin(\phi)\hat{\sigma}_y$. Applying the rotating wave approximation, and defining $\delta_g = \delta - \omega_m$, the interaction Hamiltonian can be approximated to,

$$\hat{H}_{MS} \approx \frac{i\hbar\eta\Omega}{2} \hat{\sigma}_\phi \left(\hat{a}e^{-i\delta_g t} + \hat{a}^\dagger e^{i\delta_g t} \right). \quad (4.6)$$

The trajectory of this displacement can be controlled by varying δ_g : on resonance, $\delta_g = 0$, corresponds to linear trajectories, whilst off resonance, $\delta_g \neq 0$, corresponds to cyclic trajectories where after some duration $t = 2\pi/\delta_g$, the motion returns to the initial state (with perhaps some phase shift [47]). The dynamics of the SDF may be measured by projective spin measurements, the population in $|\downarrow\rangle$, $P_\downarrow$, after applying the SDF for duration t is described by [48],

$$P_{\downarrow, \text{th}} = \frac{1}{2} \left[1 + e^{-4(\bar{n} + \frac{1}{2})|\alpha(t, \delta_g)|^2} \right], \quad (4.7)$$

where $\bar{n}$ is the initial thermal phonon occupation, and $|\alpha(t, \delta_g)| = \Omega_{\text{SDF}} \sin(\delta_g t/2)/\delta_g$, is the absolute displacement of the initial state. This control is utilised in both two-qubit entangling gate experiments, as well as in the creation of squeezed states.

Calibrating the SDF

Geometric phase gates, such as the MS interaction, are widely employed in ion trap experiments due to their robustness against variations in initial motional state. However, the SDF is sensitive to various frequency and power miscalibrations and drifts of these between calibrations. Here, we describe the workflow for calibrating and optimising the SDF behaviour.

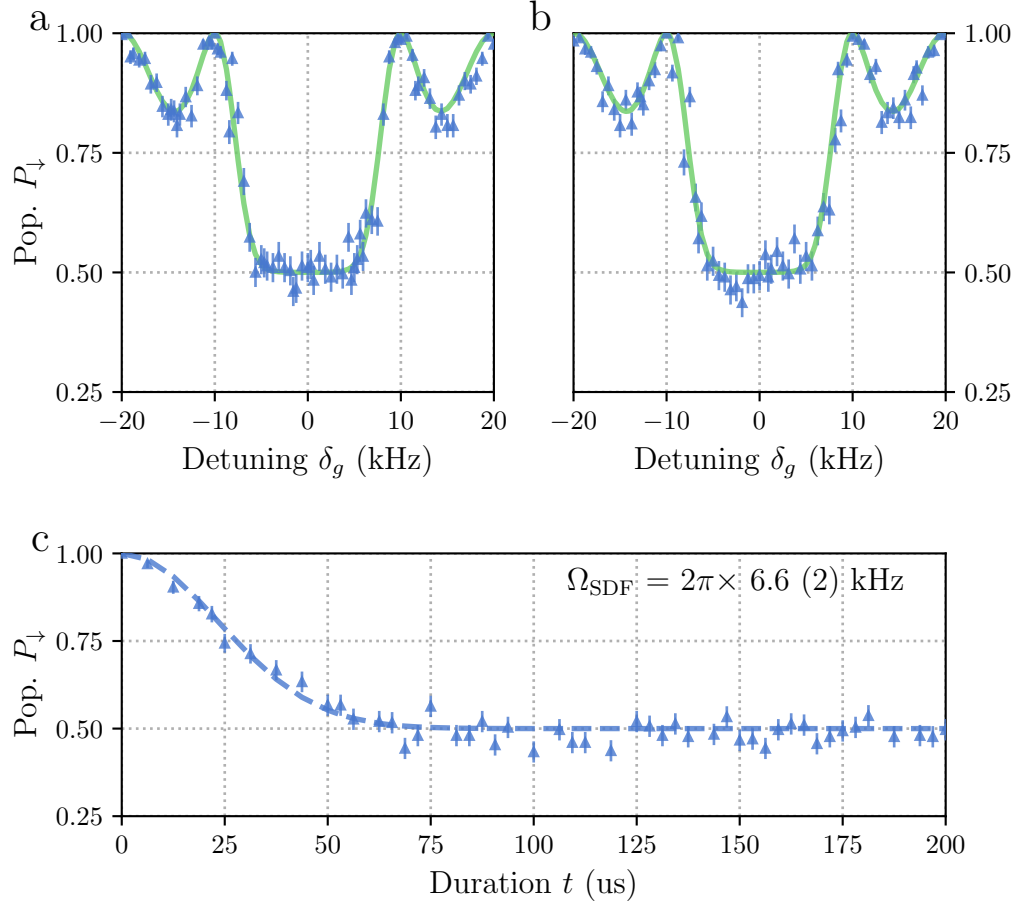


Figure 4.12: Detuning and duration scans of the spin-dependent force applied to an initial thermal state with $\bar{n} = 0.03(1)$. **a)** Detuning scan, where the detuning, δ_g , is varied, and the duration is fixed at $t = 100 \mu\text{s}$. The green line is a fit to the expected behaviour given by equation 4.7. **b)** Detuning scan with a mis-set qubit frequency of 0.5 kHz. The green line is the same fit as in *a* to guide the eye. **c)** Duration scan, where the duration, t , is varied, and the detuning is fixed at $\delta_g = 0$. The dashed line is a fit to the expected behaviour given by equation 4.7 with only Ω_{SDF} as a free parameter.

In practice, the SDF interaction involves relatively few degrees of freedom: the central qubit frequency, the motional mode frequency, the power in each tone, and the durations of the pulses must all be calibrated. The complexity comes from the fact that the ion is a multi-level system, possesses multiple motional modes, and that each experimental parameter can only be controlled with finite precision.

The effect of nearby motional modes is negated by either selecting the interaction mode to be well separated from the others either in frequency, by beam geometry, by operating the SDF near to resonance of the desired mode, or by using amplitude shaping of the pulse. Smoothing a square pulse through amplitude shaping narrows the pulse frequency bandwidth, and thus reduces unwanted off-resonant excitation of other modes.

The power balance of the SDF tones is complicated due to our use of AOMs. AOMs generate frequency shifts in the laser beam at the expense of small frequency-dependent angular shifts. After the AOM, the beam is coupled into a single mode fibre (see figure 3.10), and so the AOMs effectively introduce a frequency-dependent loss. To calibrate the power balance, a pick off of the bichromatic beam after the ion is monitored on a high bandwidth photodiode. The beatnote contrast is measured for the two tones and the power balance is optimised by maximising this contrast. The far off-resonant levels of our ion lead to light-shifts of our qubit frequency. To account for the light-shift the central qubit frequency must be calibrated at the optical power used for the interaction. This is done with the SDF interaction itself. When the qubit frequency is set incorrectly, the two tones will no longer have the same absolute detuning from their respective RSB or BSB. This manifests as a “skewness” of the SDF detuning trace. By varying the qubit frequency and inspecting these detuning scans, the desired SDF behaviour can be found.

To verify the behaviour of the calibrated SDF, the experimental data is compared with theory. Both “detuning” and “duration” scans are used. The “detuning” scan is performed by varying the detuning, δ_g , of the interaction, whilst keeping the SDF duration, t , constant, and vice-versa for the “duration” scan.

Figure 4.12 *c* shows the measured duration scan described by equation 4.7, with

$\delta_g = 0$ and the motional state assumed to be thermal with average Fock state $\bar{n} = 0.03(1)$ which was found previously from thermometry measurements. Here, the displacement, $|\alpha(t)| = \Omega_{\text{SDF}}t/2$, where Ω_{SDF} is the SDF amplitude. The dashed line fit is given by floating Ω_{SDF} only, and finds $\Omega_{\text{SDF}}/(2\pi) = 6.6(2)$ kHz. These measurements were performed with total 729-nm power of 2 mW and $\delta_{\text{LS}}/(2\pi) = 3.5$ kHz, the light shift due to the 729-nm laser relative to the unperturbed qubit frequency.

The detuning scan, shown in figure 4.12 *a*, has the same experimental parameters, but with fixed duration $t = 100$ us. The fit in green is taking the displacement to be $|\alpha(t)| = \Omega_{\text{SDF}} \sin(\delta_g t/2)/\delta_g$, given the value for Ω_{SDF} measured from the duration scan. Here, a qualitatively good agreement between the measured data and the expected behaviour is seen, with the main features being the “closure” at $\delta_g = 2\pi/t$ where the motional state returns to the initial state, and the central flat region around $P_{\downarrow, \text{th}} = 0.5$ where the two motional wave packets are non-overlapping.

Figure *b* shows an example detuning scan when the qubit frequency is mis-set by 0.5 kHz. The central region now displays a clear asymmetry around $\delta_g = 0$. The green line here is the fit from *a* to guide the eye.

4.4.2 Two-Qubit Entangling Gates

We perform two-qubit entangling gates using the Mølmer-Sørensen (MS) interaction [6]. This interaction is the same described SDF from the previous section, but applied globally to two ions. The MS interaction relies on the spin dependent geometric phase accumulated during the motional displacement [39]. To create a two-qubit entangled state, a differential geometric phase of $\pi/2$ must be accumulated between the two-qubit basis states [39]. To ensure there is no residual motional entanglement, the final motional state must return to the initial state. In practise, using an SDF detuned by δ_g , this is achieved by applying the MS interaction for a duration $t_g = 2n\pi/\delta_g$, where n is the number of “loops”. The MS gate is a universal two-qubit gate, and along with only single qubit gates, constitutes a universal gate set for discrete quantum algorithms.

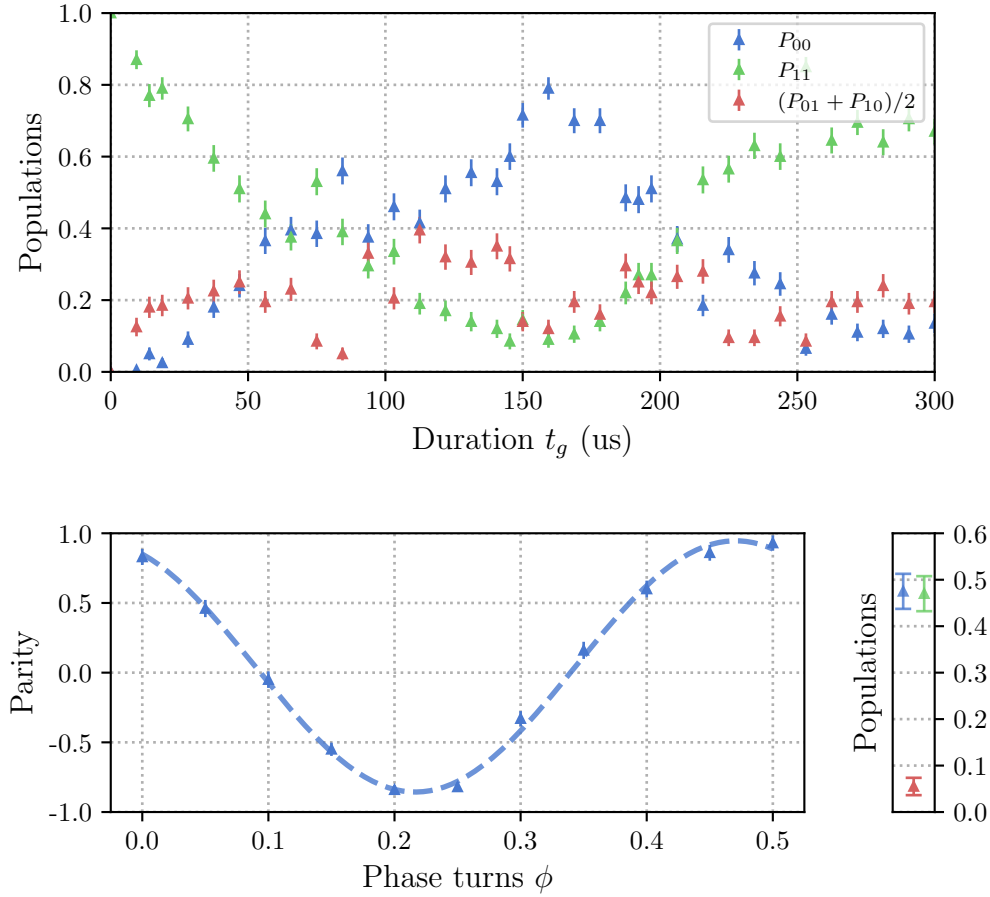


Figure 4.13: Experimental data for $\mathcal{F} = 92(2)\%$ Mølmer-Sørensen (MS) two-qubit entangling gate. **a)** Populations versus duration scan of the MS interaction. Each point corresponds to the average of 200 shots, and the error bars are given by the standard deviation of these averages. The blue points are the populations of both ions being in the $|0\rangle$ state, P_{00} , the green points are the populations of both ions being in the $|1\rangle$ state, P_{11} , and the red points are the ions being the the opposite states, $(P_{01} + P_{10})/2$. The desired entangling gate corresponds here to $t_g = 75\mu\text{s}$. **b)** Parity oscillations after the MS interaction. The parity is defined as $P_{00} + P_{11} - P_{01} - P_{10}$. The fitted contrast here is $0.90(2)$. **c)** Populations at the desired entangling gate duration, $t_g = 75$ us.

Here we quote the current fidelity of experimentally demonstrated two-qubit gates on our system. The fidelity serves to quantify the similarity of two density matrices. Here for a two-qubit entangling gate, we target the creation of the Bell state $|\Phi_{\phi_0}^+\rangle = 1/\sqrt{2}(|00\rangle + e^{i\phi_0}|11\rangle)$, from an initial state of $|00\rangle$, where ϕ_0 is some fixed phase. The fidelity between our mixed state ρ , and the pure Bell state may be given by,

$$\mathcal{F} = \langle \Phi_{\phi_0}^+ | \rho | \Phi_{\phi_0}^+ \rangle = \frac{1}{2}(\rho_{00,00} + \rho_{11,11}) + \frac{1}{2}(e^{i\phi_0}\rho_{11,00} + e^{-i\phi_0}\rho_{00,11}), \quad (4.8)$$

where $\rho_{ij,kl} = \langle ij | \rho | kl \rangle$. To extract the fidelity experimentally, a partial state tomography protocol is used [49], where the first bracketed term of equation 4.8 is measured by performing projective measurements of the two ions after the gate sequence to extract populations, and the second bracketed term, known as the coherence terms, are measured by applying a global analysis $R_\phi(\pi/2)$ pulse to the two ions and applying parity measurements. Parity measurements are defined as $P = P_{00} + P_{11} - P_{01} - P_{10}$, where P_{ij} is the population of the two ions in state $|i\rangle|j\rangle$ after the gate sequence and variable $\pi/2$ -pulse. Creating a Bell state is essential for two-qubit entanglement, but its basis can vary as long as it is consistent across experiments. This allows the Bell state phase, ϕ_0 , to float, which in practice means using only the amplitude of the fitted parity oscillations to compute fidelity, ignoring any phase offset.

The best two-qubit entangling gate fidelity currently achieved on our system is $\mathcal{F} = 92(2)\%$. As shown in figure 4.13, the magnitude of the parity scan was measured to be $0.90(2)$, while the populations $(\rho_{00,00} + \rho_{11,11}) = 0.95(2)$. Each population point in these figures are found by taking the average of 200 shots of the gate sequence.

These results serve as both a proof of principle for full spin control on our system, but also as a benchmark to which we can compare future system improvements. For future work, we will require the use of this entangling gate either as Bell state preparation for input to analogue simulation experiments [20], or as a primitive gate for the spin component of hybrid algorithms [11, 12]. In both these cases, especially

any use that requires multiple concatenated entangling gates, we will likely require improved gate fidelities. Current gate fidelities are limited by the motional dephasing. The gate error due to motional dephasing follows (see section 4.4.6 in [50])

$$\epsilon_{\text{DEPH}} = 0.686 \frac{t_g}{\tau_{\text{DEPH}}}, \quad (4.9)$$

where for these experiments we estimate a motional coherence time of approximately $\tau_{\text{DEPH}} = 1$ ms. Given the gate duration of $t_g = 75$ μs , this gives a dephasing error of $\epsilon_{\text{DEPH}} = 0.05$, or approximately 5%. This is the dominant error source in our current gate fidelity, and can be improved by decreasing the gate duration, or by improving the mode frequency stability. We have not yet attempted to optimise these gates, nor have we attempted to fully account for the observed errors. However, this work serves as a benchmark for future improvements, and a demonstration of the full spin control capabilities of our system.

5

k-copy Randomised Benchmarking

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Overview (abstract?) of method.

Revealing Correlated Errors through Symmetry in Parallel Randomised Benchmarking Structured noise in quantum processors can invalidate the independence assumptions underlying many noise models and error-correction protocols, and as such constitutes a critical process to characterise in large-scale fault-tolerant quantum computation. Explicitly accounting for such correlations typically requires characterisation techniques that scale poorly with system size. We present a scalable method for quantifying correlated noise using parallel (“k-copy”) Clifford randomised benchmarking. By applying identical Clifford sequences across multiple qubits, the protocol enforces a symmetry that decomposes the dynamics into invariant subspaces, enabling direct separation of correlated and uncorrelated error components from two-qubit measurement outcomes. We experimentally demonstrate this method on a trapped-ion quantum processor, resolving two-body correlations across up to 8 ions. Parity measurements further reveal the underlying symmetry-resolved dynamics, with correlated noise driving population into invariant subspaces that can be observed without full process reconstruction. This approach provides a practical tool for diagnosing structured noise in quantum devices and for informing the design of device-specific error mitigation and error-correction strategies. As quantum processors scale, fault-tolerance, and error mitigation become necessary to overcome decoherence and systematic control errors [shor_fault-tolerant_1996, aharonov_fault-tolerant_2008, cai_quantum_2023]. Crucially, errors which induce correlations across multiple qubits require mitigation and correction strategies built on noise models that faithfully represent their correlated structure; assuming simple white noise models can lead to systematic underestimation of logical error rates [aharonov_fault-tolerant_2006], or systematic bias of the corrected observable outcome [foldager_can_2024].

It is often assumed that localised gate errors are scrambled throughout complex circuits to more amenable depolarisation, or white noise channels, as was shown in the case of sufficiently deep random circuits [dalzell_random_2024, arute_quantum_2019]. However, for near-term shallow circuits, and highly structured deep circuits, this assumption does not hold [foldager_can_2024]. Noise channels can be coerced to more favourable models via twirling or random circuit compilation [geller_efficient_2013, wallman_noise_2016, hashim_benchmarking_2020], however, these methods must be optimised and verified on device. Determining an appropriate noise model in a given device therefore demands characterization protocols that directly diagnose correlated noise, ideally with minimal experimental overhead. Many such characterization methods exist, based on gate and process tomography [rudinger_experimental_2021], the study of collective spin phenomena [gulacsi_characterizing_2025], noise learning [harper_efficient_2020, harper_learning_2023, nielsen_efficient_2021], random circuits [sarovar_detecting_2020] and several randomised benchmarking (RB) techniques [mckay_correlated_2020, erhard_characterizing_2019, gambetta_characterization_2012]. These methods vary drastically between high expressibility of the measured error channel (e.g. tomographic methods), to high scalability of the protocol (e.g. RB methods), but often must compromise between these two aspects. We suggest a scalable RB method, named *k-copy RB*, that retains the ability to discern the quality of correlated noise, e.g. if the origin is due to fluctuating global fields, an entangling channel, or correlated non-unital errors. A related protocol, termed “Correlated RB” [mckay_correlated_2020] (based on “Simultaneous RB” [gambetta_characterization_2012]), uses independent local twirling of multiple system copies to estimate a distance of the measured channel to a product channel (non-correlated). In contrast, “k-copy RB” uses identical local twirling to preserve symmetry sectors, so spatially correlated noise appears as population trapped in, or leaking between, identifiable multi-qubit subspaces. Previously, the technique of observing subspace decay rates has been used to benchmark leakage errors during two-qubit entanglement [sutherland_subspace_2025], and

in suppressing the detrimental effects of laser decoherence in correlation spectroscopy measurements [clements_lifetime-limited_2020].

5.1 Standard RB

Perhaps in theory section?

Need to cover:

- Twirling over a group
- Liouville representation of a CPTP map
- Irreps of the Clifford group
- Clifford approximates twirling over the unitary group (Haar measure)
- How Liouville representation is transformed under twirling
- RB protocol for single qubit Clifford group
- How to extract error per Clifford from decay rate

5.2 Noise channels

As will also be used in virtual distillation, may move to theory sections

A description of the noise channels we consider in this work. Explicit CPTP maps for these channels.

5.2.1 Why isn't everything white noise?

5.2.2 Unitary noise

Both continuous and probabilistic.

5.2.3 Depolarising noise

An ensemble of unitary noise channels, with a uniform distribution over the Bloch sphere. Get this from twirling over Clifford group!

5.2.4 Spatially correlated noise

Common noise An ensemble of unitary noise channels, with a uniform distribution over the Bloch sphere, but the same unitary is applied to multiple qubits. Can enforce some symmetry to this channel by twirling over the k -copy Clifford group. This is the channel we will use to model correlated noise in our experiments.

Entangling noise In SC ZZ entangling noise is common.

5.2.5 Can we make everything white noise?

Have a look at Balint paper, deep random circuits, etc.

5.3 Single qubit Clifford applied in parallel to multiple qubits

5.3.1 Irreps of the 2-copy Clifford group

This is in Helsen paper.

5.3.2 Twirling over the 2-copy Clifford group

This naturally arises in error mitigation techniques. This also gives a sense of why Pauli twirling is important.

5.4 Pairwise decay channels

Another analytics section giving the expected decay curves as well as maybe numerics. Include some data here for the asymptotes and parity oscillations.

5.4.1 Asymptotes

For considered noise channels, what are the expected final density matrices, and the asymptotes. Further, how does the irreps play into this.

5.4.2 Parity oscillations

Probing residual coherences for isolating decay rates of the different irreps. How to do this fast!

5.4.3 Invariant subspaces

If we can chose a starting state, can we isolate a decay sector? Would potentially be nice to do this on experiment?

5.5 Efficient fitting of decay channels

Numerics for what is a good way to fit the decay curves.

5.6 Injecting errors

Data section!

5.6.1 Independent noise using single addressing

Random stark shifts on each qubit.

5.6.2 Correlated noise

Detuning the 729 laser. Random phase offsets.

5.7 Characterising spatial correlations on 8-ion chain

Data section!

5.8 Possible extensions

5.8.1 Extending to >2 qubits

5.8.2 Extending to time correlations

Protocol—Our method distinguishes correlated noise through parallel, “k-copy”, application of the single-qubit Clifford group, $Cl^{\otimes k}$ [wallman_randomized_2014, helsen_representations_2018]. Here, the enforced symmetry preserves a set of two-qubit subspaces (tabulated in supplemental 5.1). These subspaces provide a natural basis to study the action of the generic noise channel. For a subset of noise channels, which we consider here, the subspaces may be considered independent of each other. However, in general, a noise channel can couple subspaces together (if they are isotypic representations). Through observing the dynamic populations of these subspaces, e.g. measuring decay rates, the noise type and magnitude can be determined. For two qubits, these noise channels can be [sarovar_detecting_2020]: independent depolarising channels, common single-qubit rotations, two-qubit entanglement, correlated non-unital, and two-qubit swaps.

“k-copy” RB is executed experimentally by applying identical single qubit Clifford RB sequences to all qubits, as seen for one random sequence of length M in figure 5.1 (a). As with standard Clifford benchmarking [dankert_exact_2009, magesan_scalable_2011], r random Clifford sequences are generated with length M to ensure that the ensemble of random circuit approximates twirling over the group. To build statistics of the final state populations, each sequence is repeated s times. Classical post processing is required for calculating pairwise correlated populations from per-shot data. The correlated populations for a given pair (i, j) are denoted by $\mathcal{P}_{SS}^{(i,j)}, \mathcal{P}_{SF}^{(i,j)}, \mathcal{P}_{SS}^{(i,j)}, \mathcal{P}_{FF}^{(i,j)}$, with S and F representing success and failure of measuring the target outcome, respectively. The target outcome is a computational eigenstate, which is randomised between $|0\rangle$ and $|1\rangle$ for each generated sequence, to account for asymmetry in readout and state preparation [knill_randomized_2008].

As an example of what “k-copy” reveals, figure 5.1 (b) shows schematically the final state after only a dominant common rotation error affects qubits $i = 3$ and $i = 4$, while other qubits are dominated by independent noise. If the common rotations are probabilistic, or the channel is twirled over $Cl^{\otimes k}$, these common rotations lead to spatial correlations. These correlations are revealed through repeating the RB circuit for different sequences of the same total length, M . In figure 5.1 (c) 5 final states before measurement are shown overlaid on the same Bloch spheres for qubits $i = 1, 2, 3, 4$. The black arrows along Z are the depolarised density matrix, ρ_i , inferred if standard RB is used. It can be seen that ρ_i for all 4 qubits are identical, suggesting each has the same magnitude single-qubit error per Clifford. However, if we compare qubits pairwise, we can see there is a symmetry between qubits dominated by correlated noise (highlighted by dashed line pairing sequences), while the uncorrelated case has none. This extra symmetry between qubit pairs allows us to break down the total error per Clifford into spatially correlated and uncorrelated components. These components are quantified by fitting the correlated populations, which are in general multiple exponential decays (if the Liouville form is diagonalisable). This is similar to other non-standard RB techniques [sutherland_subspace_2025, helsen_new_2019], where decay rates relate to populations trapped or slowly leaking from invariant subspaces.

The analytic model considered here describes correlated random-unitary single-qubit noise [sarovar_detecting_2020], here termed *common noise*: each RB step applies a tensor product of single-qubit rotations whose random variables may be shared between qubits. This class captures correlated but non-entangling control and environmental fluctuations, which are particularly relevant to trapped-ion and neutral atom qubits, exposed to common fluctuations of qubit drive phase, amplitude, and detuning; magnetic-field noise and gradients; beam pointing and intensity drift; AC Stark shift fluctuations; motional-state-dependent over-rotations; and gate duration calibration errors. It does not describe entangling coherent errors, correlated non-unital processes, leakage, or general non-Markovian noise.

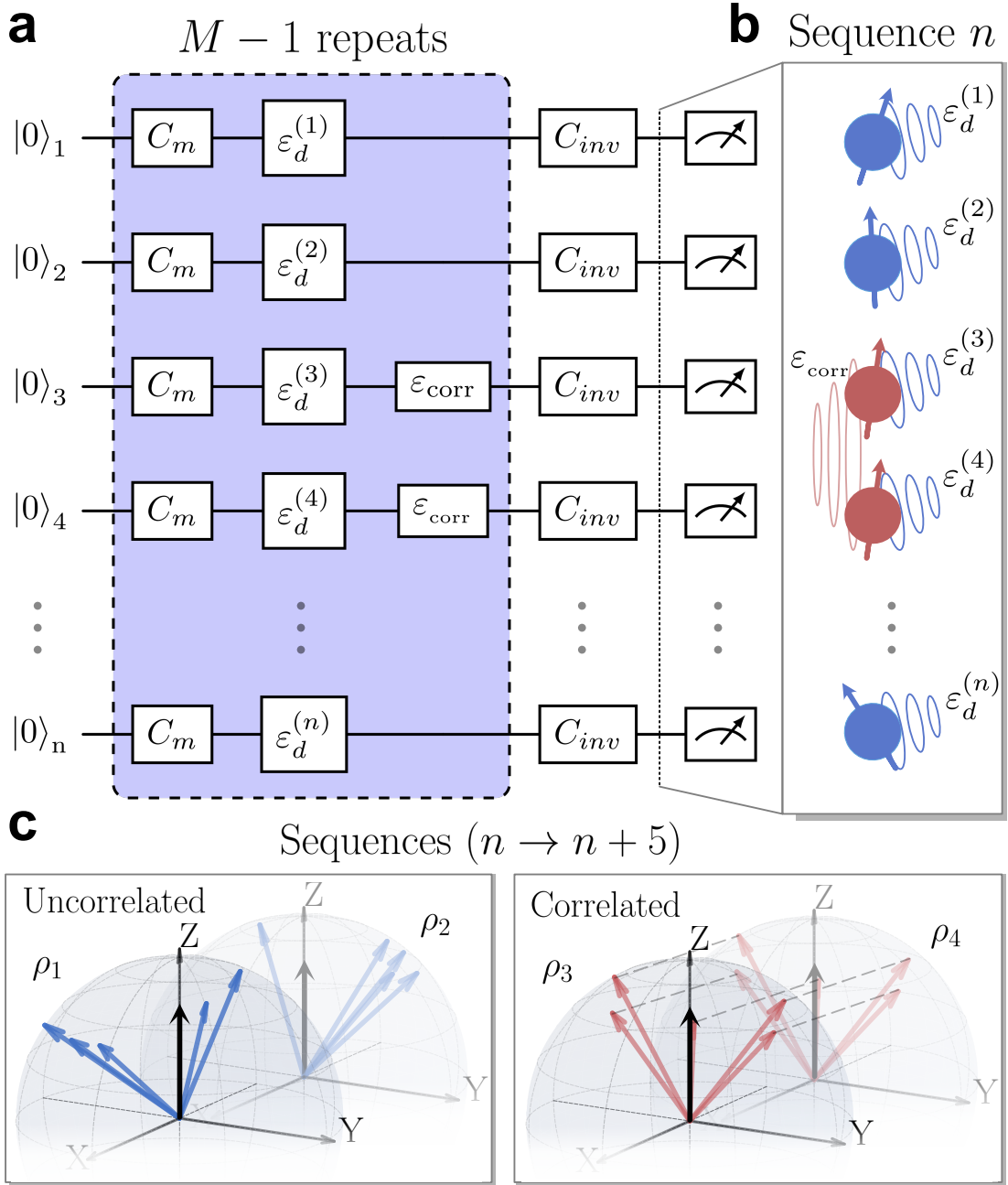


Figure 5.1: The k -copy protocol simply extends single qubit Clifford RB sequences to parallel, identical sequences on k -qubits. **(a)** Circuit diagram for a single RB sequence of length M . Circuit consists of $M - 1$ Cliffords sampled randomly from the single qubit Clifford group applied in parallel to all k qubits. After each Clifford there is some error process, here shown as independent depolarising errors, $\epsilon_d^{(i)}$, for each qubit labelled i , and a common rotation error, ϵ_{corr} , between qubits $i = 3$ and $i = 4$. A final inverting Clifford, C_{inv} , returns the single qubit system (ignoring errors) to the original state $|0\rangle$. **(b)** A schematic of the final state of each qubit before measurement for random sequence n . **(c)** Ensembles of final states after 5 random sequences of length M , for qubits $i = 1, 2$ (blue arrows uncorrelated), and qubits $i = 3, 4$ (red arrows correlated). Black arrow along Z is the resulting depolarised density matrix, ρ_i , inferred from standard single-qubit RB.

A discussion on entangling coherent errors, is given in the supplemental. With these assumptions, the probability of both qubits successfully returning to the initial state is given by,

$$\mathcal{P}_{SS} = \frac{1}{4} + \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \quad (5.1)$$

where m is the Clifford sequence length, and α_i are the scalar decay rates for each sector corresponding to the sub-representations of the irreducible representations (irreps) of the 2-copy single qubit Clifford group [**helsen__representations__2018**]. α_L and α_R describe the decays of the local subspaces for the left and right qubit respectively, α_{dt} describes the decay for the “diagonal traceless” 2-dimensional irrep, and α_{tr} is the decay rate of the 1-dimensional “symmetric trace” subspace. Similar relations for $\mathcal{P}_{SF}$, $\mathcal{P}_{FS}$, and $\mathcal{P}_{FF}$ are used for fitting experimental data, and given in the supplemental. These scalar decay rates are linked to experimentally relevant parameters by considering a collection of specific error channels. We consider independent depolarisation channels, parametrised by $\varepsilon_L, \varepsilon_R$, and, the physically motivated error model of *common noise*. This noise channel is given by,

$$\rho \mapsto \mathcal{E}_{U \otimes U}^A(\rho) = \sum_a p_a (U_a \otimes U_a) \rho (U_a \otimes U_a)^\dagger, \quad (5.2)$$

where the channel is an ensemble over distribution A of unitaries applied with probability p_a to the initial state, with $U_a(\theta_a, \hat{n}_a) = e^{-i(\theta_a/2)\hat{n}_a \cdot \hat{\sigma}}$, with rotation angle, θ_a , axis, $\hat{n}_a$, and Pauli vector, $\hat{\sigma}$. To first order the decays of each sector are then,

$$\alpha_L = 1 - \varepsilon_L - \varepsilon_{\text{corr}}, \quad \alpha_R = 1 - \varepsilon_R - \varepsilon_{\text{corr}}, \quad (5.3)$$

$$\alpha_{tr} = 1 - \varepsilon_L - \varepsilon_R, \quad \alpha_{dt} = 1 - \varepsilon_L - \varepsilon_R - 3\varepsilon_{\text{corr}}, \quad (5.4)$$

where the correlated error per Clifford, $\varepsilon_{\text{corr}}$, quantifies *common noise* with,

$$\varepsilon_{\text{corr}} = \frac{1}{3} \sum_a p_a \theta_a^2. \quad (5.5)$$

The trace subspace is seen to be invariant under *common noise* leading to the asymptote $\mathcal{P}_{SS} \rightarrow 1/3$ for $\varepsilon_L = \varepsilon_R = 0$, $\varepsilon_{\text{corr}} > 0$. If there is some finite local error rates for each qubit, then the asymptote corresponds to $\mathcal{P}_{SS} \rightarrow 1/4$. In the

realistic case that both errors are present, and correlated errors dominate over independent errors, a clear intermediate asymptote corresponding to the symmetric trace subspace can be seen in the correlated population decay curves, before a slow decay into the fully depolarised two-qubit state. The two-qubit mixed state, upon reaching the symmetric trace asymptote, is given by,

$$\rho_{sym} = \frac{1}{3}(|00\rangle\langle 00| + |11\rangle\langle 11| + |\Psi^+\rangle\langle \Psi^+|), \quad (5.6)$$

where $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ is a Bell state, and displays induced coherence between the two qubits. This coherence may be measured through parity oscillations of the two-qubit state after “k-copy” twirling. We note that the parity contrast as a function of Clifford sequence length has dependency on sector decay terms,

$$C = \frac{1}{3}(\alpha_{tr}^m - \alpha_{dt}^m), \quad (5.7)$$

which may be conveniently combined with correlated population measurements to isolate individual sector decay rates via fitting single exponential decays. A short discussion of which can be found in the supplemental.

Experimental results—Experimental verifications of the “k-copy” RB protocol were performed using trapped $^{40}\text{Ca}^+$ ions. Optical qubit states were chosen, with $|0\rangle \equiv |4S_{1/2}, m_j = -\frac{1}{2}\rangle$, and $|1\rangle \equiv |3D_{5/2}, m_j = -\frac{5}{2}\rangle$. State preparation and single qubit rotations were performed using a 729-nm Ti-sapphire laser. Up to 8 ions were confined in a single harmonic potential with a mean ion spacing of 5 μm . Both a global addressed elliptical 729-nm beam, as well as single addressed beams were used. The 24 single qubit Clifford gates were decomposed into single qubit rotations of the form $\hat{R}_x(\pi/2)$, and $\hat{R}_z(\pi/2)$. Rotations around $\hat{\sigma}_x$ were performed natively using the 729-nm light on resonance with the optical qubit transition. The relative phase between rotations was set by modulating the 729-nm beam using an Acousto-optic modulator (AOM). Rotations around $\hat{\sigma}_z$ were performed virtually by updating the phase of subsequent $\hat{\sigma}_x$ rotations.

Figure 5.2 (a) shows two-qubit correlated populations as a function of Clifford sequence length when *common noise*, is the dominant error source. Noise was injected

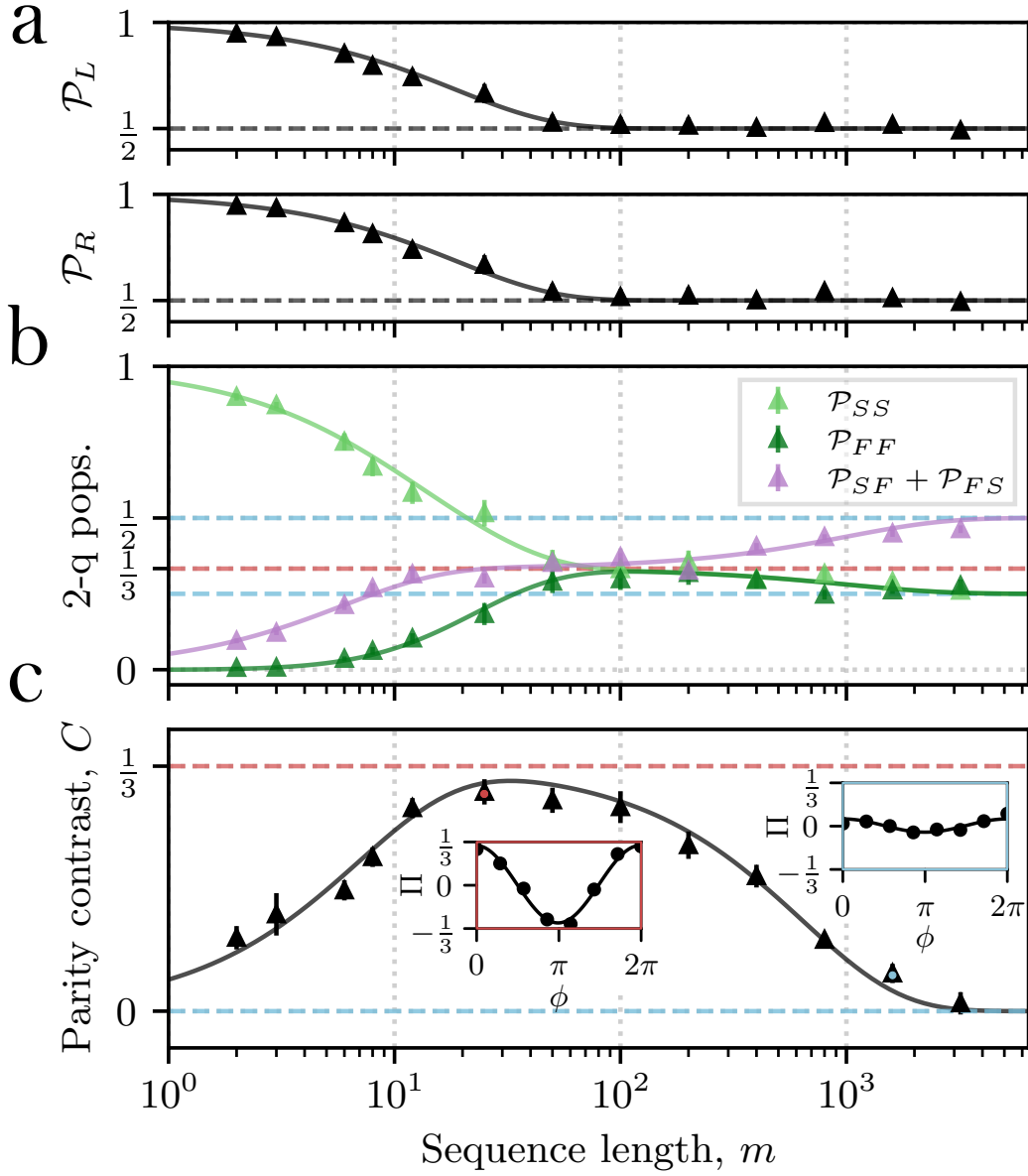


Figure 5.2: Correlated two-qubit probabilities after “ k -copy” RB discern correlated and uncorrelated error channels. The single-qubit subspaces, familiar from standard Clifford RB, measure a total error rate only, shown in (a) as a decay of success as a function of sequence length m . By observing correlated success and failures for the two qubits, as shown in (b), noise dependent decay structure emerges. Furthermore, the symmetric state occupation, given in equation 5.6, can be directly probed through measuring residual coherence between the two qubits. The parity contrast as a function of m is shown in (c). All experimental data (points) correspond to an average over 200 random Clifford sequences and 50 repeats of each sequence. Error bars denote the standard error. The solid lines are single exponential decays in (a-b), and multi-exponentials for c and d, given by equations 5.1), and 5.7 respectively. Dashed lines give the expected asymptotes. The blue (red) horizontal lines are due to depolarising (correlated) noise.

through a fixed 10 kHz frequency detuning of the 729-nm laser. The populations exhibit a clear intermediate asymptote at $m \approx 100$ Cliffords, corresponding to the initial state $|00\rangle$ converging to ρ_{sym} . The final asymptote is seen for $m \approx 1000$, corresponding to the fully depolarised two-qubit state. Extracted parameters of, $\varepsilon_{corr} = 0.051(3)$, $\varepsilon_L = 0.001(1)$, and $\varepsilon_R = 0.000(1)$ were found.

Population decays alone indicate an intermediate asymptote; parity contrasts verify that this asymptote corresponds specifically to occupation of the symmetric trace sector. A partial tomography method is used to quantify overlap of the final state with the Bell state, $|\Psi^+\rangle$. Differential rotations, of the form $\hat{R}_{+\phi}(\pi/2) \otimes \hat{R}_{-\phi}(\pi/2)$, are applied to the two qubits. The rotation axis, $\hat{\sigma}_\phi$, lies in the XY -plane of the Bloch sphere. ϕ is varied and the contrast of the resulting trace is found, examples of which are shown inset in figure 5.2 (c). The expected asymptote $C \rightarrow 1/3$ contrast, given by equation 5.7, is seen in (c) between $m \in [10, 100]$ corresponding to the two-qubit state approaching ρ_{sym} . The parity contrast data is fit (solid line) according to equation 5.7 to extract error per Cliffords of, $\varepsilon_{corr} = 0.043(3)$, $\varepsilon_L = \varepsilon_R = 0.0016(1)$. The increased depolarising rate is attributed to imperfect $R_{\pm\phi}(\pi/2)$ execution on the single addressing system.

Distinguishing correlated and uncorrelated noise is useful for optimising global gate parameters for multi-qubit systems such as gate durations and shared control field detuning. Here we demonstrate the optimisation of laser frequency in figure 5.3. The correlated and local errors per Clifford are measured as a function of laser detuning away from the qubit resonance frequency. Each point corresponds to a full “k-copy” RB experiment consisting of 11 sequence lengths, log sampled up to $m = 1024$ Cliffords, examples of which are shown in the upper panels. Each length consists of $r = 50$ randomised sequences and $s = 50$ shots. The correlated population decay curves were fit to the model given in equation 5.1, and the parameters $\{\varepsilon\}$, were extracted and plotted against the known laser offset detuning, shown in the bottom panel. The correlated error increases with detuning offset, while the single qubit depolarisation rates remain constant, confirming the expectation that detuning of a common field, and twirling over the Clifford group, leads to a *common noise*

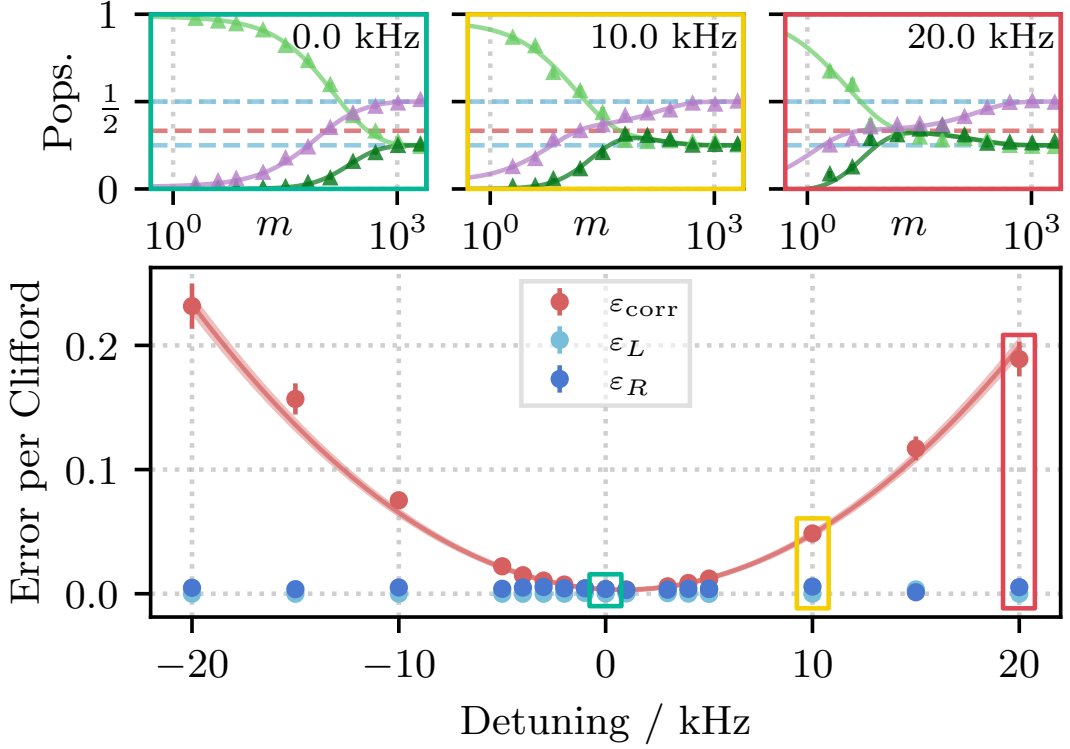


Figure 5.3: Correlated errors are injected via detuning of a shared drive laser. Figures (a)-(c) show example correlated populations, similar to figure 5.2, but for various laser detunings. Dashed lines are fits to equations 5.1 and 5.3 to extract spatially correlated and single-qubit depolarising error per Cliffords. These extracted errors are shown as a function of qubit laser detuning in (d), with red points indicated correlated errors, and blue points indicating the depolarising errors on each qubit. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations. The dashed red line is fit according to equation ??, while the single-qubit depolarising errors remain constant within errors. The red shaded region signifies the 68% confidence, generated through covariance-based parameter sampling.

channel. The detuning offset is parameterised as $\delta = \delta_L - \delta_0$, where δ_L is the laser detuning from the bare qubit frequency, and δ_0 is the frequency offset when the ion is dressed by the laser induced Stark shift. The correlated error per Clifford and the laser offset detuning are related by,

$$\varepsilon_{corr} = \varepsilon_0 + k \left(\frac{\delta_L - \delta_0}{\Omega} \right)^2, \quad (5.8)$$

where ε_0 is the residual correlated error not originating from a detuning offset, Ω is the Rabi frequency, and k is a scalar which depends on the decomposition of the Clifford gates into hardware native gates. Fitting the equation 5.8, with Rabi frequency set to $\Omega = 2\pi \times 100(1)$ kHz (measured independently), yields the solid red

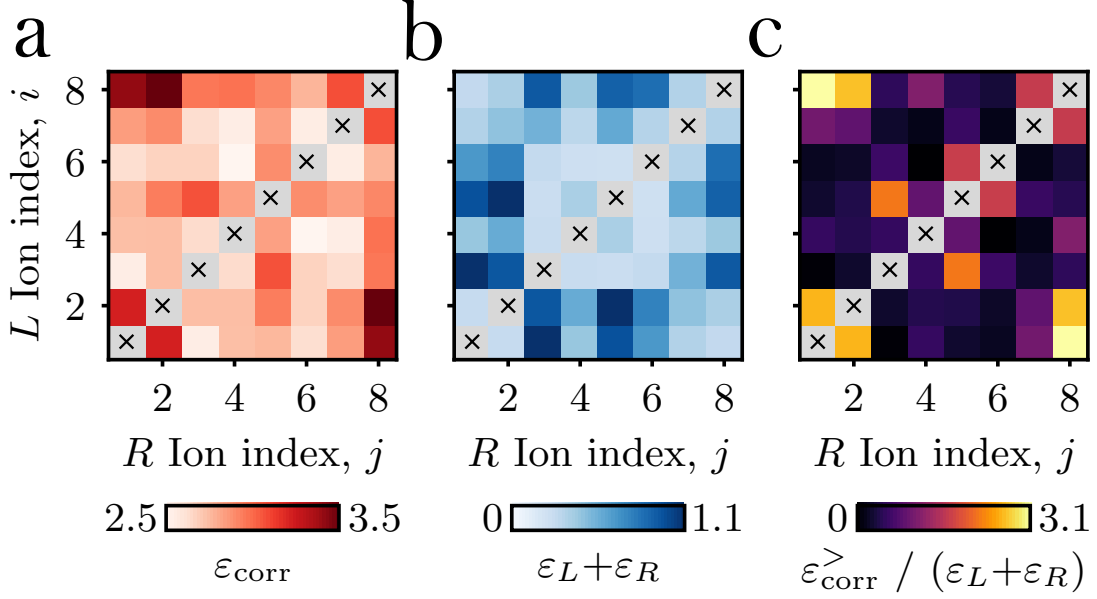


Figure 5.4: Correlated and local error per Cliffords measured for all qubit pairs after applying “k-copy” RB to an 8 ion chain. Noise is dominated by position dependent duration of $R_x(\pi/2)$ gates, due to a finite width global addressing beam. TODO Independently measured ideal gate durations are shown in XXX. Panel (a) displays the extracted correlated errors, $\varepsilon_{\text{corr}}^{(i,j)}$ for each qubit pair, (i, j) . Panel (b) displays the sum of both independent depolarising errors, $(\varepsilon_L + \varepsilon_R)^{(i,j)}$. Panel (c) displays the ratio of excess correlated noise, $(\varepsilon_{\text{corr}}^{>0})^{(i,j)} = \varepsilon_{\text{corr}}^{(i,j)} - \varepsilon_{\text{corr}}^{(0)}$ to the sum of local errors given in (b). This plot highlights the outer qubits are dominated by spatially correlated errors.

line, and fitted parameters $\varepsilon_0 = 3.2(1) \times 10^{-3}$, $k = 15.9(7)$, and $\delta_0 = 2\pi \times 0.80(5)$ kHz. Here, we have experimentally shown the extraction of a spatially correlated noise component from a total error per Clifford when either *common noise* or, depolarising noise dominates. A wider parameter space of correlated and uncorrelated noise ratios is considered both numerically and through other noise injection techniques in the supplemental material.

“K-copy” RB may be simply scaled to > 2 qubits by fitting all pairwise correlated populations to 5.1. This provides a practical and scalable diagnostic that captures pairwise spatial correlations in a large qubit register; however, it does not uniquely determine higher-order (≥ 3 -body) correlated error terms. A discussion on ≥ 3 -body modelling of the correlated population decays is given in TODO: need to add this here, as important ???. The computational overhead of scaling “k-copy” pairwise RB to k qubits is limited exclusively to the classical post-processing of the measured

shot outcomes, scaling as $O(k^2)$. The technical overhead of applying synchronized identical sequences to the qubit register, should be a trivial application for a general quantum processor.

We demonstrate this practical scaling of pairwise “k-copy” RB to resolve spatial noise correlations on an 8 ion chain, with correlation matrices shown in figure 5.4. An elliptical addressing beam is used to execute parallel single qubit rotations on all 8 qubits. The beam has major axis parallel to the ion chain axis with waist radius of $52(3) \mu\text{m}$ [gate_duration]. An axial confinement of 0.5 MHz was used, resulting in a chain extent of $35 \mu\text{m}$ [james_quantum_1998]. Parallel ion addressing without composite pulses means a single calibrated gate duration must be chosen. The optimal gate duration was calibrated using central qubit $i = 4$ through repeated $\pi/2$ pulses. A Rabi frequency error of 11(1)% was expected on the outer qubits due to the ion spacing and beam waist. The pairwise experimental data displays this expected structure, with large correlated errors between both neighbouring and distant outer qubits. For clarity, consider qubits ($i = 1, j = 8$), which although distant, have spatially correlated errors due to suffering from similar under rotation error magnitudes. To ensure sample efficient measurement of the correlated and uncorrelated errors, the requirement $\varepsilon_{\text{corr}} > (\varepsilon_L + \varepsilon_R)$ is preferable. This was ensured here by introducing random phase offsets of the laser field for each implemented $R_x(\pi/2)$. The single qubit gate, g , is thus implemented as $R_{\phi_g}(\pi/2)$, where $\hat{\sigma}_{\phi_g} = \cos(\phi_g)\hat{\sigma}_x + \sin(\phi_g)\hat{\sigma}_y$ where ϕ_g is drawn from a normal distribution with standard deviation, s_ϕ . Here, $s_\phi = 0.19$ rad, resulting in a correlated error offset of $\varepsilon_{\text{corr}}^{(0)} = 0.025$.

Conclusions—“k-copy” RB efficiently discerns correlated and uncorrelated error channels acting on a multi-qubit system. Furthermore, due to the symmetry enforced through twirling over the diagonal Clifford representation, $Cl^{\otimes k}$, structured errors can lead to the decay of correlated populations into invariant subspaces. Given an error model, these correlated decays can be used to directly quantify the magnitude of the individual error channels.

We have proposed, and experimentally demonstrated this method and extended

its application to an 8 qubit register through pairwise comparisons. This approach allows for highly efficient parallel data collection on the quantum register, where the time on device is independent of qubit number, and requires exclusively local operations followed by classical post-processing of correlated survival rates.

TODO: Add time dependent discussion points

5.9 Supplemental

5.9.1 *k-copy Clifford analysis*

We begin this section by considering the representations of the single-qubit Clifford group acting diagonally on two copies, $Cl^{\otimes 2}$ (see ref. [helsen__representations__2018] for in detail analysis). The irreducible representations give a natural (Schur) basis to consider how possible error channels are simplified upon the “*k-copy*” RB protocol. We finish by deriving the correlated populations given in equation 5.1, and observed experimentally when the system is subject to common random-unitary single-qubit noise.

Representations of the 2-copy Clifford group

Twirling over group $\mathcal{G}$ enforces symmetry on a general error channel, Λ , given in Liouville form,

$$\Lambda \mapsto \tilde{\Lambda} = \frac{1}{|\mathcal{G}|} \sum_{\mathcal{G}} G \Lambda G^\dagger, \quad (5.9)$$

where G are the vectorised representations of group elements of $\mathcal{G}$. The twirled channel now satisfies the commutative relation,

$$G \tilde{\Lambda} G^\dagger = \tilde{\Lambda} \quad \forall G \in \mathcal{G}. \quad (5.10)$$

The representation of the space, $\mathcal{H}$ decomposes to irreps λ as,

$$\mathcal{H} = \bigoplus_{\lambda} (V_{\lambda} \otimes \mathcal{C}^{m_{\lambda}}), \quad (5.11)$$

where V_{λ} is the irreducible representation space, and $\mathcal{C}^{m_{\lambda}}$ is the multiplicity space with dimension of the multiplicity m_{λ} . From Shur’s lemma, equation 5.10 implies the commuting twirled error channel has the form,

$$\tilde{\Lambda} = \bigoplus_{\lambda} (I_{\lambda} \otimes \mathcal{A}_{\lambda}), \quad (5.12)$$

where $\mathcal{A}_{\lambda}$ is the multiplicity matrix of dimensions $m_{\lambda} \times m_{\lambda}$.

The 2-copy single-qubit Clifford group, analysed in detail in references [helsen__representations__2018, wallman__randomized__2014], decomposes the general 16-dimensional two qubit

Sub-representation	irrep	Basis operators
Identity	1	II
Left qubit	3	XI, YI, ZI
Right qubit	3	IX, IY, IZ
Trace	1	$(XX + YY + ZZ)/\sqrt{3}$
Diagonal traceless	2	$(XX + YY - 2ZZ)/\sqrt{6},$ $(XX - YY)/\sqrt{2}$
Symmetric off-diagonal	3'	$(XY + YX)/\sqrt{2},$ $(XZ + ZX)/\sqrt{2},$ $(YZ + ZY)/\sqrt{2}$
Antisymmetric	3	$(XY - YX)/\sqrt{2},$ $(XZ - ZX)/\sqrt{2},$ $(YZ - ZY)/\sqrt{2}$

Table 5.1: Sub-representations of the $Cl^{\otimes 2}$ group. The irrep, dimension of each sub-representation, and a set of operators in the Pauli basis which span the subspace dimension is given. The irrep basis vectors are from equation 56 in reference [wallman_randomized_2014].

irrep	e	(12)	(12)(34)	(123)	(1234)
Identity/Trace	1	1	1	1	1
Left/Right/Antisymmetric	3	0	-1	1	1
Diagonal traceless	2	-1	2	0	0
Symmetric off-diagonal	3	0	-1	-1	-1

Table 5.2: Character table for the Clifford group given the irreps in table 5.1, and with conjugacy classes: identity, transposition, double transposition, 3-cycle, and 4-cycle.

space into 4 irreducible representations, irreps, with dimensions $\{1, 2, 3, 3'\}$, (the notation $3'$ is used to contrast 3), and multiplicities $\{2, 1, 3, 1\}$. The sub-representations are summarised in table 5.1, and the character table is given in 5.2. Given the structure of a general twirled map, enforced by equation 5.12, and further constraining the map to be trace preserving, we find that in general 13 independent scalars are needed to describe $\tilde{\Lambda}$. The diagonal terms, here labelled $\vec{\alpha} = \{\alpha_L, \alpha_R, \alpha_{tr}, \alpha_{dt}, \alpha_{so}, \alpha_{as}\}$, describe a decay rate contribution from each sub-representation, whilst off-diagonal terms, $\vec{m} = \{m_{id \rightarrow tr}, m_{L \rightarrow R}, m_{R \rightarrow L}, m_{L \rightarrow as}, m_{R \rightarrow as}, m_{as \rightarrow L}, m_{as \rightarrow R}, \}$ describe coupling between the multiplicity spaces.

To further reduce the number of parameters describing the error map, it can be instructive to consider a collection of possible error channels, parametrised canonically. This links sector decay rates and couplings to either physically

Coupling	irrep	Mechanism
$m_{id \rightarrow tr}$	1	Decays (non-unital)
$m_{L \rightarrow R},$ $m_{R \rightarrow L}$	3	Qubit Swap, Correlated decays (non-unital)
$m_{L \rightarrow as},$ $m_{R \rightarrow as},$ $m_{as \rightarrow L},$ $m_{as \rightarrow R}$	3	Entanglement

Table 5.3: Possible mechanisms that result in the coupling of sub-representation sectors of the $Cl^{\otimes 2}$ twirled error channel.

relevant error mechanisms, or to highlight possible errors that are detrimental to the performance of a quantum processor on completing a certain task, e.g. executing a code which can correct only certain error channels. Possible mechanisms that can couple the subspaces corresponding to multiplicities of the same irrep are summarised in table 5.3.

Error models

It can be seen that the error channels that lead to couplings neatly discern the physical origin of the noise channel. For example, in superconducting qubit platforms, always on ZZ -couplings may be of experimental interest, and so the couplings between single qubit subspaces and the antisymmetric subspace can be quantified; in the neutral atom platform, correlated non-unital errors may be observed if the qubit spacing is of order the qubit transition wavelength [rivas_quantifying_2015]; however, in ion trap platforms, environmental noise is often dominated by magnetic and electric field fluctuations, which may act globally on all qubits if they are stored in the same trapping potential. These field fluctuations induce both common and independent single-qubit rotations, but cannot lead to any couplings $\{m\}$. The structure of two random single-qubit rotation noise channel generally acts independently on each sub-representation, and so is parametrised by at most the 6 scalars, $\{\alpha\}$.

As such, the Liouville form in the Schur basis (ordered by table 5.1) of the general

error map without multiplicity coupling is given by diagonal matrix,

$$\tilde{\Lambda} = \text{diag}^{\mathcal{S}} \left(I_1, \alpha_L I_3, \alpha_R I_3, \alpha_{tr} I_1, \alpha_{dt} I_2, \alpha_{so} I_3, \alpha_{as} I_3 \right), \quad (5.13)$$

where $\text{diag}^{\mathcal{S}}(\cdot)$ refers to a block diagonal transfer matrix given in the Schur, $\mathcal{S}$, basis. Let us consider the effect of noise due to common single-qubit rotations acting on two-qubits, with error channels of a general form,

$$\rho \mapsto \mathcal{E}_{U \otimes U}^{(M)}(\rho) = \sum_m p_m (U_m \otimes U_m) \rho (U_m \otimes U_m)^\dagger, \quad (5.14)$$

where the sum is over a distribution, M , of single qubit unitaries each applied with probability p_m , with $\sum_m p_m = 1$ to satisfy trace preservation. In general, this error channel cannot be described by the separable map,

$$\mathcal{E}(\rho_1 \otimes \rho_2) = \mathcal{E}_1(\rho_1) \otimes \mathcal{E}_2(\rho_2), \quad (5.15)$$

where ρ_i is the density matrix of qubit i , as such, the channel generally induces spatial correlations.

The single qubit rotation is parametrised by rotation angle, θ , and the axis of rotation $\vec{n}$.

$$U(\theta, \hat{n}) = e^{-i\frac{\theta}{2}\hat{n}\cdot\hat{\sigma}}. \quad (5.16)$$

We now consider the channel twirled over a discrete group $\mathcal{G}$ with elements G , with twirling defined by equation 5.9. Considering the form of equation 5.2, the twirled channel may be written as,

$$\tilde{\mathcal{E}} = \sum_m p_m \tilde{\mathcal{E}}_m, \quad (5.17)$$

where

$$\tilde{\mathcal{E}}_m(\rho) = \frac{1}{|\mathcal{G}|} \sum_{\mathcal{G}} G(U \otimes U) \rho (U \otimes U)^\dagger G^\dagger, \quad (5.18)$$

is the channel for diagonally applied coherent unitary rotations.

The goal is now to find the scalar parameters $\{\alpha\}$, which describe this channel. It can be seen from the basis vectors, that only the “Trace” sector is preserved under global single-qubit rotations, $\alpha_{tr} = 1$. It is in fact this preservation that leads to the decay into the symmetric state given in the main text equation 5.6.

The other scalar parameters can be calculated “mechanically” by constructing the Pauli-transfer matrix (PTM) representation and twirling over $Cl^{\otimes 2}$, however, we note that the implicit symmetry of each sub-representation can be used to define tensor contractions acting on the $SO(3)^{\otimes 2}$ space spanned by two rotations,

$$\begin{aligned}
s_{tr} &= \sum_{i,j,k,l} \delta_{ij} \delta_{kl} A_{ki} B_{ij} = \text{Tr}(A^T B), \\
s_{id} &= \sum_{i,j,k,l} \delta_{ik} \delta_{jl} A_{ki} B_{ij} = \text{Tr}(A) \text{Tr}(B), \\
s_{swap} &= \sum_{i,j,k,l} \delta_{il} \delta_{jk} A_{ki} B_{ij} = \text{Tr}(AB), \\
s_{diag} &= \sum_{i,j,k,l} (\delta_{ik} \delta_{jl}) \delta_{ij} A_{ki} B_{ij} = \sum_i (A_{ii} B_{ii}),
\end{aligned} \tag{5.19}$$

where A and B are the $SO(3)$ representation of independent unitaries acting on the left and right qubits respectively. The scalar parameters $\{\alpha\}$ are then found by,

$$\begin{aligned}
\alpha_{tr} &= \frac{1}{3} s_{tr}, \\
\alpha_{dt} &= \frac{1}{2} s_{diag} - \frac{1}{6} s_{tr}, \\
\alpha_{so} &= \frac{1}{6} (s_{id} + s_{swap}) - \frac{1}{3} s_{diag}, \\
\alpha_{as} &= \frac{1}{6} (s_{id} - s_{swap}).
\end{aligned} \tag{5.20}$$

In the case of a global rotation, $A = B$, we find,

$$\begin{aligned}
\alpha_L &= \alpha_R = \alpha_{as} = \frac{1 + 2 \cos \theta}{3}, \\
\alpha_{tr} &= 1, \\
\alpha_{dt} &= \frac{1}{2} \left(1 + \cos(2\theta) + 8n_{an} \sin(\theta/2)^4 \right), \\
\alpha_{so} &= \frac{1}{3} \left(2 \cos(\theta) + \cos(2\theta) - 8n_{an} \sin(\theta/2)^4 \right),
\end{aligned} \tag{5.21}$$

with an anisotropic term, $n_{an} = n_y^4 + n_z^4 + n_y^2 n_z^2 - n_y^2 - n_z^2$. Taking these scalar products to second order in error angle θ , we find that the anisotropic term may be ignored for small errors,

$$\begin{aligned}
\alpha_L &= \alpha_R = \alpha_{as} = 1 - \theta^2/3, \\
\alpha_{tr} &= 1, \\
\alpha_{dt} &= \alpha_{so} = 1 - \theta^2.
\end{aligned} \tag{5.22}$$

As such, the dynamics are entirely described by the error angle magnitude. The ensemble of twirled common unitaries across some arbitrary distribution, M , given in equation 5.2, thus combine to give scalar parameters describing the contribution from each subspace to the observed decay,

$$\alpha_i^{(M)} = \sum_m p_m \alpha_i^{(m)}. \quad (5.23)$$

We further define the correlated error per Clifford,

$$\varepsilon_{corr} = \frac{1}{3} \sum_m p_m \theta_m^2, \quad (5.24)$$

to conform with other RB work. Comparing to equation 5.13, the general ensemble of common single-qubit rotations channel, after twirling over $Cl^{\otimes 2}$ is given by the diagonal matrix in the Schur basis,

$$\tilde{\Lambda}_{U \otimes U} = \text{diag}^S \left(I_1, (1 - \varepsilon_{corr}) I_6, I_1, (1 - 3\varepsilon_{corr}) I_2, (1 - \varepsilon_{corr}) I_6 \right). \quad (5.25)$$

Ultimately, environmental noise will not perfectly couple symmetrically to both qubits, and so independent local single-qubit error channels must also be considered. Assuming an independent noise channel acting on each qubit, Clifford twirling leads to a fully depolarising noise channel of the form,

$$\rho^{(1q)} \mapsto \mathcal{E}_{dp,a}(\rho^{(1q)}) = (1 - \varepsilon_a) \rho^{(1q)} + \varepsilon_a \frac{I}{2} \text{Tr}(\rho^{(1q)}), \quad (5.26)$$

where the superscript denotes that this is for a single qubit subspace, and the subscript a signifies which qubit we are considering. For the "left" and "right" qubits the decay rates are parametrised by ε_L and ε_R , respectively. Comparing to equation 5.13, the independent depolarisation channel may be expressed in Liouville form as the diagonal matrix in the Pauli basis,

$$\Lambda_{dp} = \text{diag}^{\mathcal{P}} \left(1, (1 - \varepsilon_L) I_3, (1 - \varepsilon_R) I_3, (1 - \varepsilon_L)(1 - \varepsilon_R) I_9 \right), \quad (5.27)$$

where $\text{diag}^{\mathcal{P}}(\cdot)$ represents the block diagonal matrix in the ordered two-qubit Pauli basis $\mathcal{P} = \{I \otimes I, \vec{\sigma} \otimes I, I \otimes \vec{\sigma}, \vec{\sigma} \otimes \vec{\sigma}\}$, with $\vec{\sigma} = \{X, Y, Z\}$. Combining the depolarisation channels with the twirled common single-qubit rotation channel,

$$\Lambda_T(\varepsilon_L, \varepsilon_R, \varepsilon_{corr}) = \Lambda_{dp}(\varepsilon_L, \varepsilon_R) \tilde{\Lambda}_{corr}(\varepsilon_{corr}), \quad (5.28)$$

we find scalar parameters,

$$\begin{aligned}
\alpha_L &= 1 - \varepsilon_L - \varepsilon_{corr}, \\
\alpha_R &= 1 - \varepsilon_R - \varepsilon_{corr}, \\
\alpha_{tr} &= 1 - \varepsilon_L - \varepsilon_R, \\
\alpha_{dt} &= \alpha_{so} = 1 - \varepsilon_L - \varepsilon_R - 3\varepsilon_{corr}, \\
\alpha_{as} &= 1 - \varepsilon_L - \varepsilon_R - \varepsilon_{corr},
\end{aligned} \tag{5.29}$$

which fully describe the error channels we consider in the main text.

During the proposed RB experiment, the initial state is always prepared to be $|0, 0\rangle$, and the final correlated populations correspond to projecting onto the states $\{|0, 0\rangle, |0, 1\rangle, |1, 0\rangle, |1, 1\rangle\}$. In the main text, we consider “success” and “failure” rather than “0”, and “1”, as we allow for target state randomisation, however this detail is excluded here, and we assume that the noise-less RB sequence results in the identity operation. The correlated populations may be given analytically using the derived Liouville operators, and by the vectorized initial and final states,

$$\mathcal{P}_{L,R} = \langle\langle L, R | \Lambda_T^m | 00 \rangle\rangle, \tag{5.30}$$

where L, R denote the final projected computational state for the “left” and “right” qubit (either $|0\rangle$ or $|1\rangle$) respectively, and m is the number of Cliffords in the applied sequence. Using this equation, and the explicit transfer matrices, the final populations decays are found,

$$\begin{aligned}
\mathcal{P}_{0,0} &= \frac{1}{4} + \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m, \\
\mathcal{P}_{0,1} &= \frac{1}{4} + \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \\
\mathcal{P}_{1,0} &= \frac{1}{4} - \frac{1}{4}(\alpha_L^m - \alpha_R^m) - \frac{1}{6}\alpha_{dt}^m - \frac{1}{12}\alpha_{tr}^m, \\
\mathcal{P}_{1,1} &= \frac{1}{4} - \frac{1}{4}(\alpha_L^m + \alpha_R^m) + \frac{1}{6}\alpha_{dt}^m + \frac{1}{12}\alpha_{tr}^m.
\end{aligned} \tag{5.31}$$

Considering the final state of a single qubit after “k-copy” RB, we find the expected single exponential decay,

$$\mathcal{P}_0^{(L)} = \mathcal{P}_{0,0} + \mathcal{P}_{0,1} = \frac{1}{2}(1 + (1 - \varepsilon_L - \varepsilon_{corr})^m), \tag{5.32}$$

where the typical “error-per-Clifford” parameter, ε_c , in standard Clifford RB is here separated into a correlated and independent error component,

$$\varepsilon_c^{(L)} = \varepsilon_L + \varepsilon_{corr}, \quad (5.33)$$

which through fitting the correlated populations may be quantified.

Standard Clifford RB also reveals a state-preparation and measurement, “SPAM”, error as an initial scaling factor of the exponential decay. The effects of SPAM are discussed in section 5.9.5.

5.9.2 Invariant subspaces

In the main text, we considered the initial prepared state $|0, 0\rangle$, however, if we extend to the preparation of arbitrary initial states, we can isolate subsets of the scalar parameters $\{\alpha\}$. This can be especially useful to simplify the decays curves from multiple exponentials to instead a single exponential decay. This can dramatically reduce the number of samples required for accurate fitting.

For example, consider the singlet state,

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \quad (5.34)$$

with density matrix in the Pauli basis,

$$\rho_s = \frac{1}{4}(II - XX - YY - ZZ). \quad (5.35)$$

By a change of basis to the irrep vectors given in table 5.1, it is clear to see that ρ_s has initial overlap with only the identity and “trace” sub-representations. Therefore, under the noise model considered in section 5.9.1, we can immediately see that correlated populations decay with a single exponential,

$$\begin{aligned} \mathcal{P}_{0,0} = \mathcal{P}_{1,1} &= \frac{1}{4} - \frac{1}{4}(1 - \varepsilon_L - \varepsilon_R)^m, \\ \mathcal{P}_{0,1} = \mathcal{P}_{1,0} &= \frac{1}{4} + \frac{1}{4}(1 - \varepsilon_L - \varepsilon_R)^m, \end{aligned} \quad (5.36)$$

which corresponds to the scalar parameter α_{tr} .

5.9.3 Parity analysis

Figure 5.2 (b) in the main text shows the contrast of parity measurements after a differential $\frac{\pi}{2}$ -pulse is applied to the two qubits. This measurement directly probes coherences $|01\rangle\langle 10|$ and $|10\rangle\langle 01|$. The analytic model of the parity contrast (black dashed line in figure), can be simply calculated by extending equation 5.30, to include a final differential rotation, which we refer to as the parity pulse, in transfer matrix form, Λ_{Π} ,

$$\begin{aligned}\Pi(\phi) &= \langle\langle\pi|\Lambda_{\Pi}(\phi)\Lambda_T^m|00\rangle\rangle, \\ &= C \cos(2\phi),\end{aligned}\tag{5.37}$$

where Π is the parity signal, $|\pi\rangle\rangle = |00\rangle\rangle + |11\rangle\rangle - |01\rangle\rangle - |10\rangle\rangle$, ϕ describes the rotation axis of the differential pulse, and C is the parity contrast. The differential parity pulses are parallel single qubit rotations $\hat{R}_{+\phi}^{(L)}(\pi/2)$ applied to the left qubit, and $\hat{R}_{-\phi}^{(R)}(\pi/2)$ to the right qubit, with rotation axis $\sigma_{\phi} = \cos(\phi)\sigma_x + \sin(\phi)\sigma_y$. The parity contrast, which effectively measures the overlap between the final mixed state after RB, and the pure Bell state $|\Psi^+\rangle$, (this is a typical reduced tomography method, see ref. [sackett_experimental_2000]), has the form,

$$C = \frac{1}{3}(\alpha_{tr}^m - \alpha_{dt}^m)\tag{5.38}$$

with α_{tr} and α_{dt} given in equation 5.29 in terms of $\{\varepsilon_L, \varepsilon_R, \varepsilon_{corr}\}$. The maximum contrast seen in figure 5.2 (b), $C_{\max} = 1/3$ after noise incident on the initial state $|00\rangle$ can be understood by directly inspecting the space of possible pure two qubit states reached through common only rotations. The general pure state after common rotations is the separable identical two-qubit states,

$$|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle\tag{5.39}$$

with $|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$ for a single qubit general state. Depolarising into the symmetric separable two-qubit states is equivalent to averaging over the full Bloch sphere,

$$\begin{aligned}\bar{\rho}_{\text{sym}} &= \frac{1}{4\pi} \int_0^\pi \int_0^{2\pi} (\sin(\theta) |\Psi\rangle\langle\Psi|) d\phi d\theta \\ &= \frac{1}{3} |00\rangle\langle 00| + \frac{1}{3} |11\rangle\langle 11| + \frac{1}{3} |\Psi^+\rangle\langle\Psi^+|\end{aligned}\tag{5.40}$$

where the third term is the Bell state $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$, with the expected $\frac{1}{3}$ population.

Although here the measured outcome, $\Pi(\phi)$, suggests that multiple samples across ϕ are needed to fit a cosine and extract the contrast C , in fact, only a single RB experiment with fixed $\phi = 0$, executing a global rotation, can be used to recover C .
 TODO Add discussion on $\Pi + C$ to find single exponential decays.

5.9.4 Sample cost

TODO Currently have numerics on this, however Jordan may be able to look into this in more detail.

5.9.5 State preparation errors

Similar to previous work on subspace leakage benchmarking [sutherland_subspace_2025], state preparation errors can lead to non-trivial modifications of the correlated population decay curves. First, we consider independent state preparation on 2 qubits. We model independent errors as probabilistic bit flips. To first order in error, on a single qubit we have,

$$\rho_{sp}^{(i)} = (1 - \varepsilon_{sp}^{(i)}) |0\rangle \langle 0| + \varepsilon_{sp}^{(i)} |1\rangle \langle 1|, \quad (5.41)$$

where the superscript refers to which qubit, L , or R , the preparation error is relevant to. This error may be included in Liouville form $\Lambda_{sp}^{(i)}$ to equation 5.30, by acting on the initial state,

$$\mathcal{P}_{L,R} = \langle \langle L, R | \Lambda_T^m \Lambda_{sp}^{(L)} \Lambda_{sp}^{(R)} | 00 \rangle \rangle. \quad (5.42)$$

In this picture, the state preparation errors scale only the overlap of the initial state with the sub-representations. The overlaps are multiplied by,

$$A_{sp}^{(i)} = (1 - \varepsilon_{sp}^{(i)}), \quad (5.43)$$

if the qubit is involved non-trivially in the subspace projector, i.e. for the left and right qubit subsectors $A_{sp} = A_{sp}^{(i)}$. Relevant to this work, the correlated survival

populations with state preparation errors included are,

$$\mathcal{P}_{S,S} = \frac{1}{4} + \frac{A_{sp}^{(L)}}{4}\alpha_L^m + \frac{A_{sp}^{(R)}}{4}\alpha_R^m + \frac{A_{sp}^{(L)}A_{sp}^{(R)}}{6}\alpha_{dt}^m + \frac{A_{sp}^{(L)}A_{sp}^{(R)}}{12}\alpha_{tr}^m. \quad (5.44)$$

If state preparation errors are not independent... TODO extend section.

5.9.6 Noise injection methods

Deriving equation ?? used to fit detuning data. Applying detuning of 729-nm laser δ from the carrier frequency. The propagator for a detuned carrier pulse is,

$$\hat{U}_\delta(t) = \exp\left(\frac{-i}{2}t(\Omega\hat{\sigma}_x - \delta\hat{\sigma}_z)\right), \quad (5.45)$$

where Ω is the Rabi frequency. For a $\pi/2$ pulse set duration, $\tau = \frac{\pi}{2\Omega}$,

$$\hat{U}_\delta(\tau) = \exp\left(\frac{-i\pi}{4}(\hat{\sigma}_x - \eta\hat{\sigma}_z)\right), \quad (5.46)$$

where we have set $\eta = \delta/\Omega$. Calculate fidelity of $\pi/2$ pulse with this detuning over Haar measure on Bloch sphere

$$\mathcal{F}_{avg} = \frac{|\text{Tr}(U_{id}^\dagger U_\delta)|^2 + 2}{6}, \quad (5.47)$$

where $\hat{U}_{id}(\tau) = \exp\left(\frac{-i\pi}{4}\hat{\sigma}_x\right)$, Evaluating for the above expressions, and assuming $\eta \ll 1$, we find fidelity,

$$\mathcal{F}_{avg} = 1 - \frac{\eta^2}{3} + O(\eta^4). \quad (5.48)$$

The error per $R_\phi(\pi/2)$ gate is $\varepsilon = 1 - \mathcal{F}_{avg}$. Assuming ideal twirling, the error of consecutive pulses should be proportional to sequence length. In the chosen decomposition of Clifford gates, there are on average $k = 2.17$ $R(\pi/2)$ pulses (both virtual and physical). The detuning offset is reparameterised as $\delta = \delta_0 + \delta_L$, where δ_L is the setting of the laser detuning from the bare qubit frequency, and δ_0 is the frequency offset when the qubit is dressed by the laser induced Stark shift. Finally, accounting for a correlated error source that does not originate from a frequency offset, ε_0 , the desired expression for correlated error per Clifford is found,

$$\varepsilon_{corr} = \varepsilon_0 + k\left(\frac{\delta_L - \delta_0}{\Omega}\right)^2. \quad (5.49)$$

- Describe goal of noise injection (explore parameter space)
- List methods of noise injection: stark shift, phase inject, detuning, gate duration balancing.
- Describe experimental calibration of noise injection.
- Give analytic form for expected noise quantity.
- Highlight here regime of $p_{\text{diff}} > p_{\text{com}}$

- 1) Motivate data by saying we want to explore different error regimes.
- 2) Inject non-correlated noise by random addressing of single qubits.
 - 2a) Give simple linear equation of injected vs measured, assuming some base error.
 - 2b) Highlight that correlated noise does not increase.
 - 2c) Highlight limitation of method $\varepsilon_{\text{com}} > \varepsilon_{\text{diff}}$.
- 3) Inject correlated noise by parallel addressing of both qubits.
 - 3a) Use same simple linear equation of injected vs measured as 2a.

5.9.7 Non-Markovian noise

“k-copy” RB may also be used to probe non-Markovian (time correlated) noise, which is of fundamental interest for QEC schemes [**<empty citation>**]. Repeating identical Clifford RB sequences at two distinct times on a single qubit and measuring correlated survival populations is analogous to parallel application of “k-copy” RB on a pair of qubits. The analysis of the correlated populations is identical; however, common noise no longer discerns spatial correlations but rather indicates temporal correlations. A simple example of temporally correlated noise is shown for a single

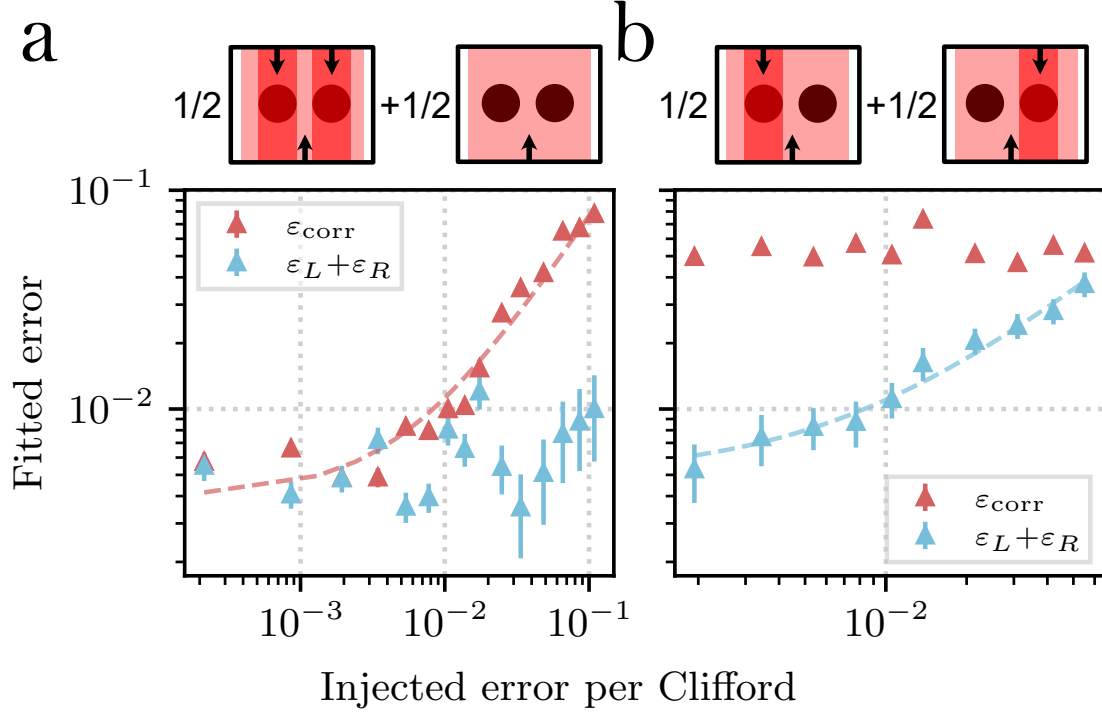


Figure 5.5: Correlated and uncorrelated errors are injected via a single-addressed Stark-shift beam, and quantified through fitting the correlated success populations. To inject correlated errors, the Stark-shift beams randomly address either both or neither qubit after each Clifford gate with equal probability, whilst uncorrelated errors are generated by randomly addressing only one ion at a time. These four addressing configurations are depicted visually above the plot. The wide beam signifies the on-resonance global drive which executes the parallel single-qubit rotations, while the narrow beams represent the off-resonant Stark shift beams addressing single-qubits (black circles). Both panels show the extracted correlated and depolarising type errors per Clifford as function of injected error. In (a) the errors are correlated between qubits, while panel (b) shows the extracted errors after independent errors are injected. A constant correlated error offset is injected in the independent error experiment via random phase jitters of the drive field. This is done to reduce the number of samples required for error extraction with low systematic offsets. The dashed lines are fits to equation ???. Error bars are the extracted standard deviations from a non-linear Levenberg–Marquardt fit of the underlying correlated populations.

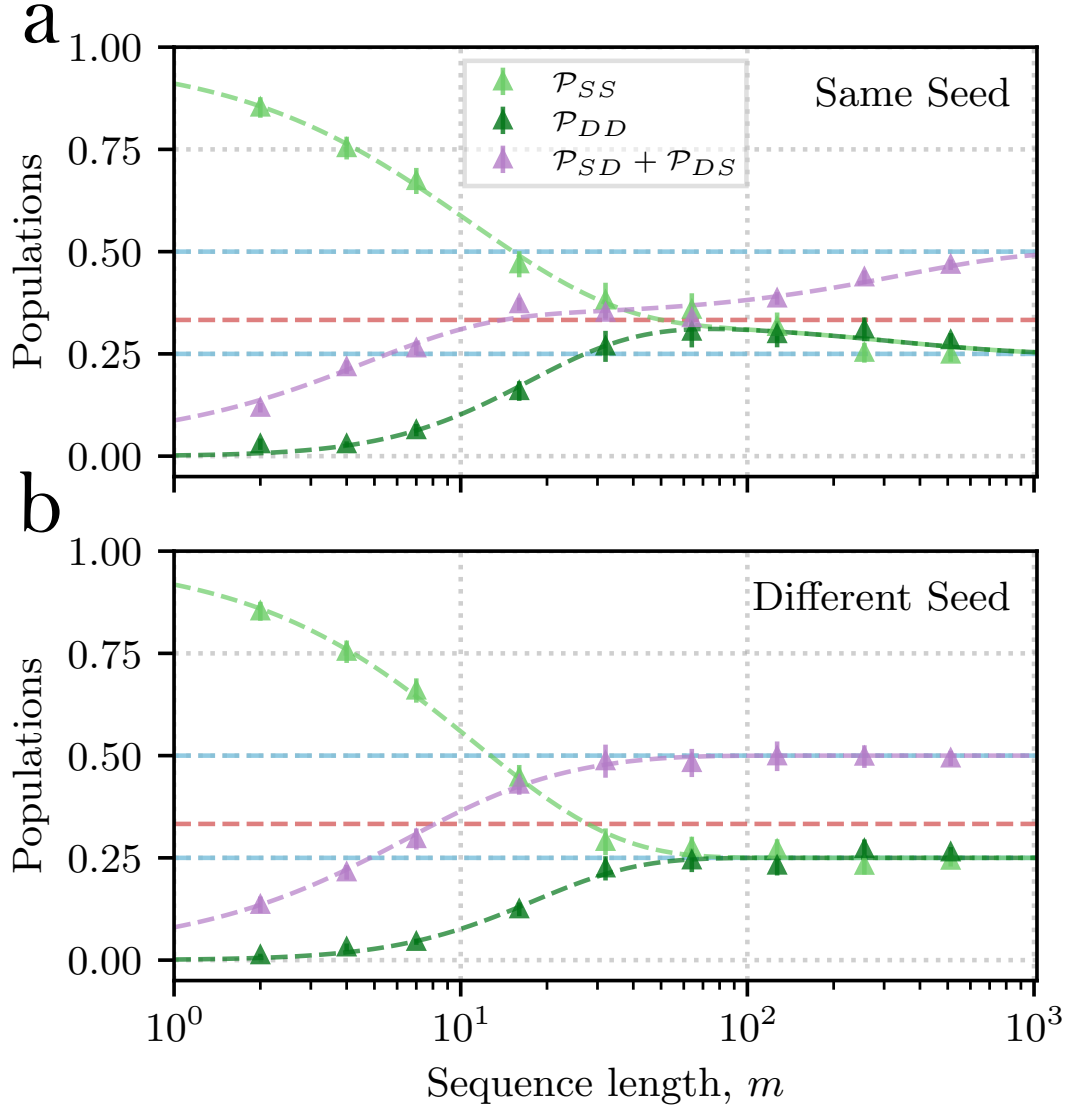


Figure 5.6: Discerning time dependent noise through repeating Clifford RB on a single qubit at two different times. (a) shows correlated populations when identical errors are present at different times, while (b) has independent errors at the both times.

trapped Calcium ion with noise injected via random phase jitter of the 729-nm laser. Identical Clifford RB sequences are executed one minute apart for three repeats. For the first two repeats, the random phase offsets are generated using the same random seed, and the final repeat, a different random seed is used. The time correlated populations can be seen in figure 5.6 for both identical and non-identical errors in time. For identical errors, $\varepsilon_{\text{com}} =, \varepsilon_L =, \varepsilon_R =,$ while for non-identical errors, correlated errors are converted to locally independent: $\varepsilon_{\text{com}} =, \varepsilon_L =, \varepsilon_R =.$

6

Motional Interactions

Unsure yet if this will be a chapter, or a section in the outlook?

Presentation of the experimental results demonstrating motional control

- Describe Two-SDF toolkit
- Describe tomography method for motional states. Could show Fock state here.
- Results:
 - Squeezed state
 - Beam splitter
 - Parity operation
 - Fredkin gate

7

Virtual Distillation

Presentation of the experimental results demonstrating virtual distillation.

- two copy, B matrix demonstration
- Virtual distillation on spatially correlated noise
- any-to-any two-qubit gate
- $\pi/4$ calibration
- Motional virtual distillation. What errors does this mitigate? How to implement it? Results?

8

Conclusion and Outlook

Appendices



Appendix

A.1 Generating Ions

A.2 Extracting Laser Offset and Magnetic Field

A.3 Experimental Control

A.4 Creating Squeezed States

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